EXAM PACK

EEE421

Solid State Devices



Exam Pack

Session 2020–21 *Solutions and formula sheet of the 4th Year 1st Semester Final Examination, 2024*

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Question Paper Solutions 2020–21 — pages 3--110 — 108 pages

2 FORMULA SHEET
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Prepared by Nur Mohammad

November 2026

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Question Paper Solutions 2020–21

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PAST PAPER SOLUTIONS

EEE421

Solid State Devices

Question Paper Solutions

Session 2020–21

4th Year 1st Semester Final Examination, 2024

QUESTION MAP

Q1	Q2	Q3	Q4	Q5	Q6
25 marks	25 marks	25 marks	25 marks	25 marks	25 marks

EACH ANSWER

Theory → Answer → Keyword Explanation → Notes and Remarks → Final Concise Answer

Prepared by Nur Mohammad

November 2026

BACK OF THE COVER

The paper at a glance

3 h	100	4 of 6	45 min
Total time	Total marks	Questions to answer	Per question

Time = marks × 1.8 min (180 min for 100 marks). Tick the questions you will answer; each sub-part links to its solution.

Q	Sub-part	Marks	Time	Pick
Part-A · Answer any two of the following questions				
Q1	(a) Fermi Level Position in Si Versus Doping	5	9 min	
	(b) Minority Carriers, Diffusion and Poisson's Equation	5	9 min	☐
	(c) Number of Quantum States Near the Conduction Band Edge	5	9 min	
	(d) Carrier Concentrations and Donor Ionization at 300 K and 30 K	10	18 min	
Q2	(a) Energy Band Diagram of a Compensated Semiconductor	5	9 min	
	(b) The Einstein Relation	5	9 min	☐
	(c) Width of the Depletion Region	5	9 min	
	(d) Fermi Level Probability and Electron Concentration	10	18 min	
Q3	(a) Electron and Hole Concentrations from Charge Neutrality	5	9 min	
	(b) Energy Band Diagram of a PN Junction in Equilibrium	5	9 min	☐
	(c) Drift Mobility and Carrier Scattering	5	9 min	
	(d) Steady-State Excess Carriers in p-Type Silicon	10	18 min	
Part-B · Answer any two of the following questions				
Q4	(a) Effect of Temperature on Mobility	5	9 min	
	(b) MOS Energy Band Diagrams and Channel Formation	5	9 min	☐
	(c) Four Operating Modes of a Bipolar Transistor	5	9 min	
	(d) Carrier Concentrations at the Edge of a Biased PN Junction	10	18 min	
Q5	(a) Drift and Diffusion of an Excess Carrier Pulse	5	9 min	
	(b) Minority Carrier Distribution in an NPN Transistor	5	9 min	☐
	(c) The Tunnel Diode	5	9 min	
	(d) The Hall Effect	10	18 min	
Q6	(a) Double-Gate MOS Structure	5	9 min	
	(b) Complete Ionization and Freeze-Out	5	9 min	☐
	(c) Formation of the Inversion Layer	5	9 min	
	(d) Built-In Potential, Space Charge Width and Peak Field	10	18 min	

Formulas, symbols and constants follow Neamen, Semiconductor Physics and Devices (4th ed.).

Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering
4th Year 1st Semester Final Examination, 2024
Course Code: EEE 421, Session: 2020-2021
Course Title: Solid State Devices

Full Marks: 100 **Credit: 3.0** **Total Time: 3 Hours**

Part-A

*[Answer any **two** of the following questions]*

1. (a) Sketch the position of Fermi level in Si at 300 K as a function of doping concentration. 5

   (b) How do you define a minority carrier? Why does it diffuse? What is Poisson's equation and why is it important in PN junction? 5

   (c) Determine the number of quantum states(cm^{-3}) in silicon between E_c and E_c+kT at $T = 300$ K. 5

   (d) A Si sample is doped with $N_d = 1 \times 10^{15}$ cm^{-3} of As atoms. Calculate the electron and hole concentrations and the position of Fermi level position (w.r.t. E_c/E_v) at 300 K, assuming full ionization. Justify the assumption by calculating the probability of doner [*sic*: should be donor] states being occupied by electrons and therefore not ionized (assume, $E_C - E_D = 50$ meV). Repeat the above calculations for $T = 30$ K and comment on the values and assumptions. [$N_c = 2.8 \times 10^{19}$ cm^{-3}, for Si]. 10

2. (a) Draw energy band diagram of a compensated semiconductor. 5

   (b) Derive the Einstein relation relating the mobility and diffusion constant. Also, discuss the significance of the Einstein relation. 5

   (c) Using the expression for V_{bi}, find the width of the depletion region.

   $$W = \left\{ \frac{2\epsilon_s V_{bi}}{e} \left[\frac{N_a + N_d}{N_a N_d} \right] \right\}^{\frac{1}{2}}$$ 5

   (d) Answer the following questions on Fermi level: 10

   i. At what energy level above Fermi level (in unit of kT) the probability of finding an electron is 10% ?

   ii. Find the electron concentration(n_0) for Si at room temperature if Fermi level is 0.5 eV below the conduction band. (at room temperature, $N_c = 2.8 \times 10^{19}$ cm^{-3}).

3. (a) Starting from charge neutrality condition $n_0 + N_a^- = p_0 + N_d^+$, derive the expressions for n_0 and p_0 in terms of N_a, N_d, and n_i. 5

   (b) Sketch Energy-band diagram of a PN junction in thermal equilibrium and show P region, N region, E_C, E_V, E_F, E_{Fi}, $e\phi_{Fn}$, $e\phi_{Fp}$ and eV_{bi}. 5

   (c) Define drift mobility. Explain the mechanism of carrier scattering in semiconductor materials and significance of τ. 5

   (d) For the following questions: 10

   i. Solve the ambipolar transport equation for a semiconductor with $N_a = 5 \times 10^{16}$ cm^{-3} for $x > 0$ at steady state with zero electric field where excess charges are created only at $x = 0$. ($\delta n_0 = 5 \times 10^{15}$ cm^{-3}).

   ii. Find the distance where the excess charge will drop to 10×10^{13} cm^{-3}. (Assume $\tau_{n0} = 7 \times 10^{-7}$ s, $\tau_{p0} = 5 \times 10^{-7}$ s, $D_n = 30$ cm^2 s^{-1}, $D_p = 25$ cm^2 s^{-1})

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iii. Plot n and p as function of x in log scale.

Part-B

*[Answer any **two** of the following questions]*

4. (a) Explain the effects of temperature on carrier mobility inside a semiconductor. 5

   (b) Draw the energy band diagram of a Metal-Oxide-Semiconductor structure for the three modes of operation of MOSFET and thus explain how the channel is formed. 5

   (c) Show the four operating modes of a bipolar transistor as a function of V_{BE} and V_{CB}. Draw the minority carrier distribution of the four modes of a NPN transistor with proper labelling of all the quantities. 5

   (d) Consider a PN junction at room temperature with $n_i = 1.5 \times 10^{10}$ cm^{-3}, assume doping in N side is 1×10^6 cm^{-3} [*sic*: should be 1×10^{16} cm^{-3}]. What will be the minority carrier hole concentrations at the edge in N side if forward bias and reverse bias of 0.7 V is applied? Do the same for majority carrier electron concentration in N side. 10

5. (a) Write the ambipolar equation for $g' = 0$ for n type semiconductor with constant electric field E_0. At time $t = 0$, a number of electron-hole pair is generated at $x = 0$. Sketch excess electron concentration as a function of distance at different time. 5

   (b) For NPN transistor, plot minority carrier concentration in base, emitter, and collector as a function of space. What are the minority carrier concentrations at the boundaries of base, emitter, and collector? 5

   (c) Explain the working principle of Tunnel diode. Draw its characteristics curve. 5

   (d) Current is flowing through an extrinsic semiconductor placed in a magnetic field as shown in Fig. 1. 10

   i. What is the Hall voltage?
   ii. What is type of carrier?
   iii. What is the carrier concentration?
   iv. What is the mobility?

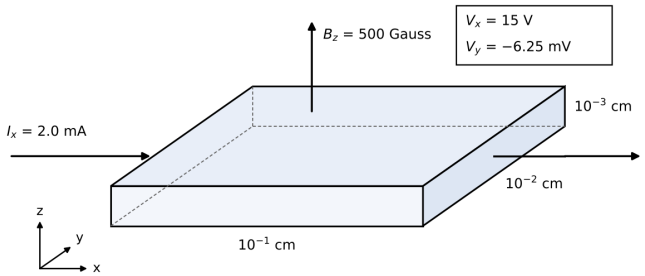


Fig. 1: A bar of an extrinsic semiconductor in a magnetic field (redrawn from the paper)

6. (a) Draw the schematic of a double-gate MOS structure. What are the benefits of such structure over single-gate MOS? 5

   (b) Briefly explain the terms complete ionization and freeze out. Can we achieve complete ionization in room temperature? Prove your answer. 5

   (c) Explain production of inverse layer at oxide and semiconductor layer with proper sketches of energy band diagram. 5

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(d) A silicon PN junction at $T = 300$ K with zero applied bias has doping concentration of $N_d = 5 \times 10^{16}$ cm^{-3} and $N_a = 5 \times 10^{15}$ cm^{-3}. Determine built-in potential, space charge width and electric field at metallurgical region ($\epsilon_t = 11.7$ [*sic*: should be $\epsilon_r = 11.7$]) 10

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1. (a) Sketch the position of Fermi level in Si at 300 K as a function of doping concentration. Marks: 5
   (b) How do you define a minority carrier? Why does it diffuse? What is Poisson's equation and why is it important in PN junction? Marks: 5
   (c) Determine the number of quantum states(cm^{-3}) in silicon between E_c and E_c+kT at $T = 300$ K. Marks: 5
   (d) A Si sample is doped with $N_d = 1 \times 10^{15}$ cm^{-3} of As atoms. Calculate the electron and hole concentrations and the position of Fermi level position (w.r.t. E_c/E_v) at 300 K, assuming full ionization. Justify the assumption by calculating the probability of doner [*sic*: should be donor] states being occupied by electrons and therefore not ionized (assume, $E_C - E_D = 50$ meV). Repeat the above calculations for $T = 30$ K and comment on the values and assumptions. [$N_c = 2.8 \times 10^{19}$ cm^{-3}, for Si]. Marks: 10

1.(a) Sketch the position of Fermi level in Si at 300 K as a function of doping concentration. Marks: 5

1. Theory

1. Energy bands: where electrons can sit (Core Idea)

In a single atom an electron can only have certain energies. In a crystal, billions of atoms sit close together, and each allowed energy spreads into a **band** of very many closely spaced levels. For a semiconductor two bands matter:

- The **valence band**, whose top edge is E_v. At 0 K it is completely full of electrons. These are the electrons of the covalent bonds.
- The **conduction band**, whose bottom edge is E_c. At 0 K it is completely empty.

Between them is the **band gap** $E_g = E_c - E_v$, where no electron states exist. For Si, $E_g = 1.12$ eV.

When an electron gains enough energy to jump from the valence band into the conduction band, two carriers appear:

- a free **electron** in the conduction band, which can move and carry current;
- an empty state left in the valence band, the **hole**. It behaves like a positive carrier.

The density of free electrons is written n_0 and of holes p_0 (cm^{-3}); the subscript 0 means thermal equilibrium.

2. The Fermi-Dirac function and the Fermi level

Electrons obey the Pauli principle: at most one electron per quantum state. In thermal equilibrium the probability that a state of energy E holds an electron is the **Fermi-Dirac function**

$$f_F(E) = \frac{1}{1 + \exp\left(\dfrac{E - E_F}{kT}\right)}$$

where

- E_F is the **Fermi level** (Fermi energy), in eV
- $k = 8.617 \times 10^{-5}$ eV/K is Boltzmann's constant and T the absolute temperature; $kT = 0.0259$ eV at 300 K

What it says:

- At $E = E_F$, $f_F = 1/2$. So the Fermi level is the energy where a state is **half likely** to be filled.
- At $T = 0$ K, f_F is a step: every state below E_F is full, every state above is empty.
- At $T > 0$ the step is rounded over a few kT. A few states above E_F are filled and a few below are empty.

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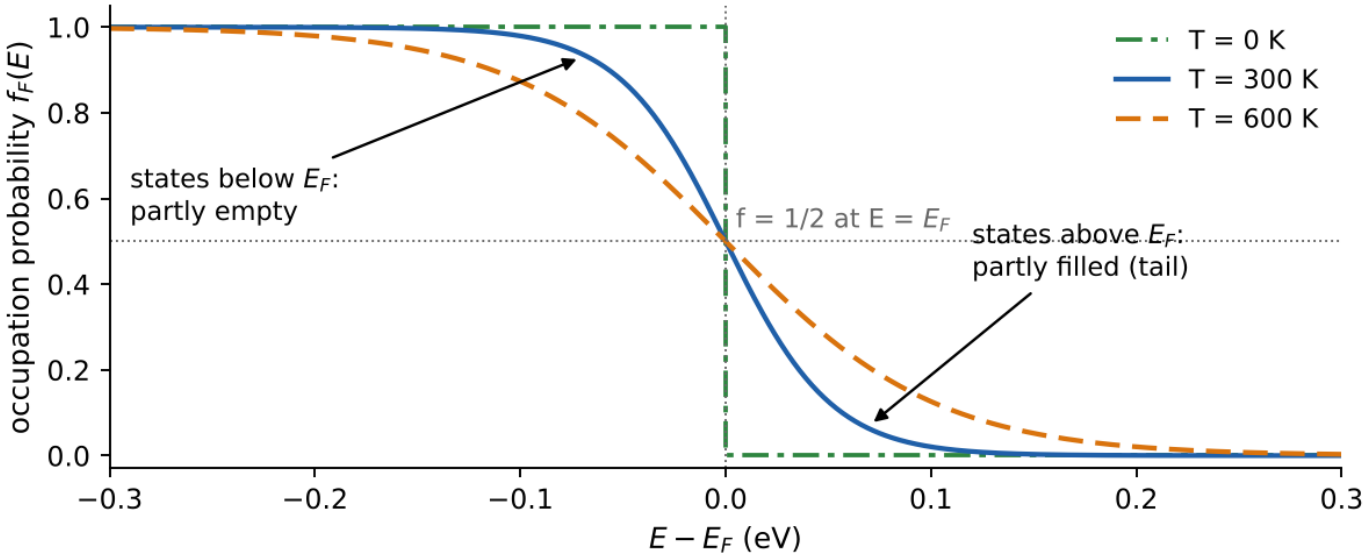


Figure 1: The Fermi-Dirac function at 0 K, 300 K and 600 K

Reading the figure. The dash-dot green line is the sharp step at 0 K. The solid blue curve (300 K) and the dashed orange curve (600 K) all pass through $f_F = 1/2$ at $E = E_F$. The hotter the crystal, the wider the region where the probability changes from 1 to 0. Already $3kT$ (0.078 eV) above E_F, f_F is only 0.047 at 300 K.

Boltzmann approximation. When $E - E_F \gg kT$ (in practice more than about $3kT$), the 1 in the denominator can be dropped:

$$f_F(E) \approx \exp\left[-\frac{E - E_F}{kT}\right]$$

At $E - E_F = 3kT$ this gives 0.0498 against the exact 0.0474: about 5 % error, and less further away.

3. The electron and hole concentrations

The conduction band has many states ($g_c(E)$ per unit energy and volume, counted in 1(c)). Multiply the number of states by the chance each is full, and add up over the band. If E_F is more than a few kT below E_c, the Boltzmann form applies and the result is (Neamen)

$$n_0 = N_c \exp\left[-\frac{E_c - E_F}{kT}\right], \qquad N_c = 2\left(\frac{2\pi m_n^* kT}{h^2}\right)^{3/2}$$

In the same way, a hole is a state that is **empty**, with probability $1 - f_F(E)$, so

$$p_0 = N_v \exp\left[-\frac{E_F - E_v}{kT}\right], \qquad N_v = 2\left(\frac{2\pi m_p^* kT}{h^2}\right)^{3/2}$$

where

- N_c, N_v are the **effective densities of states** of the conduction and valence band; for Si at 300 K, $N_c = 2.8 \times 10^{19}$ and $N_v = 1.04 \times 10^{19}$ cm^{-3}
- m_n^*, m_p^* are the density-of-states effective masses of electrons and holes

The picture is simple: **the closer E_F is to E_c, the more electrons; the closer it is to E_v, the more holes.** Each $kT \ln 10 = 0.0596$ eV that E_F moves towards a band edge multiplies that carrier density by 10.

4. The intrinsic semiconductor and the intrinsic Fermi level

In a pure (**intrinsic**) crystal, every electron in the conduction band left a hole behind, so $n_0 = p_0 = n_i$, the **intrinsic carrier concentration**. For Si at 300 K, $n_i = 1.5 \times 10^{10}$ cm^{-3} (Neamen). Multiplying the two formulas above, E_F cancels:

$$n_0 p_0 = N_c N_v \exp\left(-\frac{E_g}{kT}\right) = n_i^2$$

This **mass-action law** holds for any non-degenerate semiconductor in equilibrium, doped or not. The Fermi level of the intrinsic crystal is called the **intrinsic Fermi level** E_{Fi}. Setting $n_0 = p_0$ gives

$$E_{Fi} - E_{midgap} = \frac{1}{2}kT \ln\left(\frac{N_v}{N_c}\right) = \frac{1}{2}(0.0259)\ln\left(\frac{1.04}{2.8}\right) = -0.0128 \text{ eV}$$

So in Si, E_{Fi} is only 0.013 eV below the centre of the gap: it is practically at midgap. Measured from E_{Fi}, the band edges are $E_c - E_{Fi} = 0.573$ eV and $E_{Fi} - E_v = 0.547$ eV.

5. Doping moves the Fermi level

Doping adds impurity atoms on purpose:

- **Donors** (P, As in Si) give one extra electron each. With N_d donors per cm^3, all ionized at room temperature, $n_0 \approx N_d$. The material is **n-type**.
- **Acceptors** (B in Si) take one electron each from the valence band and create holes, $p_0 \approx N_a$. The material is **p-type**.

Writing n_0 and p_0 relative to the intrinsic case gives the most convenient form:

$$n_0 = n_i \exp\left(\frac{E_F - E_{Fi}}{kT}\right), \qquad p_0 = n_i \exp\left(\frac{E_{Fi} - E_F}{kT}\right)$$

Putting $n_0 = N_d$ in the first and $p_0 = N_a$ in the second:

$$E_F - E_{Fi} = kT \ln\left(\frac{N_d}{n_i}\right) \quad \text{(n-type)}, \qquad E_{Fi} - E_F = kT \ln\left(\frac{N_a}{n_i}\right) \quad \text{(p-type)}$$

The same results measured from the band edges are $E_c - E_F = kT\ln(N_c/N_d)$ and $E_F - E_v = kT\ln(N_v/N_a)$.

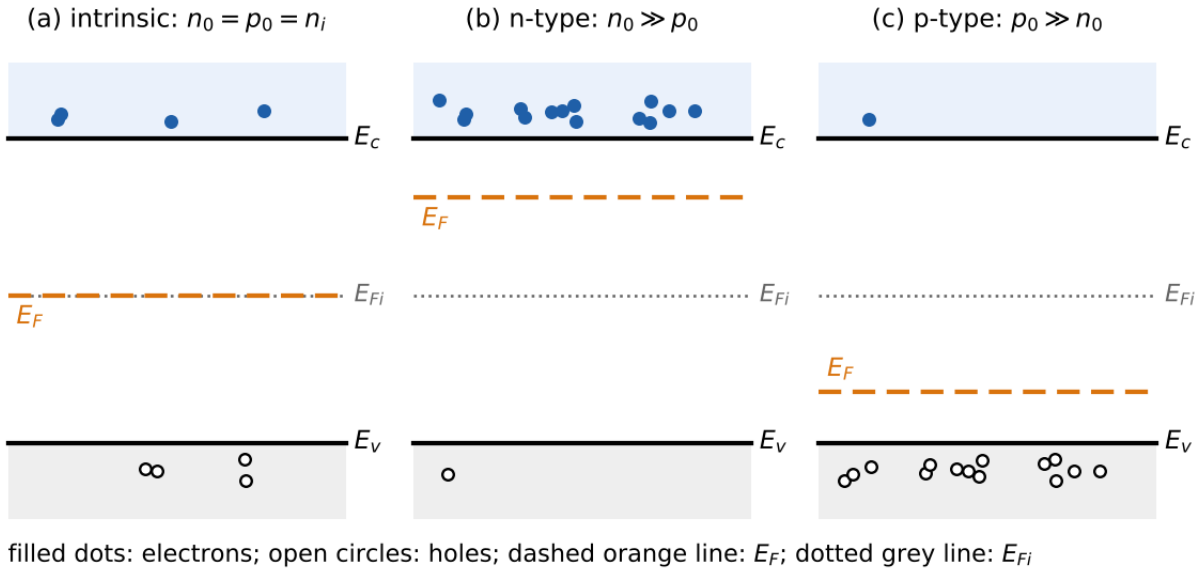


Figure 2: Band diagrams of intrinsic, n-type and p-type silicon with the Fermi level

Reading the figure. The solid black lines are E_c and E_v, the dotted grey line is E_{Fi} and the dashed orange line is E_F. In panel (a) the crystal is intrinsic: $E_F = E_{Fi}$ and there are as many electrons (filled dots) as holes (open circles). In panel (b) donors push E_F up towards E_c: many electrons, very few holes. In panel (c) acceptors pull E_F down towards E_v: many holes, very few electrons.

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6. Limits of the simple formula

The formula is a straight line on a log axis, but it stops being exact at both ends:

- **Very low doping** (N close to n_i): then $n_0 \neq N_d$; use $n_0 = N_d/2 + \sqrt{(N_d/2)^2 + n_i^2}$. E_F goes smoothly to E_{Fi}.
- **Very high doping** ($N \gtrsim 10^{18}$ cm^{-3}): E_F comes within a few kT of the band edge, the Boltzmann approximation fails and the material becomes **degenerate**. The donors are also not all ionized any more. The real E_F then lies a little below the simple line.

2. Answer

Formulas (Si, 300 K, complete ionization, non-degenerate):

$$E_F - E_{Fi} = kT \ln\left(\frac{N_d}{n_i}\right) \quad \text{(n-type)}, \qquad E_{Fi} - E_F = kT \ln\left(\frac{N_a}{n_i}\right) \quad \text{(p-type)}$$

where $kT = 0.0259$ eV, $n_i = 1.5 \times 10^{10}$ cm^{-3}, and E_{Fi} is the intrinsic Fermi level (0.013 eV below midgap; $E_c - E_{Fi} = 0.573$ eV, $E_{Fi} - E_v = 0.547$ eV).

Computed points:

Doping N (cm^{-3})	10^{12}	10^{14}	10^{15}	10^{16}	10^{17}	10^{18}	10^{19}
$\lvert E_F - E_{Fi}\rvert$ (eV)	0.109	0.228	0.288	0.347	0.407	0.467	0.526

Table 1: Summary

Sketch:

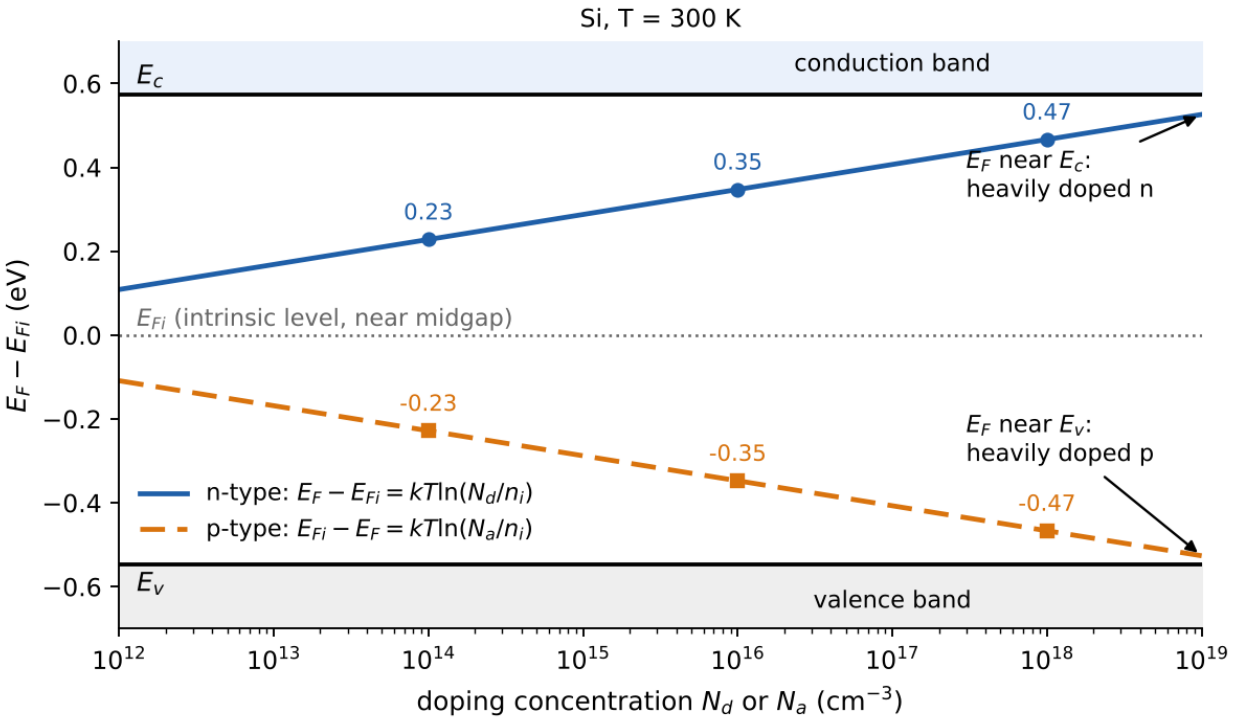


Figure 3: Position of the Fermi level in silicon at 300 K against donor and acceptor concentration

Reading the figure. The solid blue line is E_F in n-type Si (above E_{Fi}), the dashed orange line is E_F in p-type Si (below E_{Fi}). The solid black lines are E_c and E_v, the dotted grey line is E_{Fi}. Both lines are straight on the logarithmic doping axis, with a slope of $kT \ln 10 = 0.0596$ eV per decade.

Key Points:

- Undoped (or $N \ll n_i$) Si has $E_F = E_{Fi}$, close to midgap.

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- Donors raise E_F towards E_c; acceptors lower it towards E_v. The shift is the same size for the same N_d or N_a.
- The shift grows only **logarithmically**: each tenfold increase of doping moves E_F by 0.06 eV.
- At about 10^{19} cm^{-3}, E_F is within 0.05 eV of E_c (n-type) or 0.02 eV of E_v (p-type). Near and beyond this the semiconductor becomes degenerate and the simple formula is no longer accurate.

3. Keyword Explanation

1. Fermi level

The energy E_F at which a state has a 50 % chance of holding an electron. In equilibrium it is constant through the whole sample.

2. Intrinsic Fermi level

The Fermi level E_{Fi} of the undoped crystal, where $n_0 = p_0 = n_i$. In Si it is practically at midgap.

3. Effective density of states

N_c (N_v): the number that, multiplied by the Boltzmann factor $\exp[-(E_c - E_F)/kT]$, gives n_0 (p_0). It acts like all band states squeezed into the band edge.

4. Degenerate semiconductor

A semiconductor so heavily doped that E_F is within a few kT of a band edge, or inside the band. The Boltzmann approximation then fails.

4. Notes and Remarks

- **A small inconsistency in the textbook constants.** With Neamen's N_c, N_v and E_g, $\sqrt{N_c N_v}\exp(-E_g/2kT) = 6.95 \times 10^9$ cm^{-3}, not the quoted $n_i = 1.5 \times 10^{10}$ cm^{-3}. So $E_c - E_F$ found from N_c (0.265 eV at 10^{15} cm^{-3}, used in 1(d)) and from E_{Fi} (0.573 − 0.288 = 0.285 eV) differ by about 0.02 eV. Both are accepted; for one material always use one route.
- **Temperature:** as T rises, n_i grows quickly and E_F moves back towards E_{Fi} (Neamen Fig. 4.19). At very low T the donors freeze out and E_F rises above E_d, see 1(d).
- **Neamen's Fig. 4.18** plots the same curves; some books draw E_F against the band edges, using $E_c - E_F = kT\ln(N_c/N_d)$. The shape is the same.
- The formula $E_F - E_{Fi}$ is used again to find the built-in potential of a PN junction, $V_{bi} = |\phi_{Fn}| + |\phi_{Fp}|$, in 3(b).

1.(b) How do you define a minority carrier? Why does it diffuse? What is Poisson's equation and why is it important in PN junction? Marks: 5

1. Theory

1. Majority and minority carriers (Core Idea)

A semiconductor has two kinds of carriers, electrons and holes (1(a)). Doping makes one kind far more common than the other:

- In **n-type** material, electrons are the **majority carriers** and holes are the **minority carriers**.
- In **p-type** material, holes are the majority and electrons are the minority.

The mass-action law $n_0 p_0 = n_i^2$ fixes the minority density. For n-type Si with $N_d = 10^{15}$ cm^{-3} at 300 K:

$$n_0 = N_d = 10^{15}\ \text{cm}^{-3}, \qquad p_0 = \frac{n_i^2}{N_d} = \frac{(1.5\times10^{10})^2}{10^{15}} = 2.25\times10^5\ \text{cm}^{-3}$$

So there are 10^{15} electrons but only about 2×10^5 holes per cm^3: ten orders of magnitude fewer. Even so, the minority carriers control how diodes and transistors work. The reason is that a small injected number of minority carriers changes their own density by a huge factor, while hardly changing the majority density.

2. Two ways carriers move: drift and diffusion

- **Drift:** an electric field pushes the carriers. Its current is $J_{drf} = e(n\mu_n + p\mu_p)E$ (built in 3(c)).
- **Diffusion:** carriers spread from where they are many to where they are few. No force is needed.

3. Why does diffusion happen?

Every carrier is in constant random **thermal motion**: it moves at about 10^7 cm/s, collides, changes direction, and so on. No direction is preferred.

Now imagine an imaginary plane across the sample, with more carriers on its left than on its right. Each carrier is equally likely to cross the plane either way. But there are more carriers on the left, so **more cross from left to right than from right to left**. The result is a net flow from high to low concentration. Nothing pushes the carriers; the flow is just statistics of random motion. It stops only when the concentration is uniform.

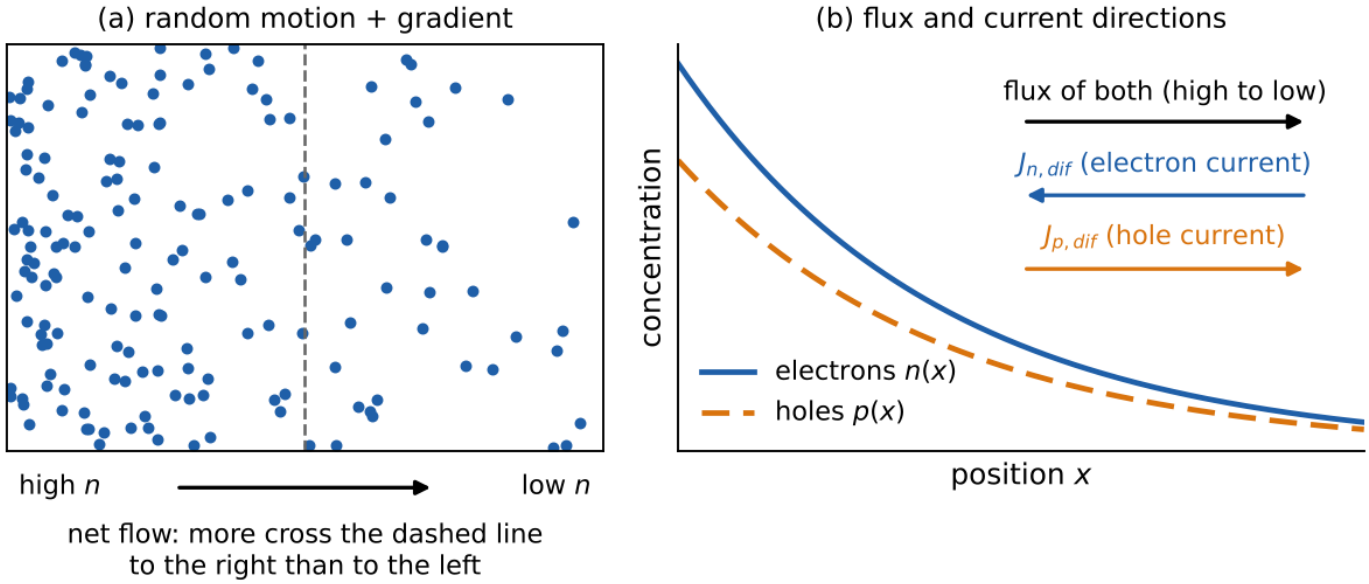


Figure 4: Diffusion: random motion of many carriers with a concentration gradient gives a net flow from high to low concentration

Reading the figure. In panel (a) the carriers (filled dots) are dense on the left and sparse on the right. Each moves at random, but more cross the dashed line to the right than to the left (black arrow). In panel (b) the solid blue curve is an electron profile $n(x)$ and the dashed orange curve a hole profile $p(x)$, both falling with x. The black arrow shows that both kinds of particles flow to $+x$ (high to low). The electron current (blue arrow) points to $-x$, because electrons are negative; the hole current (orange arrow) points to $+x$.

4. Fick's law and the diffusion current

The particle flux (particles per cm^2 per second) is proportional to the concentration gradient (**Fick's law**):

$$F_n = -D_n \frac{dn}{dx}, \qquad F_p = -D_p \frac{dp}{dx}$$

The minus sign says the flow goes *down* the gradient. Multiplying by the charge of each particle ($-e$ for electrons, $+e$ for holes) gives the **diffusion current densities** (Neamen):

$$J_{n,dif} = eD_n \frac{dn}{dx}, \qquad J_{p,dif} = -eD_p \frac{dp}{dx}$$

where

- D_n, D_p are the **diffusion coefficients** (cm^2/s); they are tied to the mobilities by the Einstein relation $D/\mu = kT/e$ (2(b)), so $D_n = 0.0259 \times 1350 = 35$ cm^2/s in Si
- dn/dx, dp/dx are the concentration gradients (cm^{-4})

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Example

if the electron density falls linearly from 10^{16} cm^{-3} to zero over 10 μm, $dn/dx = -10^{19}$ cm^{-4} and $J_{n,dif} = (1.602 \times 10^{-19})(35)(-10^{19}) = -56$ A/cm^2. A modest gradient gives a large current.

The total current is drift plus diffusion: $J_n = en\mu_n E + eD_n\, dn/dx$, and the same for holes.

5. Why minority carriers move mainly by diffusion

In a device, minority carriers are added at one place: injected across a forward-biased junction, or generated by light. Their density is then high there and falls with distance, because they **recombine** with the many majority carriers as they go. This gradient makes them diffuse away from the source. In the neutral regions the field is very small, and the minority density is tiny, so their drift current $ep\mu_p E$ is negligible. **Minority current is almost entirely diffusion current.** The distance they travel before recombining is the diffusion length $L = \sqrt{D\tau}$ (3(d)).

6. Poisson's equation

Gauss's law in one dimension says that the electric field changes wherever there is net charge:

$$\frac{dE}{dx} = \frac{\rho(x)}{\epsilon_s}$$

The field is minus the gradient of the electric potential ϕ, $E = -d\phi/dx$. Substituting gives **Poisson's equation**:

$$\frac{d^2\phi}{dx^2} = -\frac{\rho(x)}{\epsilon_s} = -\frac{dE}{dx}$$

where

- $\rho(x) = e\left(p - n + N_d^+ - N_a^-\right)$ is the net charge density (C/cm^3): holes and ionized donors are positive, electrons and ionized acceptors negative
- $\epsilon_s = \epsilon_r \epsilon_0 = 11.7 \times 8.85 \times 10^{-14} = 1.035 \times 10^{-12}$ F/cm is the permittivity of Si

In words: **charge makes the field change, and the field makes the potential change.** Integrate the charge once to get the field; integrate the field once more to get the potential.

7. Poisson's equation in a PN junction

When p- and n-type regions meet, electrons diffuse into the p side and holes into the n side. They leave behind the fixed ionized dopants: negative acceptors on the p side ($-x_p < x < 0$) and positive donors on the n side ($0 < x < x_n$). This is the **space charge region** (depletion region). In the **depletion approximation** it holds no free carriers, so $\rho = -eN_a$ on the p side and $\rho = +eN_d$ on the n side.

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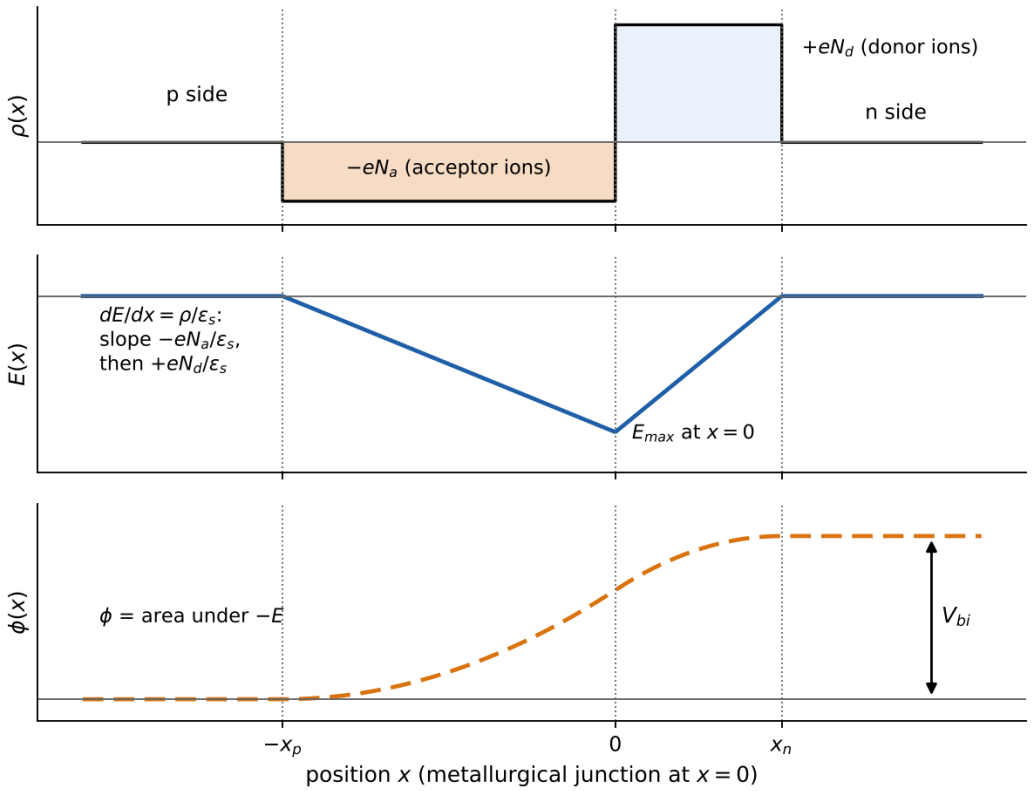


Figure 5: Charge density, electric field and potential of an abrupt PN junction from Poisson's equation

Reading the figure. The top panel is the space charge: a negative block (the shaded acceptor region) and a positive block (the shaded donor region) of equal total charge, $N_a x_p = N_d x_n$. The middle panel (solid blue line) is the field found by integrating once: it is zero outside, falls linearly through the p side, and rises back to zero through the n side. Its peak magnitude E_{max} is at the junction. The bottom panel (dashed orange curve) is the potential, found by integrating again: two parabolas joined at $x = 0$. Its total step is the built-in potential V_{bi}.

Integrating gives the key junction results (derived in full in 2(c)):

$$|E_{max}| = \frac{eN_d x_n}{\epsilon_s} = \frac{eN_a x_p}{\epsilon_s}, \qquad W = x_n + x_p = \sqrt{\frac{2\epsilon_s(V_{bi} + V_R)}{e}\left(\frac{N_a + N_d}{N_a N_d}\right)}$$

2. Answer

Minority carrier: the carrier type that is present in the **smaller** number in a doped semiconductor: holes in n-type, electrons in p-type. Its equilibrium density follows from $n_0 p_0 = n_i^2$. For example, in n-type Si with $N_d = 10^{15}$ cm^{-3}, $p_{n0} = n_i^2/N_d = 2.25 \times 10^5$ cm^{-3}, against 10^{15} cm^{-3} electrons.

Why it diffuses:

- Every carrier is in random thermal motion. Where there is a concentration gradient, more carriers cross any plane from the crowded side than from the sparse side, so there is a net flow from high to low concentration. This is **diffusion**; no force is needed.
- Minority carriers are injected (at a forward-biased junction) or generated (by light) at one place, and recombine as they move away. So their concentration always has a strong gradient, and they flow by diffusion.
- In the neutral regions the field is almost zero and the minority density is tiny, so their drift current is negligible: minority current is essentially a **diffusion current**:

$$J_{n,dif} = eD_n \frac{dn}{dx}, \qquad J_{p,dif} = -eD_p \frac{dp}{dx}$$

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where D_n, D_p are the diffusion coefficients (cm^2/s).

Poisson's equation: it relates the electric potential ϕ (and the field E) to the net charge density ρ:

$$\frac{d^2\phi}{dx^2} = -\frac{\rho(x)}{\epsilon_s} = -\frac{dE}{dx}, \qquad \rho = e\left(p - n + N_d^+ - N_a^-\right)$$

where ϵ_s is the permittivity of the semiconductor.

Why it is important in a PN junction:

- In the space charge region the charge is known: $-eN_a$ on the p side, $+eN_d$ on the n side (depletion approximation).
- Integrating Poisson's equation once gives the **electric field** (triangular, maximum at the junction, $|E_{max}| = eN_d x_n/\epsilon_s$).
- Integrating again gives the **potential** across the junction, whose total step is the **built-in potential** V_{bi} (plus any reverse bias V_R).
- From these come the **depletion width** $W = \sqrt{2\epsilon_s(V_{bi}+V_R)(N_a+N_d)/(eN_aN_d)}$, the **junction capacitance** $C' = \epsilon_s/W$ and the **breakdown** condition (E_{max} reaching the critical field). So all the electrostatics of the junction come from Poisson's equation (sketch of ρ, E and ϕ in the Theory).

3. Keyword Explanation

1. Majority carrier

The carrier present in the larger number: electrons in n-type, holes in p-type. Its density is about the doping, $n_0 \approx N_d$ or $p_0 \approx N_a$.

2. Diffusion coefficient

The constant D (cm^2/s) in Fick's law that measures how quickly carriers spread out; $D = \mu kT/e$.

3. Space charge region

The region around the junction emptied of free carriers, so the fixed dopant ions give a net charge.

4. Depletion approximation

The assumption that the space charge region has no free carriers at all and ends sharply at $-x_p$ and x_n.

4. Notes and Remarks

- **Sign of the diffusion currents:** electrons and holes diffusing the same way give currents in opposite directions, because their charges are opposite. Hence the $+$ in $J_{n,dif}$ and the $-$ in $J_{p,dif}$.
- **Equilibrium junction:** in the space charge region the diffusion current (down the concentration gradient) and the drift current (from the built-in field) of each carrier are large but cancel exactly, so the net current is zero.
- **Electric field symbol:** we write the field as E (as Neamen does). It is never the same as an energy level E_c, E_F in the formulas, which always carry a subscript.

1.(c) Determine the number of quantum states(cm^{-3}) in silicon between E_c and E_c+kT at $T = 300$ K. Marks: 5

1. Theory

1. What is a quantum state? (Core Idea)

Inside a crystal an electron behaves like a wave. Only waves that fit the crystal are allowed, so only certain wave patterns are possible. Each allowed pattern, with one spin direction, is one **quantum state**. By the Pauli principle each state can hold at most one electron.

To know how many electrons the conduction band can hold near its edge, we need to count these states. The **density of states** $g_c(E)$ is the number of states per unit energy per unit volume ($\text{cm}^{-3}\ \text{eV}^{-1}$, or

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$\mathrm{J}^{-1}\ \mathrm{m}^{-3}$ in SI). The number of states between two energies is the area under $g_c(E)$:

$$N = \int_{E_1}^{E_2} g_c(E)\, dE$$

2. Counting states in k-space

Take a cube of crystal of side a. An electron wave must fit the cube, so its wave number in each direction can only be

$$k_x = \frac{n_x \pi}{a}, \quad k_y = \frac{n_y \pi}{a}, \quad k_z = \frac{n_z \pi}{a}, \qquad n_x, n_y, n_z = 1, 2, 3, \ldots$$

So the allowed states are points on a grid in k-space, one point in every volume $(\pi/a)^3$. The number of points between k and $k+dk$ is the volume of a thin spherical shell, $4\pi k^2 dk$, taken in the positive octant only (a factor 1/8, since only positive n are allowed), times 2 for the two spins, divided by $(\pi/a)^3$:

$$g_T(k)\, dk = 2 \cdot \frac{1}{8} \cdot \frac{4\pi k^2\, dk}{(\pi/a)^3} = \frac{k^2 a^3}{\pi^2}\, dk$$

3. From k to energy: the density of states

Near the bottom of the conduction band the electron energy is

$$E = E_c + \frac{\hbar^2 k^2}{2m_n^*} \quad \Rightarrow \quad k = \frac{\sqrt{2m_n^*(E - E_c)}}{\hbar}, \qquad dk = \frac{1}{\hbar}\sqrt{\frac{m_n^*}{2(E - E_c)}}\, dE$$

where m_n^* is the density-of-states effective mass of the electron ($1.08\, m_0$ in Si) and $\hbar = h/2\pi$. Substituting, dividing by the volume a^3 and using $\hbar = h/2\pi$ gives the result (Neamen):

$$g_c(E) = \frac{4\pi\,(2m_n^*)^{3/2}}{h^3}\sqrt{E - E_c} \qquad (E \geq E_c)$$

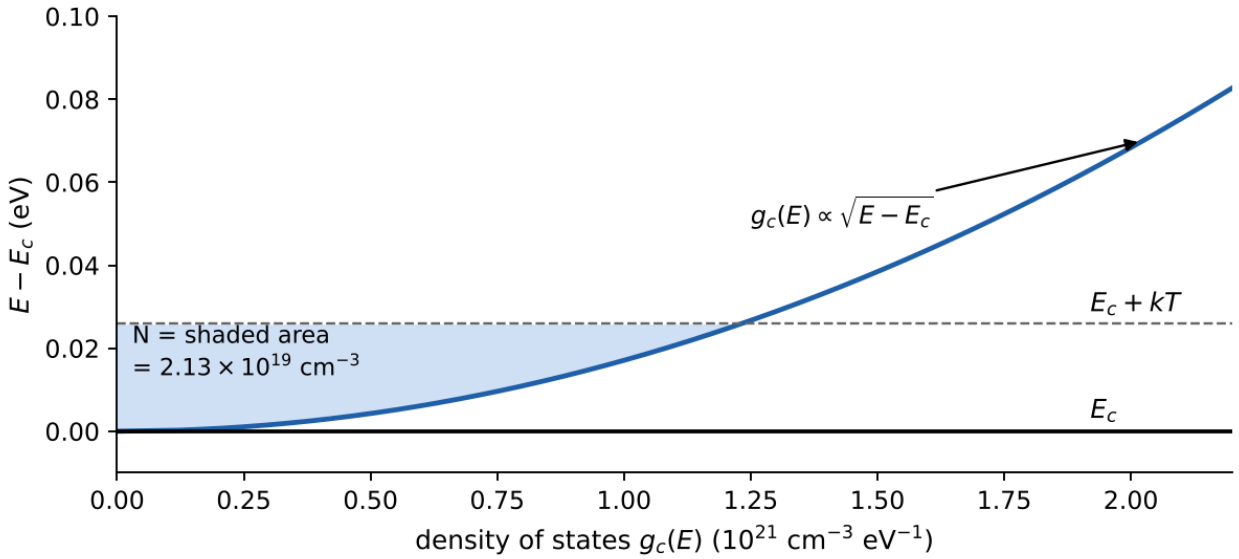


Figure 6: Density of states in the conduction band of Si; the shaded area between E_c and $E_c + kT$ is the number of states asked for

Reading the figure. The solid blue curve is $g_c(E)$ for Si, with energy upwards. It starts at zero at E_c (solid black line) and grows as $\sqrt{E - E_c}$. The shaded area between E_c and the dashed grey line $E_c + kT$ is the number of states asked for in this question.

4. Integrating over a band of width kT

The integral of $\sqrt{E - E_c}$ from E_c to $E_c + kT$ is $\frac{2}{3}(kT)^{3/2}$, so

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$$N = \int_{E_c}^{E_c+kT} g_c(E)\,dE = \frac{4\pi\,(2m_n^*)^{3/2}}{h^3} \cdot \frac{2}{3}(kT)^{3/2}$$

Unit check: $(2m_n^*)^{3/2}$ has units $\text{kg}^{3/2}$, h^3 has $\text{J}^3\,\text{s}^3$, and $(kT)^{3/2}$ has $\text{J}^{3/2}$. With $\text{J} = \text{kg}\,\text{m}^2/\text{s}^2$, the result is m^{-3}: a number of states per unit volume, as it should be. Work in SI, then convert with $1\ \text{m}^{-3} = 10^{-6}\ \text{cm}^{-3}$.

2. Answer

Given:

- Si, $T = 300$ K, $kT = 0.0259$ eV $= 0.0259 \times 1.602 \times 10^{-19} = 4.149 \times 10^{-21}$ J
- $m_n^* = 1.08\,m_0 = 1.08 \times 9.11 \times 10^{-31} = 9.839 \times 10^{-31}$ kg
- $h = 6.625 \times 10^{-34}$ J s

Formula: the number of states per unit volume between E_c and $E_c + kT$ is

$$N = \int_{E_c}^{E_c+kT} \frac{4\pi\,(2m_n^*)^{3/2}}{h^3}\sqrt{E - E_c}\,dE = \frac{4\pi\,(2m_n^*)^{3/2}}{h^3} \cdot \frac{2}{3}(kT)^{3/2}$$

where $g_c(E) = 4\pi(2m_n^*)^{3/2}\sqrt{E - E_c}/h^3$ is the conduction band density of states.

Step 1: The prefactor

$$(2m_n^*)^{3/2} = \left(1.968 \times 10^{-30}\right)^{3/2} = 2.760 \times 10^{-45}\ \text{kg}^{3/2}, \qquad h^3 = \left(6.625 \times 10^{-34}\right)^3 = 2.908 \times 10^{-100}\ \text{J}^3\text{s}^3$$

$$\frac{4\pi\,(2m_n^*)^{3/2}}{h^3} = \frac{4\pi\,(2.760 \times 10^{-45})}{2.908 \times 10^{-100}} = 1.193 \times 10^{56}\ \text{J}^{-3/2}\,\text{m}^{-3}$$

Step 2: The energy factor

$$(kT)^{3/2} = \left(4.149 \times 10^{-21}\right)^{3/2} = 2.673 \times 10^{-31}\ \text{J}^{3/2}$$

Step 3: Number of states

$$N = \left(1.193 \times 10^{56}\right)\left(\frac{2}{3}\right)\left(2.673 \times 10^{-31}\right) = 2.126 \times 10^{25}\ \text{m}^{-3}$$

$$\boxed{N = 2.13 \times 10^{19}\ \text{cm}^{-3}}$$

There are about 2.1×10^{19} quantum states per cm^3 in the first kT (26 meV) of the conduction band. This is of the same order as $N_c = 2.8 \times 10^{19}\ \text{cm}^{-3}$ ($N/N_c = 0.76$), and much more than any normal doping, so there is plenty of room for the electrons.

Final Answer:

- $N = 2.13 \times 10^{25}\ \text{m}^{-3} = 2.13 \times 10^{19}\ \text{cm}^{-3}$ quantum states between E_c and $E_c + kT$ at 300 K

3. Keyword Explanation

1. Quantum state

One allowed wave pattern of the electron, with one spin direction. It holds at most one electron.

2. Density of states

$g_c(E)$: the number of quantum states per unit energy per unit volume at energy E; it grows as $\sqrt{E - E_c}$ near the band edge.

3. Density-of-states effective mass

The mass m_n^* that makes the parabolic formula give the right number of states. For Si it is $1.08\,m_0$, because the six conduction band valleys are counted together.

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4. Notes and Remarks

- **Comparison with N_c:** N_c is the integral of $g_c(E)$ weighted by the Boltzmann factor over the whole band. Our N counts states without weighting, but only up to kT. The two are close by coincidence of scale: the exact ratio is $4/(3\sqrt{\pi}) = 0.752$; with Neamen's rounded N_c it is 0.759.
- **Units trap:** using kT in eV with h in J s gives a wrong answer. Convert kT to joules first, or use h in eV s and m^* consistently.
- Neamen solves this exact problem as an example (Section 3.4) and also gets 2.12×10^{19} cm^{-3}; the last digit depends only on rounding of kT.
- The density of states of the valence band has the same form, with m_p^* and $\sqrt{E_v - E}$.

1.(d) A Si sample is doped with $N_d = 1 \times 10^{15}$ cm^{-3} of As atoms. Calculate the electron and hole concentrations and the position of Fermi level position (w.r.t. E_c/E_v) at 300 K, assuming full ionization. Justify the assumption by calculating the probability of doner [*sic*: should be donor] states being occupied by electrons and therefore not ionized (assume, $E_C - E_D = 50$ meV). Repeat the above calculations for $T = 30$ K and comment on the values and assumptions. [$N_c = 2.8 \times 10^{19}$ cm^{-3}, for Si]. Marks: 10

1. Theory

1. The donor level (Core Idea)

An arsenic atom in Si has five valence electrons. Four form bonds with the neighbouring Si atoms. The fifth is held only weakly by the As^+ ion, like the electron of a hydrogen atom but in a dielectric with a large effective permittivity. The energy needed to free it is small, so its state is drawn as a **donor level** E_d just below E_c. Here $E_c - E_d = 50$ meV, while $E_g = 1.12$ eV.

- If the fifth electron is still bound, the donor is **neutral** (not ionized).
- If it has gone into the conduction band, the donor is a fixed positive ion, N_d^+: **ionized**.

At room temperature $kT = 26$ meV is comparable to 50 meV, and there are far more conduction band states than donors, so almost all donors give up their electron. This is **complete ionization**, the usual assumption $n_0 = N_d$.

2. Probability that a donor holds an electron

The donor level is filled by the Fermi-Dirac rule, but with a factor 1/2, because a donor can hold one electron of either spin but not two (Neamen):

$$f_d = \frac{n_d}{N_d} = \frac{1}{1 + \frac{1}{2}\exp\left(\frac{E_d - E_F}{kT}\right)}$$

where

- n_d is the density of electrons still on donors (neutral donors), so $N_d^+ = N_d - n_d$
- f_d is the probability that a donor state is occupied, i.e. *not* ionized

If E_F is several kT below E_d, f_d is tiny: the donors are ionized. If E_F is above E_d, f_d is close to 1: the donors keep their electrons.

3. Fraction of electrons in donor states

Dividing n_d by all the donor electrons, $n_d + n_0$, and using $n_0 = N_c \exp[-(E_c - E_F)/kT]$, the Fermi level cancels (Neamen, Section 4.4):

$$\frac{n_d}{n_d + n_0} = \frac{1}{1 + \frac{N_c}{2N_d}\exp\left[-\frac{E_c - E_d}{kT}\right]}$$

This is the tool to test complete ionization: if the fraction is small, almost every donor electron is in the conduction band.

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4. Charge neutrality and the Fermi level

In equilibrium the crystal is neutral: $n_0 + N_a^- = p_0 + N_d^+$. For donors only, with p_0 negligible:

$$n_0 = N_d^+ \quad \Rightarrow \quad N_c \exp\left[-\frac{E_c - E_F}{kT}\right] = \frac{N_d}{1 + 2\exp\left[\dfrac{E_F - E_d}{kT}\right]}$$

This has one unknown, E_F. With complete ionization the right side is just N_d and $E_c - E_F = kT \ln(N_c/N_d)$. In general it is solved numerically. The holes then follow from $p_0 = n_i^2/n_0$ (1(a)).

5. Effect of temperature: freeze-out

When the temperature falls, two things change:

- kT becomes much smaller than $E_c - E_d$, so the thermal energy to free the fifth electron is missing;
- $N_c \propto T^{3/2}$ falls, so there are fewer conduction band states to go to.

The electrons then stay on the donors: the donors **freeze out**. E_F moves up above E_d, between E_d and E_c, and n_0 drops far below N_d. At $T = 0$ K no donor is ionized and $n_0 = 0$. At high temperature, on the other hand, n_i eventually exceeds N_d and the material becomes intrinsic. Between the two is the useful **extrinsic** (complete ionization) range. Freeze-out at room temperature for higher doping is examined in 6(b).

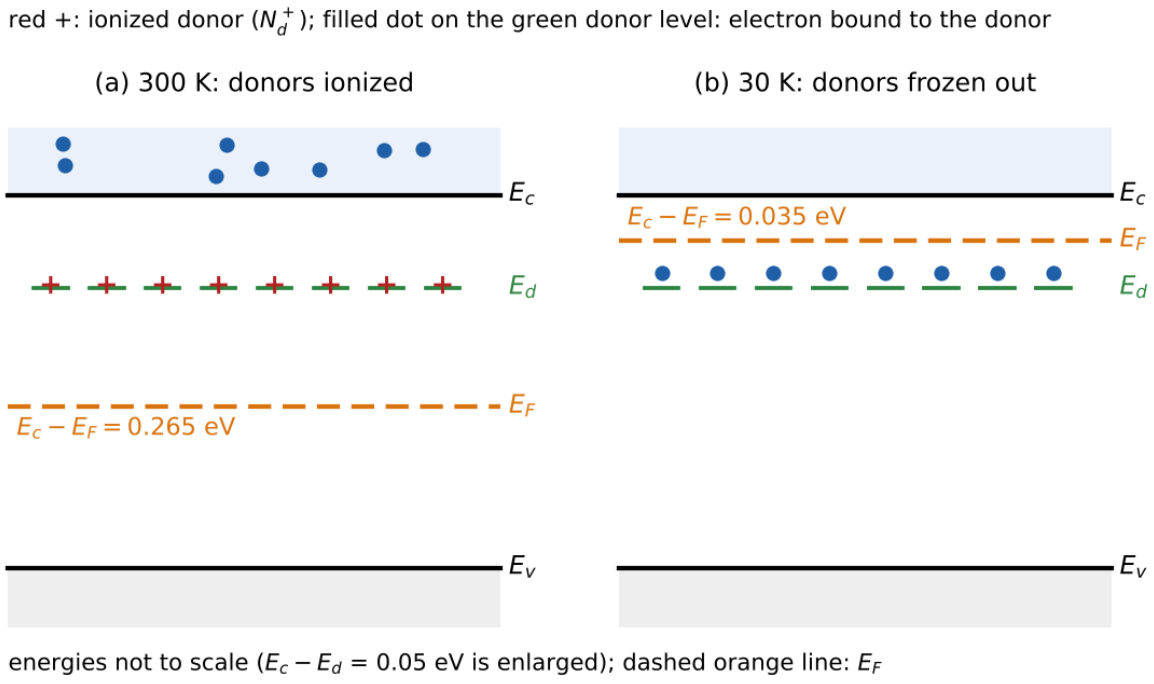


Figure 7: Band diagrams of the As-doped sample at 300 K, with donors ionized, and at 30 K, with donors frozen out

Reading the figure. The short solid green lines are the donor states at E_d, the dashed orange line is E_F. In panel (a), at 300 K, E_F is far below E_d: every donor is an ion (red +) and its electron is in the conduction band (filled dots above E_c). In panel (b), at 30 K, E_F is between E_d and E_c: the electrons sit on the donors (filled dots on the green lines) and the conduction band is almost empty.

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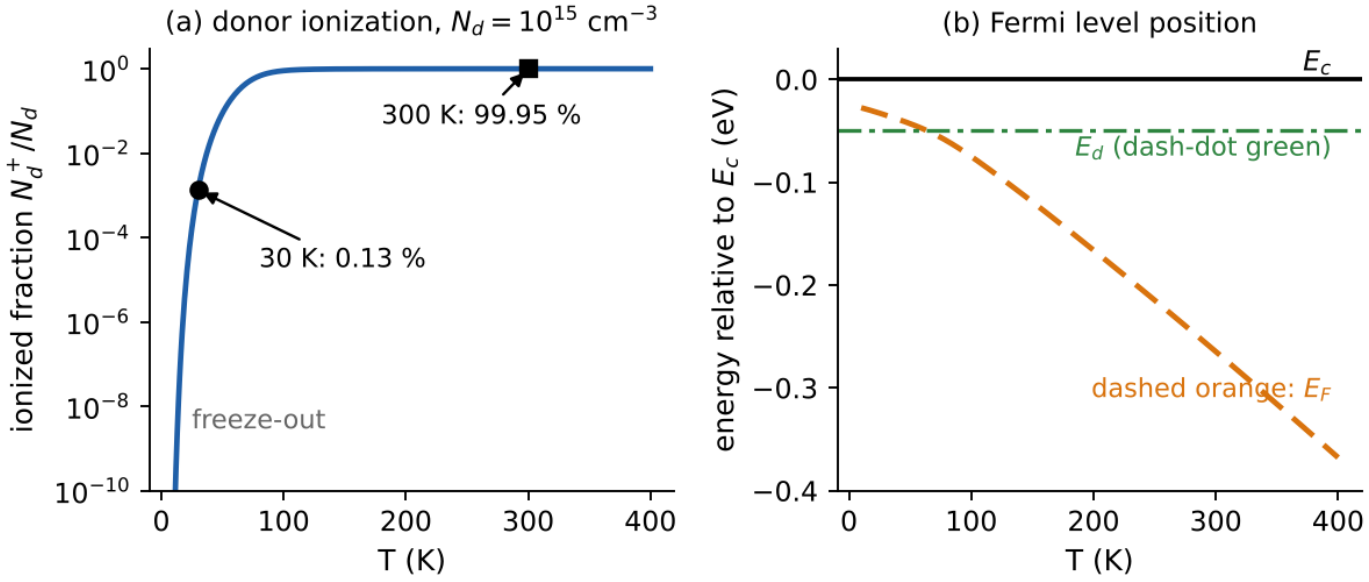


Figure 8: Ionized fraction of the donors and position of the Fermi level against temperature for $N_d = 10^{15}$ cm^{-3}

Reading the figure. In panel (a) the solid blue curve is N_d^+/N_d on a log scale. It is almost 1 above about 150 K (99.95 % at 300 K, filled square) but falls steeply at low temperature (0.13 % at 30 K, filled circle). In panel (b) the dashed orange curve is E_F measured from E_c (solid black line). As T falls it rises and crosses E_d (dash-dot green line) at about 64 K, where one third of the donors are ionized.

2. Answer

Given:

- Si doped with As: $N_d = 1 \times 10^{15}$ cm^{-3}, $N_a = 0$
- $E_c - E_d = 50$ meV $= 0.05$ eV
- $N_c = 2.8 \times 10^{19}$ cm^{-3} at 300 K, $N_v = 1.04 \times 10^{19}$ cm^{-3}, $n_i = 1.5 \times 10^{10}$ cm^{-3}, $E_g = 1.12$ eV
- $kT = 0.0259$ eV at 300 K; $kT = (8.617 \times 10^{-5})(30) = 0.002585$ eV at 30 K

Relationships:

$$n_0 = N_c \exp\left[-\frac{E_c - E_F}{kT}\right], \qquad p_0 = \frac{n_i^2}{n_0}, \qquad \frac{n_d}{n_d + n_0} = \frac{1}{1 + \dfrac{N_c}{2N_d}\exp\left[-\dfrac{E_c - E_d}{kT}\right]}$$

where n_d is the density of electrons still bound to donors (un-ionized donors).

Step 1: Carrier concentrations at 300 K (full ionization)

$N_d \gg n_i$, so

$$n_0 = N_d = 1 \times 10^{15} \text{ cm}^{-3}, \qquad p_0 = \frac{n_i^2}{n_0} = \frac{(1.5 \times 10^{10})^2}{1 \times 10^{15}}$$

$$\boxed{n_0 = 1 \times 10^{15} \text{ cm}^{-3}, \qquad p_0 = 2.25 \times 10^{5} \text{ cm}^{-3}}$$

Step 2: Fermi level at 300 K

$$E_c - E_F = kT \ln\left(\frac{N_c}{n_0}\right) = 0.0259 \ln\left(\frac{2.8 \times 10^{19}}{1 \times 10^{15}}\right) = 0.0259 \times 10.240$$

$$\boxed{E_c - E_F = 0.2652 \text{ eV}, \qquad E_F - E_v = 1.12 - 0.2652 = 0.8548 \text{ eV}}$$

E_F is $0.2652 - 0.05 = 0.2152$ eV $(8.3\,kT)$ below the donor level.

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Step 3: Justifying full ionization at 300 K

$$\frac{N_c}{2N_d} = \frac{2.8\times10^{19}}{2\times10^{15}} = 1.4\times10^4, \qquad \exp\left(-\frac{0.05}{0.0259}\right) = 0.1451$$

$$\frac{n_d}{n_d+n_0} = \frac{1}{1+(1.4\times10^4)(0.1451)} = \frac{1}{1+2031}$$

$$\boxed{\frac{n_d}{n_d+n_0} = 4.92\times10^{-4}}$$

Check with the donor occupation probability: $f_d = 1/[1+\frac{1}{2}\exp(0.2152/0.0259)] = 4.92\times10^{-4}$, the same.
Only 0.049 % of the donors still hold their electron, so **99.95 % are ionized**. The assumption $n_0 = N_d$ is fully justified at 300 K.

Step 4: 30 K, first assuming full ionization

The effective density of states scales as $T^{3/2}$:

$$N_c(30\text{ K}) = 2.8\times10^{19}\left(\frac{30}{300}\right)^{3/2} = 8.854\times10^{17}\text{ cm}^{-3}$$

If $n_0 = N_d = 10^{15}$ cm^{-3} were true:

$$E_c - E_F = 0.002585\ln\left(\frac{8.854\times10^{17}}{1\times10^{15}}\right) = 0.0175\text{ eV}$$

This puts E_F **above** E_d (since 0.0175 < 0.05 eV), which means the donor states are full, not empty. The test formula confirms it:

$$\frac{N_c}{2N_d}\exp\left(-\frac{E_c-E_d}{kT}\right) = (442.7)\exp\left(-\frac{0.05}{0.002585}\right) = (442.7)(3.98\times10^{-9}) = 1.76\times10^{-6}$$

$$\frac{n_d}{n_d+n_0} = \frac{1}{1+1.76\times10^{-6}} = 0.9999982$$

So almost *all* donor electrons should be bound to the donors. This contradicts the starting assumption: **full ionization is not valid at 30 K.**

Step 5: 30 K, correct (self-consistent) solution

Solve charge neutrality $n_0 = N_d^+$ (holes are negligible):

$$N_c\exp\left[-\frac{E_c-E_F}{kT}\right] = \frac{N_d}{1+2\exp\left[\dfrac{E_F-E_d}{kT}\right]}$$

Since almost all donors are neutral, $N_d^+ \approx \frac{N_d}{2}\exp[-(E_F-E_d)/kT]$. Multiplying this by n_0 removes E_F:

$$n_0^2 \approx \frac{N_cN_d}{2}\exp\left(-\frac{E_c-E_d}{kT}\right)$$

$$\Rightarrow \quad n_0 \approx \sqrt{\frac{(8.854\times10^{17})(10^{15})}{2}}\exp\left(-\frac{0.05}{2\times0.002585}\right) = 1.33\times10^{12}\text{ cm}^{-3}$$

The exact numerical solution gives the same:

$$\boxed{n_0 = N_d^+ = 1.33\times10^{12}\text{ cm}^{-3}\ \text{ (only 0.13 % of the donors ionized)}}$$

$$E_c - E_F = kT\ln\left(\frac{N_c}{n_0}\right) = 0.002585\ln\left(\frac{8.854\times10^{17}}{1.327\times10^{12}}\right)$$

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$$\boxed{E_c - E_F = 0.0347 \text{ eV}, \qquad E_F - E_v = 1.12 - 0.0347 = 1.085 \text{ eV}}$$

E_F lies 0.0153 eV ($5.9\,kT$) **above** E_d, so the donors are 99.87 % occupied, consistent with 0.13 % ionized.
Holes: $n_i(30\text{ K}) = \sqrt{N_c N_v}\exp(-E_g/2kT) \approx 10^{-76}$ cm^{-3}, so

$$\boxed{p_0 = \frac{n_i^2}{n_0} \approx 10^{-165} \text{ cm}^{-3} \approx 0}$$

Step 6: Comment

- **300 K:** E_F is far below E_d, 99.95 % of the donors are ionized, and $n_0 = N_d$ is an excellent assumption.
- **30 K:** the thermal energy (2.6 meV) is much less than the 50 meV binding energy, and N_c is 32 times smaller. The electrons **freeze out** onto the donors. E_F rises above E_d. The real n_0 (1.33×10^{12}) is about 750 times lower than N_d, and full ionization would give a wrong Fermi level (0.0175 eV instead of 0.0347 eV below E_c).
- At both temperatures $E_c - E_F > 3kT$ ($10.2\,kT$ at 300 K, $13.4\,kT$ at 30 K), so the Boltzmann approximation used for n_0 is valid.
- The hole concentration is negligible at 300 K and practically zero at 30 K: the sample is strongly n-type in both cases.

Final Answer:

- 300 K: $n_0 = 1 \times 10^{15}$ cm^{-3}, $p_0 = 2.25 \times 10^5$ cm^{-3}, $E_c - E_F = 0.2652$ eV, $E_F - E_v = 0.8548$ eV; fraction of electrons on donors 4.92×10^{-4}, so 99.95 % ionized: assumption valid
- 30 K, if fully ionized: $E_c - E_F = 0.0175$ eV (above E_d) and $n_d/(n_d + n_0) = 0.9999982$: contradiction, assumption invalid
- 30 K, correct: $n_0 = 1.33 \times 10^{12}$ cm^{-3} (0.13 % ionized), $p_0 \approx 10^{-165}$ cm^{-3} ≈ 0, $E_c - E_F = 0.0347$ eV, $E_F - E_v = 1.085$ eV: freeze-out

3. Keyword Explanation

1. Donor level

The energy E_d of the weakly bound fifth electron of a donor atom, a little below E_c (here 50 meV).

2. Complete ionization

The state in which every donor has given its electron to the conduction band, so $n_0 = N_d$.

3. Freeze-out

The state at low temperature in which most donor electrons stay bound to their donors, so $n_0 \ll N_d$ and E_F lies above E_d.

4. Charge neutrality

In equilibrium the total charge is zero: $n_0 + N_a^- = p_0 + N_d^+$.

4. Notes and Remarks

- **Band gap at 30 K:** we kept $E_g = 1.12$ eV (Neamen's room-temperature value) to find $E_F - E_v$ and n_i. The real gap of Si at 30 K is about 1.17 eV. This changes $E_F - E_v$ to about 1.135 eV and makes p_0 even smaller, but none of the conclusions.
- **$n_d/(n_d + n_0)$ and f_d:** the first is the fraction of all donor electrons that are still on donors; the second is the probability that one donor is occupied. When $n_d + n_0 \approx N_d$ (as at 300 K), they are equal. The question's "probability of donor states being occupied" is f_d; both give 4.92×10^{-4}.
- **The arsenic level:** Neamen (Table 4.3) lists about 0.05 eV for the ionization energy of As in Si, so the given 50 meV is realistic; for P it is 0.045 eV (6(b)).

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- **Conductivity in freeze-out** falls mainly because n_0 falls; the change of mobility with temperature is a separate effect (4(a)).
- **Very heavy doping** does not freeze out: the donors are then so close together that their levels merge into a band touching the conduction band.

5. Final Concise Answer

(a) For Si at 300 K, $E_F - E_{Fi} = kT\ln(N_d/n_i)$ in n-type and $E_{Fi} - E_F = kT\ln(N_a/n_i)$ in p-type, with E_{Fi} near midgap. On a log doping axis both are straight lines of slope 0.06 eV per decade: $|E_F - E_{Fi}| = 0.23$, 0.35, 0.47 and 0.53 eV at 10^{14}, 10^{16}, 10^{18} and 10^{19} cm^{-3}. E_F rises towards E_c with donors and falls towards E_v with acceptors, reaching the band edge (degenerate) near 10^{19} cm^{-3}.

(b) A minority carrier is the less numerous carrier (holes in n-type, electrons in p-type; $p_{n0} = n_i^2/N_d$). It diffuses because random thermal motion in a concentration gradient gives a net flow from high to low concentration; minority carriers are injected or generated at one place and recombine as they go, so their current is diffusion current, $J_{n,dif} = eD_n\, dn/dx$. Poisson's equation, $d^2\phi/dx^2 = -\rho/\epsilon_s$ ($dE/dx = \rho/\epsilon_s$), turns the space charge of a PN junction into its electric field, potential (V_{bi}), depletion width W and capacitance.

(c) $N = \int_{E_c}^{E_c+kT} g_c(E)\,dE = \frac{4\pi(2m_n^*)^{3/2}}{h^3} \cdot \frac{2}{3}(kT)^{3/2} = 2.13 \times 10^{25}$ m^{-3} $= 2.13 \times 10^{19}$ cm^{-3}.

(d) At 300 K: $n_0 = 10^{15}$ cm^{-3}, $p_0 = 2.25 \times 10^5$ cm^{-3}, $E_c - E_F = 0.2652$ eV, $E_F - E_v = 0.8548$ eV; only 4.92×10^{-4} of the donor electrons stay on donors (99.95 % ionized), so full ionization is justified. At 30 K, full ionization would put E_F 0.0175 eV below E_c, above E_d, with 99.9998 % of the donors filled: a contradiction. The correct solution is freeze-out: $n_0 = 1.33 \times 10^{12}$ cm^{-3} (0.13 % ionized), $p_0 \approx 0$, $E_c - E_F = 0.0347$ eV, $E_F - E_v = 1.085$ eV.

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2. (a) Draw energy band diagram of a compensated semiconductor. Marks: 5

(b) Derive the Einstein relation relating the mobility and diffusion constant. Also, discuss the significance of the Einstein relation. Marks: 5

(c) Using the expression for V_{bi}, find the width of the depletion region.

$$W = \left\{ \frac{2\epsilon_s V_{bi}}{e} \left[\frac{N_a + N_d}{N_a N_d} \right] \right\}^{\frac{1}{2}}$$ Marks: 5

(d) Answer the following questions on Fermi level: Marks: 10

i. At what energy level above Fermi level (in unit of kT) the probability of finding an electron is 10% ?

ii. Find the electron concentration(n_0) for Si at room temperature if Fermi level is 0.5 eV below the conduction band. (at room temperature, $N_c = 2.8 \times 10^{19}$ cm^{-3}).

2.(a) Draw energy band diagram of a compensated semiconductor. Marks: 5

1. Theory

1. What is a compensated semiconductor? (Core Idea)

A **donor** atom (for example phosphorus in Si) has one electron more than it needs for bonding. It gives that electron away easily and becomes a fixed positive ion. An **acceptor** atom (for example boron in Si) has one electron too few. It takes an electron easily and becomes a fixed negative ion.

A **compensated semiconductor** contains *both* donors and acceptors in the **same region** (Neamen). The two kinds of impurity work against each other: the acceptors "use up" electrons that the donors give, so part of the doping cancels.

- N_d is the donor concentration and N_a the acceptor concentration (cm^{-3}).
- If $N_d > N_a$ the material is a **compensated n-type** semiconductor.
- If $N_a > N_d$ it is a **compensated p-type** semiconductor.
- If $N_d = N_a$ it is **fully compensated**: it has the carrier concentrations of intrinsic material.

Where does it happen? Almost every real device has compensated regions. To make an n-type well inside a p-type wafer, donors are diffused into material that already holds acceptors. Near every diffused PN junction both dopants are present. Even "pure" starting wafers hold a little of the opposite impurity.

2. The energy levels involved

The energy band ideas (E_c, E_v, the Fermi level E_F, the intrinsic level E_{Fi}) are built in 1(a). Here we add the two impurity levels:

- The **donor level** E_d lies just below E_c (0.045 eV below it for P in Si). When the donor holds its extra electron, the electron sits at E_d. When the electron leaves, the donor is **ionized** and carries a charge $+e$.
- The **acceptor level** E_a lies just above E_v (0.045 eV above it for B in Si). When the acceptor accepts an electron, the electron sits at E_a and the acceptor carries a charge $-e$.

So in a band diagram a compensated semiconductor has *both* a row of donor states near the top of the gap and a row of acceptor states near the bottom of the gap.

3. What happens to the electrons? (Why the doping cancels)

At $T = 0$ K every electron goes to the lowest empty energy it can find. An electron at the donor level E_d sees empty acceptor states at E_a, almost a full band gap lower. So it **falls from the donor into the acceptor**. This happens long before it would be thermally excited into the conduction band.

- Each such transfer makes one positive donor ion and one negative acceptor ion. Neither of these creates a free carrier.

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- If $N_d > N_a$, all N_a acceptors are filled by donor electrons. The remaining $N_d - N_a$ donor electrons are the only ones left to go into the conduction band. At room temperature (complete ionization, see 1(d)) they all do.
- If $N_a > N_d$, all N_d donor electrons fall into acceptors. The remaining $N_a - N_d$ acceptors take their electrons from the valence band, leaving $N_a - N_d$ holes.

So the material behaves as if it had only the **net doping** $|N_d - N_a|$.

4. The carrier concentrations

Charge neutrality ($n_0 + N_a^- = p_0 + N_d^+$) together with $n_0 p_0 = n_i^2$ gives, for complete ionization (the full derivation is in 3(a)):

$$n_0 = \frac{N_d - N_a}{2} + \sqrt{\left(\frac{N_d - N_a}{2}\right)^2 + n_i^2}, \qquad p_0 = \frac{n_i^2}{n_0}$$

where

- n_0, p_0 are the thermal-equilibrium electron and hole concentrations (cm^{-3})
- $n_i = 1.5 \times 10^{10}$ cm^{-3} is the intrinsic concentration of Si at 300 K

When $N_d - N_a \gg n_i$ this is simply $n_0 \approx N_d - N_a$. For $N_a > N_d$ the same formula with the roles swapped gives $p_0 \approx N_a - N_d$.

5. Where is the Fermi level?

The Fermi level follows from the carrier concentration exactly as in uncompensated material (1(a)):

$$E_F - E_{Fi} = kT \ln\left(\frac{n_0}{n_i}\right), \qquad E_c - E_F = kT \ln\left(\frac{N_c}{n_0}\right)$$

with $kT = 0.0259$ eV at 300 K and $N_c = 2.8 \times 10^{19}$ cm^{-3}.

- $N_d > N_a$: $n_0 > n_i$, so E_F is **above** E_{Fi}, but **lower** than it would be with the donors alone.
- $N_a > N_d$: $p_0 > n_i$, so E_F is **below** E_{Fi}.
- $N_d = N_a$: $n_0 = p_0 = n_i$, so $E_F = E_{Fi}$, the same place as in intrinsic material.

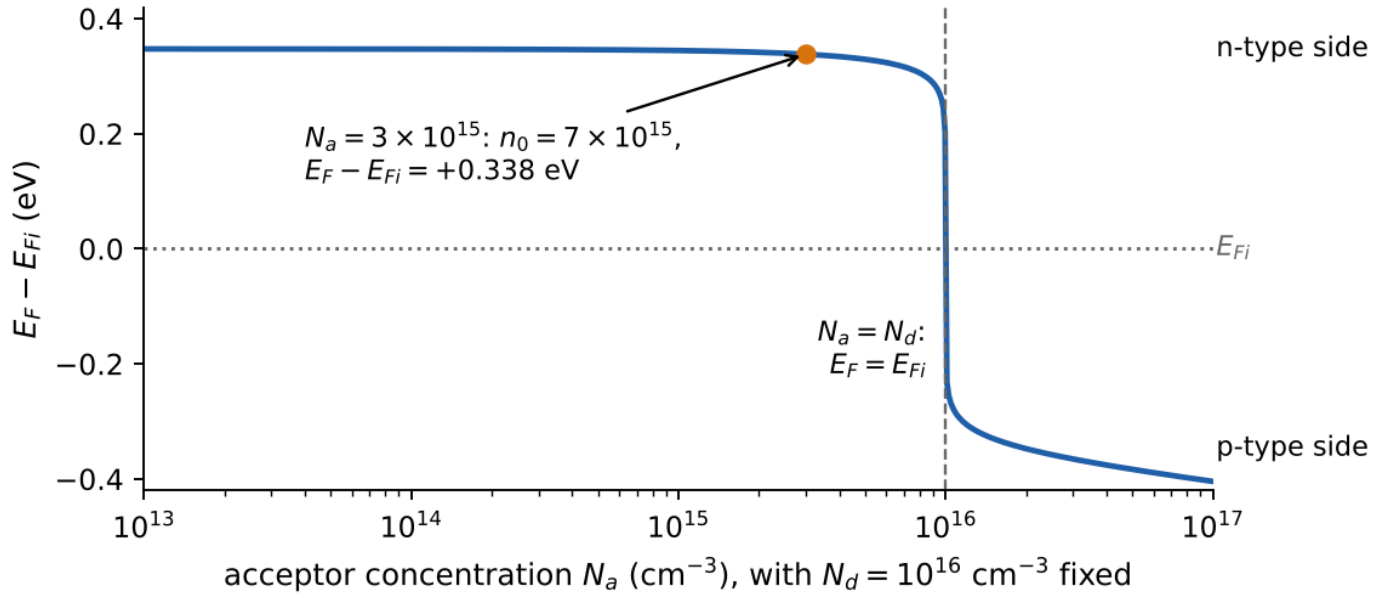


Figure 9: Fermi level position against acceptor concentration in Si with a fixed donor concentration $N_d = 10^{16}$ cm^{-3} at 300 K

Reading the figure. The donor concentration is fixed at $N_d = 10^{16}$ cm^{-3} and acceptors are added. The solid blue curve is $E_F - E_{Fi}$. While $N_a \ll N_d$ it hardly moves (about +0.35 eV). The orange dot

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marks the example of the Answer: $N_a = 3 \times 10^{15}$ cm^{-3} gives +0.338 eV. As N_a approaches N_d (dashed grey vertical line) the net doping vanishes and E_F drops very steeply through E_{Fi} (dotted grey line). Beyond it the material is p-type and E_F falls below E_{Fi}.

2. Answer

A **compensated semiconductor** contains both donor (N_d) and acceptor (N_a) impurities in the same region. Electrons from the donor level E_d fall into the lower-energy acceptor level E_a, so the two dopings cancel and only the net doping $|N_d - N_a|$ produces free carriers.

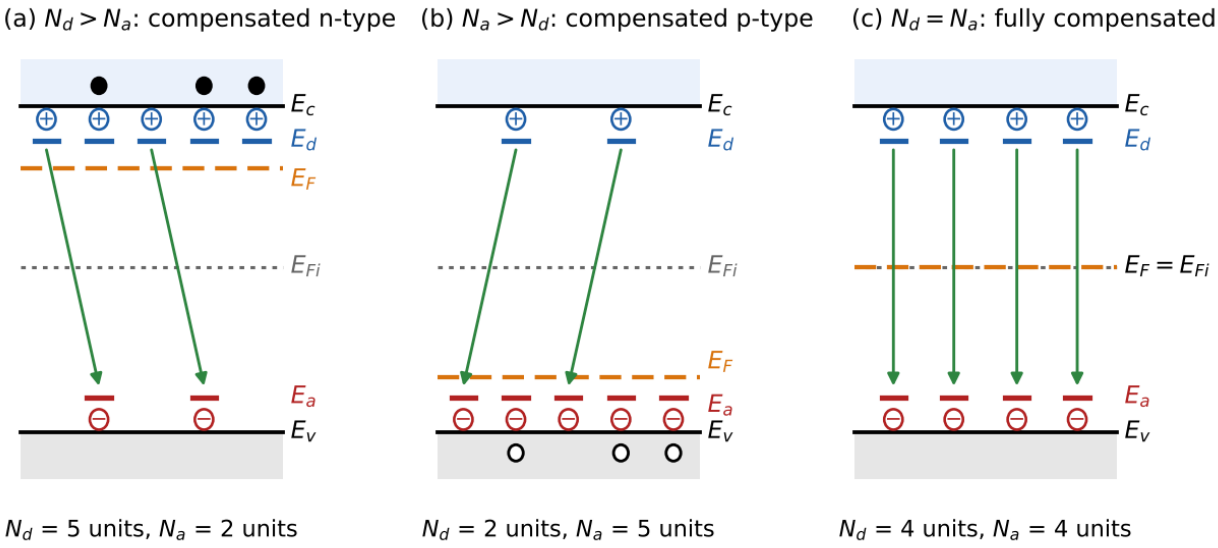


Figure 10: Energy band diagrams of a compensated semiconductor for $N_d > N_a$, $N_a > N_d$ and $N_d = N_a$ (impurity levels not to scale)

Reading the figure. The short solid blue bars are donor states at E_d; the circled + above them shows that every donor is ionized. The short solid red bars are acceptor states at E_a; the circled − below them shows that every acceptor holds an electron. The solid green arrows are electrons falling from donor states into acceptor states. Filled black dots above E_c are free electrons; open circles below E_v are holes. The dashed orange line is E_F and the dotted grey line is E_{Fi}.

Key Points:

- **(a) $N_d > N_a$ (compensated n-type):** N_a of the donor electrons fill all the acceptors; the other $N_d - N_a$ go to the conduction band. $n_0 \approx N_d - N_a$, and E_F lies above E_{Fi}, near E_c.
- **(b) $N_a > N_d$ (compensated p-type):** all donor electrons fall into acceptors; the remaining $N_a - N_d$ acceptors take electrons from the valence band. $p_0 \approx N_a - N_d$, and E_F lies below E_{Fi}, near E_v.
- **(c) $N_d = N_a$ (fully compensated):** every donor electron is captured by an acceptor; no free carriers come from the doping. $n_0 = p_0 = n_i$ and $E_F = E_{Fi}$: the material behaves like intrinsic material.

Carrier concentration (complete ionization, $N_d > N_a$):

$$n_0 = \frac{N_d - N_a}{2} + \sqrt{\left(\frac{N_d - N_a}{2}\right)^2 + n_i^2} \approx N_d - N_a$$

Example

(Si, 300 K, $N_d = 1 \times 10^{16}$, $N_a = 3 \times 10^{15}$ cm^{-3}):

$$n_0 = 7.0 \times 10^{15} \text{ cm}^{-3}, \qquad p_0 = \frac{n_i^2}{n_0} = \frac{(1.5 \times 10^{10})^2}{7.0 \times 10^{15}} = 3.21 \times 10^4 \text{ cm}^{-3}$$

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$$E_F - E_{Fi} = 0.0259 \ln\left(\frac{7.0\times10^{15}}{1.5\times10^{10}}\right) = 0.338 \text{ eV}, \qquad E_c - E_F = 0.0259 \ln\left(\frac{2.8\times10^{19}}{7.0\times10^{15}}\right) = 0.215 \text{ eV}$$

$$\boxed{n_0 \approx 7\times10^{15} \text{ cm}^{-3}, \quad E_c - E_F = 0.215 \text{ eV}}$$

Without the acceptors ($n_0 = 10^{16}$) $E_c - E_F$ would be 0.206 eV, so compensation lowers E_F by about 0.009 eV.

3. Keyword Explanation

1. Donor level

The energy E_d of the extra electron bound to a donor atom, just below E_c. An empty donor state is a fixed positive ion.

2. Acceptor level

The energy E_a of an electron captured by an acceptor atom, just above E_v. A filled acceptor state is a fixed negative ion.

3. Net doping

The difference $|N_d - N_a|$. It sets the majority carrier concentration of a compensated semiconductor.

4. Counter-doping

Adding impurities of the opposite type to an already doped region on purpose, for example to form a well or to adjust a threshold voltage.

4. Notes and Remarks

- **Intrinsic-like, but not intrinsic:** a fully compensated sample has $n_0 = p_0 = n_i$, but it still holds $N_d + N_a$ ionized impurities. Each scatters carriers, so its **mobility is lower** than that of pure material (ionized impurity scattering grows with the **total** $N_d + N_a$, see 3(c)).
- **Exact compensation is hard:** a tiny mismatch $N_d - N_a$ of only 10^{12} cm^{-3} already gives $n_0 \approx 10^{12}$ cm^{-3} and moves E_F 0.109 eV above E_{Fi} (the steep part of the curve in the Theory figure).
- **Low temperature:** in compensated n-type material the N_a electrons trapped by acceptors are not available even at freeze-out, so the donors freeze out differently from uncompensated material; the room-temperature picture above assumes complete ionization.
- **Not to scale:** the figure draws E_d and E_a further from the band edges than the real 0.045 eV, only so that the states and labels can be seen.

2.(b) Derive the Einstein relation relating the mobility and diffusion constant. Also, discuss the significance of the Einstein relation. Marks: 5

1. Theory

1. Two ways a carrier can move (Core Idea)

Charge carriers make a current in two different ways:

- **Drift:** an electric field E pushes the carriers. Electrons drift with velocity $\mu_n E$ against the field, holes with $\mu_p E$ along it. The current densities are $J_{n,drift} = en\mu_n E$ and $J_{p,drift} = ep\mu_p E$. The **mobility** μ (cm^2/V s) measures how easily a carrier moves; it is set by scattering (3(c)).
- **Diffusion:** carriers spread from where there are many to where there are few, just by their random thermal motion, even with no field. The current densities are $J_{n,diff} = eD_n\, dn/dx$ and $J_{p,diff} = -eD_p\, dp/dx$. The **diffusion coefficient** D (cm^2/s) measures how fast they spread (1(b)).

The total electron current is the sum:

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$$J_n = en\mu_n E_x + eD_n \frac{dn}{dx}$$

The question: μ and D look like two separate material numbers. The Einstein relation shows they are not independent.

2. Why should they be related?

Both motions are the **same random walk** of the carrier. Between collisions the carrier flies freely; at each collision its direction is randomised. Drift is this random walk with a small push added; diffusion is the same random walk with more carriers on one side than the other. Both depend on how often the carrier collides (the mean free time). So one number should fix the other.

3. A graded doping makes a built-in field

To find the link, Neamen uses a sample where drift and diffusion must cancel exactly: an n-type semiconductor whose donor concentration $N_d(x)$ **falls** with x, in **thermal equilibrium** (no applied voltage, no current).

- More donors on the left means more electrons on the left. Electrons **diffuse** to the right.
- When they leave, the positive donor ions they leave behind are uncovered on the left, and the electrons pile up a little on the right. This small charge separation creates an **induced electric field** E_x pointing from left to right (from + to −).
- The field pushes electrons to the left (force $-eE_x$). In equilibrium this **drift** exactly cancels the diffusion.

In equilibrium the Fermi level is **flat** (constant in x). Since $n_0 \approx N_d(x)$ changes with x, the distance $E_c - E_F$ must change: E_c rises where N_d is low. A sloping band means an electric field.

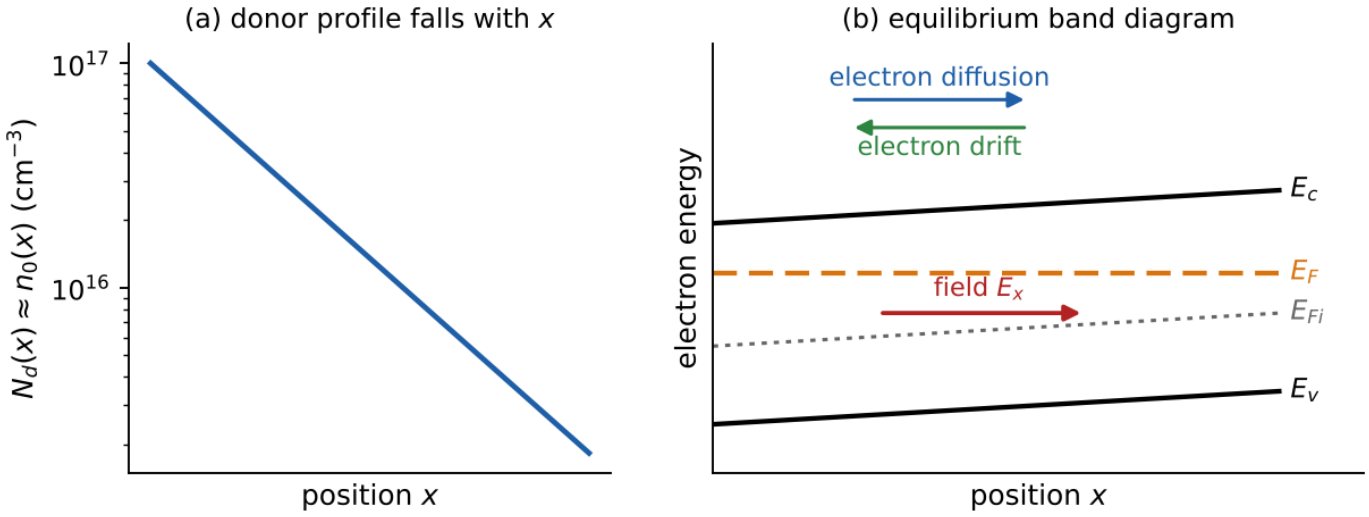


Figure 11: Graded n-type doping: donor profile and the equilibrium band diagram with the induced field

Reading the figure. Panel (a): the solid blue line is the donor concentration, falling with x (log scale). Panel (b): E_F (dashed orange) is flat; E_c and E_v (solid black) and E_{Fi} (dotted grey) all rise to the right, because there are fewer electrons there. The blue arrow shows electrons diffusing to the right (down their concentration gradient); the green arrow shows them drifting to the left, pushed by the induced field (red arrow, pointing to $+x$).

4. Field from a band slope

The electron energy is $E_c = \text{constant} - e\phi$, where ϕ is the electric potential. The field is $E_x = -d\phi/dx$. So

$$E_x = \frac{1}{e}\frac{dE_c}{dx}$$

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A band that rises to the right means a field pointing to the right. Using $n_0 = N_c \exp[-(E_c - E_F)/kT] \approx N_d(x)$ with E_F constant gives the size of the induced field:

$$E_x = -\frac{kT}{e}\,\frac{1}{N_d(x)}\,\frac{dN_d(x)}{dx}$$

Example

an exponential profile $N_d = N_0\, e^{-x/L}$ with $L = 1\ \mu\text{m} = 10^{-4}$ cm gives a constant field $E_x = (kT/e)/L = 0.0259/10^{-4} = 259$ V/cm. Small, but enough to balance the diffusion.

2. Answer

Set-up (Neamen, Sec. 5.3): take a non-uniformly doped n-type semiconductor, $N_d(x)$ decreasing with x, in thermal equilibrium (band diagram in the Theory). Electrons diffuse towards $+x$; the uncovered donor ions create an induced field E_x; the drift it causes cancels the diffusion.

Step 1: Zero electron current

In thermal equilibrium no current flows, so the drift and diffusion parts of the electron current cancel (Neamen, Eq. 5.43):

$$J_n = en\mu_n E_x + eD_n \frac{dn}{dx} = 0$$

Step 2: The electron concentration in terms of the band

In equilibrium the Fermi level E_F is constant in x, and

$$n = N_c \exp\left[\frac{-(E_c - E_F)}{kT}\right]$$

Only E_c changes with x (it follows the doping).

Step 3 (Basic): Derivative of an exponential

For $y = e^{u(x)}$ the chain rule gives $dy/dx = e^u\, du/dx = y\, du/dx$. Here $u = -(E_c - E_F)/kT$, so $du/dx = -\dfrac{1}{kT}\dfrac{dE_c}{dx}$ (E_F and kT are constants), and

$$\frac{dn}{dx} = -\frac{n}{kT}\,\frac{dE_c}{dx}$$

Step 4: The field from the band slope

The electron energy is $E_c = \text{const} - e\phi$, where ϕ is the electric potential, and the field is $E_x = -d\phi/dx$. So $dE_c/dx = -e\, d\phi/dx = eE_x$, and Step 3 becomes

$$\frac{dn}{dx} = -\frac{e\,n}{kT}\,E_x$$

Step 5: Substitute into the zero-current condition

Put Step 4 into Step 1:

$$0 = en\mu_n E_x - eD_n\,\frac{e\,n}{kT}\,E_x = enE_x\left(\mu_n - \frac{eD_n}{kT}\right)$$

The graded doping makes $E_x \neq 0$, and $n \neq 0$, so the bracket is zero:

$$\boxed{\frac{D_n}{\mu_n} = \frac{kT}{e}}$$

(Neamen, Eq. 5.46a).

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Step 6: The same derivation for holes

Take a graded p-type sample, $N_a(x)$ decreasing with x, in equilibrium. The hole current is zero:

$$J_p = ep\mu_p E_x - eD_p \frac{dp}{dx} = 0$$

With $p = N_v \exp[-(E_F - E_v)/kT]$ and only E_v depending on x, the chain rule (Step 3) gives

$$\frac{dp}{dx} = \frac{p}{kT}\frac{dE_v}{dx}$$

E_v also equals a constant minus $e\phi$, so $dE_v/dx = eE_x$ and $dp/dx = \dfrac{e\,p}{kT}E_x$. Then

$$0 = ep\mu_p E_x - eD_p \frac{e\,p}{kT}E_x = epE_x\left(\mu_p - \frac{eD_p}{kT}\right) \quad \Rightarrow \quad \boxed{\frac{D_p}{\mu_p} = \frac{kT}{e}}$$

(Neamen, Eq. 5.46b). So $D_n/\mu_n = D_p/\mu_p = kT/e = V_t$, the thermal voltage: **0.0259 V at 300 K** (Neamen, Eq. 5.47).

Steps to remember

Equilibrium: $J_n = 0 \to n = N_c e^{-(E_c - E_F)/kT}$ with E_F flat $\to dn/dx = -(n/kT)\,dE_c/dx \to dE_c/dx = eE_x \to$ substitute: $\mu_n = eD_n/kT \to$ same for holes.

Significance:

One number gives the other. Mobility is easy to measure (Hall effect, conductivity); the diffusion coefficient is not. The relation gives D directly from μ. For Si at 300 K:

$$D_n = (0.0259)(1350) = 35.0 \text{ cm}^2/\text{s}, \qquad D_p = (0.0259)(480) = 12.4 \text{ cm}^2/\text{s}$$

- **Same physics.** Drift and diffusion are two faces of the same random thermal motion, limited by the same scattering. Whatever lowers the mobility (more doping, higher temperature) lowers the diffusion coefficient in the same ratio.
- **Device equations.** It lets the diode, BJT and MOSFET equations be written with one transport parameter: diffusion lengths $L_n = \sqrt{D_n \tau_{n0}}$ and the diode current use D, which is then taken from μ.
- **Temperature:** the ratio D/μ grows linearly with T.

3. Keyword Explanation

1. Diffusion coefficient

The constant D (cm^2/s) in the diffusion current $J = \pm eD\,dn/dx$; how fast carriers spread down a concentration gradient.

2. Mobility

The drift velocity per unit field, $\mu = v_d/E$ (cm^2/V s).

3. Thermal equilibrium

No external voltage, light or current; the Fermi level is flat and every current (electron and hole separately) is zero.

4. Induced electric field

The built-in field that appears in a non-uniformly doped semiconductor, because majority carriers diffuse away from their ions.

5. Thermal voltage

$V_t = kT/e = 0.0259$ V at 300 K.

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4. Notes and Remarks

- **Neamen's form:** Neamen writes $n = n_i \exp[(E_F - E_{Fi})/kT]$ and $E_x = (1/e)\, dE_{Fi}/dx$. This is the same derivation, because E_c and E_{Fi} are parallel.
- **Validity:** the relation assumes non-degenerate statistics (Boltzmann approximation, E_F more than about $3kT$ below E_c). In degenerately doped material D/μ is larger than kT/e.
- **Why the field is small:** only a tiny charge imbalance is needed, so the graded region stays nearly neutral ($n \approx N_d$), the **quasi-neutrality** approximation used above.
- **Units check:** D/μ has units $(\text{cm}^2/\text{s})/(\text{cm}^2/\text{V s}) = \text{V}$, the unit of kT/e ✔.

2.(c) Using the expression for V_{bi}, find the width of the depletion region.

$W = \left\{ \frac{2\epsilon_s V_{bi}}{e} \left[\frac{N_a + N_d}{N_a N_d} \right] \right\}^{\frac{1}{2}}$ **Marks: 5**

1. Theory

1. What is the space charge region? (Core Idea)

A **PN junction** is one crystal with p-type doping (N_a acceptors) on one side and n-type doping (N_d donors) on the other. Take the **metallurgical junction** (the plane where the doping changes) as $x = 0$, the p side on the left.

- The p side has many holes, the n side many electrons. Holes **diffuse** into the n side and electrons into the p side.
- When a hole leaves the p side it leaves behind a **negative acceptor ion**; when an electron leaves the n side it leaves a **positive donor ion**. These ions are fixed in the lattice and cannot move.
- So a layer near the junction is left with **net charge**: negative on the p side ($-x_p < x < 0$), positive on the n side ($0 < x < x_n$). This is the **space charge region**, also called the **depletion region** because it is emptied of free carriers.
- The charge creates an **electric field** pointing from the + ions to the − ions, from n to p (towards $-x$). This field pushes holes back to the p side and electrons back to the n side. In equilibrium the drift it causes cancels the diffusion (2(b) uses the same balance).

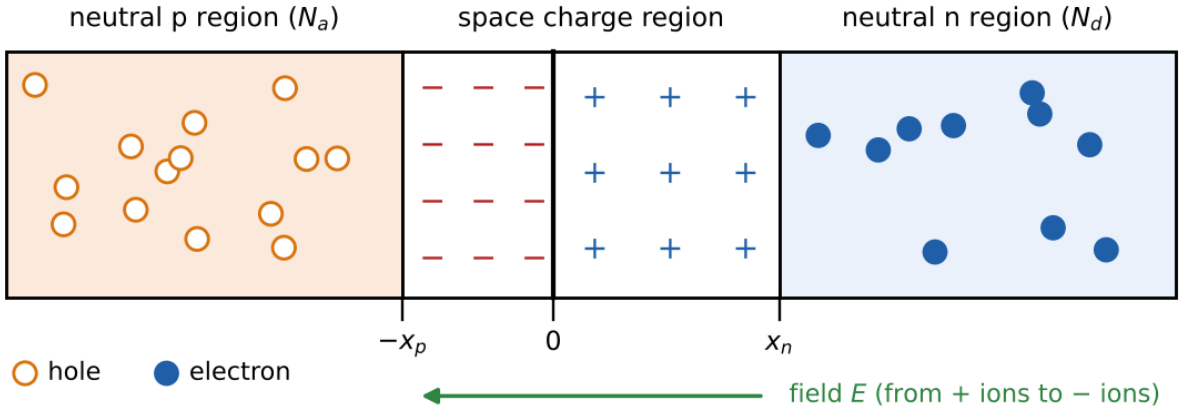


Figure 12: The PN junction in equilibrium: neutral p and n regions and the space charge region of fixed ions

Reading the figure. On the left, the neutral p region holds holes (open orange circles); on the right, the neutral n region holds electrons (filled blue dots). Between $-x_p$ and x_n there are no free carriers, only the fixed negative acceptor ions (red −) and positive donor ions (blue +). The green arrow is the electric field, from the + ions to the − ions. Here $N_a > N_d$, so the p side of the space charge region is the narrower one.

2. The depletion approximation and the charge density

We assume (Neamen) that the space charge region is **completely empty** of free carriers and ends **sharply** at $-x_p$ and x_n; outside it the material is neutral. Then the volume charge density is

$$\rho(x) = -eN_a \quad (-x_p < x < 0), \qquad \rho(x) = +eN_d \quad (0 < x < x_n), \qquad \rho(x) = 0 \text{ elsewhere}$$

Charge balance. The junction as a whole is neutral, so the negative charge per unit area on the p side equals the positive charge on the n side:

$$N_a x_p = N_d x_n$$

The more heavily doped side has the thinner layer.

3. Poisson's equation

Poisson's equation (1(b)) links charge to field and potential:

$$\frac{d^2\phi(x)}{dx^2} = -\frac{\rho(x)}{\epsilon_s} = -\frac{dE(x)}{dx}$$

where

- ϕ is the electric potential (V) and $E = -d\phi/dx$ the electric field (V/cm)
- $\epsilon_s = \epsilon_r \epsilon_0 = 11.7 \times 8.85 \times 10^{-14} = 1.035 \times 10^{-12}$ F/cm for Si

So $dE/dx = \rho/\epsilon_s$: a **constant** charge gives a field that changes **linearly**, and a linear field gives a **parabolic** potential.

4. The built-in potential

The n side sits at a higher potential than the p side by the **built-in potential barrier** V_{bi}. It is found from the band diagram (built in 3(b)):

$$V_{bi} = |\phi_{Fn}| + |\phi_{Fp}| = V_t \ln\left(\frac{N_a N_d}{n_i^2}\right)$$

with $V_t = kT/e = 0.0259$ V. Typical values for Si are 0.6 to 0.9 V. The same V_{bi} must also equal the potential difference across the space charge region that Poisson's equation gives. Setting the two equal fixes the width. That is the derivation the question asks for.

2. Answer

Assumptions: abrupt (step) junction at $x = 0$, p side $-x_p < x < 0$, n side $0 < x < x_n$; depletion approximation; thermal equilibrium; complete ionization.

Charge density (Neamen, Eqs. 7.12a and 7.12b):

$$\rho(x) = -eN_a \quad (-x_p < x < 0), \qquad \rho(x) = +eN_d \quad (0 < x < x_n)$$

Step 1: Electric field

Poisson's equation (Gauss's law in one dimension: the field changes by the charge it passes) gives $dE/dx = \rho/\epsilon_s$ (Neamen, Eq. 7.11). On the p side, integrate once:

$$E(x) = \int \frac{-eN_a}{\epsilon_s}\,dx = -\frac{eN_a}{\epsilon_s}x + C_1$$

The field is zero in the neutral region, so $E(-x_p) = 0$, which gives $C_1 = -eN_a x_p/\epsilon_s$:

$$E(x) = -\frac{eN_a}{\epsilon_s}(x + x_p), \qquad -x_p \le x \le 0$$

(Neamen, Eq. 7.14.)

On the n side, integrate $+eN_d/\epsilon_s$ once:

$$E(x) = \int \frac{eN_d}{\epsilon_s}\,dx = \frac{eN_d}{\epsilon_s}x + C_2$$

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With $E(x_n) = 0$, $C_2 = -eN_d x_n/\epsilon_s$:

$$E(x) = -\frac{eN_d}{\epsilon_s}(x_n - x), \qquad 0 \le x \le x_n$$

(Neamen, Eq. 7.16.)
The field must be continuous at $x = 0$. Putting $x = 0$ in both expressions:

$$\frac{eN_a x_p}{\epsilon_s} = \frac{eN_d x_n}{\epsilon_s} \quad \Rightarrow \quad N_a x_p = N_d x_n$$

which is the charge balance (Neamen, Eq. 7.17). The field is a triangle; its largest magnitude is at the junction: $|E_{max}| = eN_a x_p/\epsilon_s = eN_d x_n/\epsilon_s$.

Step 2: Potential

The potential is $\phi = -\int E\,dx$. On the p side:

$$\phi(x) = \int \frac{eN_a}{\epsilon_s}(x + x_p)\,dx = \frac{eN_a}{2\epsilon_s}(x + x_p)^2 + C_3$$

Take $\phi(-x_p) = 0$ as the reference, which gives $C_3 = 0$:

$$\phi(x) = \frac{eN_a}{2\epsilon_s}(x + x_p)^2, \qquad -x_p \le x \le 0$$

(Neamen, Eq. 7.21.)
At the junction this gives $\phi(0) = eN_a x_p^2/2\epsilon_s$. On the n side:

$$\phi(x) = \int \frac{eN_d}{\epsilon_s}(x_n - x)\,dx = \frac{eN_d}{\epsilon_s}\left(x_n x - \frac{x^2}{2}\right) + C_4$$

The potential is continuous at $x = 0$, so $C_4 = \phi(0) = eN_a x_p^2/2\epsilon_s$:

$$\phi(x) = \frac{eN_d}{\epsilon_s}\left(x_n x - \frac{x^2}{2}\right) + \frac{eN_a}{2\epsilon_s}x_p^2, \qquad 0 \le x \le x_n$$

(Neamen, Eq. 7.25.)

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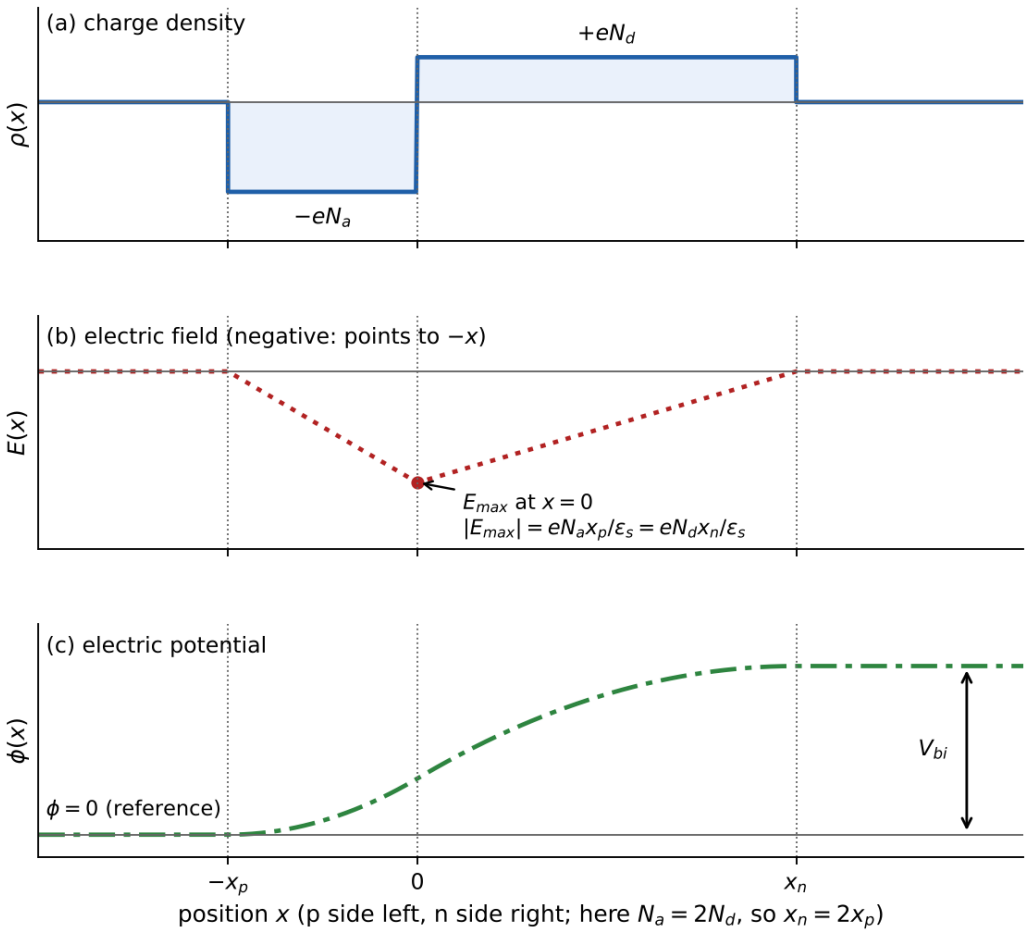


Figure 13: Charge density, electric field and electric potential across an abrupt PN junction in equilibrium

Reading the figure. Panel (a): the solid blue line is $\rho(x)$, a negative rectangle $-eN_a$ on the p side and a positive rectangle $+eN_d$ on the n side, of equal area. Panel (b): the dotted red line is $E(x)$, a triangle, zero at both edges and most negative at $x = 0$. Panel (c): the dash-dot green curve is $\phi(x)$, two parabolas joined at $x = 0$, rising from 0 to V_{bi}.

Step 3: Built-in potential from the potential

The total potential difference across the region is $V_{bi} = \phi(x_n) - \phi(-x_p) = \phi(x_n)$. Put $x = x_n$ in the n-side expression. The bracket becomes $x_n^2 - x_n^2/2 = x_n^2/2$:

$$V_{bi} = \phi(x_n) = \frac{e}{2\epsilon_s}\left(N_d x_n^2 + N_a x_p^2\right)$$

(Neamen, Eq. 7.26.)

Step 4: Solve for the edges

Put $x_p = N_d x_n / N_a$ (charge balance). The second term becomes $N_a x_p^2 = N_a (N_d x_n/N_a)^2 = N_d^2 x_n^2 / N_a$:

$$V_{bi} = \frac{eN_d x_n^2}{2\epsilon_s}\left(1 + \frac{N_d}{N_a}\right) = \frac{eN_d x_n^2}{2\epsilon_s} \cdot \frac{N_a + N_d}{N_a}$$

Solve for x_n, then use $x_p = N_d x_n / N_a$:

$$x_n = \left\{\frac{2\epsilon_s V_{bi}}{e}\left[\frac{N_a}{N_d}\right]\left[\frac{1}{N_a + N_d}\right]\right\}^{1/2}, \qquad x_p = \left\{\frac{2\epsilon_s V_{bi}}{e}\left[\frac{N_d}{N_a}\right]\left[\frac{1}{N_a + N_d}\right]\right\}^{1/2}$$

(Neamen, Eqs. 7.28 and 7.29.)

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Step 5: Total width

$$W = x_n + x_p = \left\{\frac{2\epsilon_s V_{bi}}{e(N_a+N_d)}\right\}^{1/2}\left(\sqrt{\frac{N_a}{N_d}}+\sqrt{\frac{N_d}{N_a}}\right)$$

Put the bracket over a common denominator:

$$\sqrt{\frac{N_a}{N_d}}+\sqrt{\frac{N_d}{N_a}} = \frac{N_a+N_d}{\sqrt{N_aN_d}}$$

Then, moving this factor inside the square root,

$$W = \left\{\frac{2\epsilon_s V_{bi}}{e(N_a+N_d)}\cdot\frac{(N_a+N_d)^2}{N_aN_d}\right\}^{1/2}$$

One factor (N_a+N_d) cancels:

$$\boxed{W = \left\{\frac{2\epsilon_s V_{bi}}{e}\left[\frac{N_a+N_d}{N_aN_d}\right]\right\}^{1/2}}$$

with $V_{bi} = V_t \ln(N_aN_d/n_i^2)$ (Neamen, Eqs. 7.31 and 7.10). This is the required result.
Check: the area of the field triangle is $\frac{1}{2}|E_{max}|W$, and it must equal V_{bi}, so $|E_{max}| = 2V_{bi}/W$ ✔.

Steps to remember

ρ: $-eN_a$ and $+eN_d$ → integrate Poisson once with $E=0$ at both edges → triangle field, continuity gives $N_ax_p = N_dx_n$ → integrate again → two parabolas → $V_{bi} = \phi(x_n)$ → put in $x_p = N_dx_n/N_a$ → x_n, x_p → add for W.

Example

for Si with $N_d = 5\times10^{16}$, $N_a = 5\times10^{15}$ cm^{-3} at 300 K, $V_{bi} = 0.718$ V and $W = 0.452$ μm ($x_n = 0.041$ μm, $x_p = 0.411$ μm); worked in 6(d).

3. Keyword Explanation

1. Space charge (depletion) region

The layer around the metallurgical junction that has lost its free carriers and holds only fixed ionized dopants.

2. Depletion approximation

The model in which the space charge region is fully depleted with sharp edges, so that $\rho = -eN_a$ or $+eN_d$ inside it and zero outside.

3. Built-in potential barrier

The potential difference V_{bi} between the neutral n and p regions in equilibrium; it holds the majority carriers back.

4. Poisson's equation

$d^2\phi/dx^2 = -\rho/\epsilon_s$: the link between a charge distribution and the potential it makes.

4. Notes and Remarks

- **Reverse bias:** with an applied reverse voltage V_R the barrier becomes $V_{bi}+V_R$, so $W = \{2\epsilon_s(V_{bi}+V_R)(N_a+N_d)/(eN_aN_d)\}^{1/2}$; the region widens.
- **One-sided junction:** if $N_a \gg N_d$ (a p$^+$n junction), $W \approx \{2\epsilon_s V_{bi}/eN_d\}^{1/2}$ and almost all of it lies on the lightly doped n side.

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- **Sign of the field:** the field is negative (it points towards $-x$, from n to p). Many books quote only its magnitude.
- **Reference point:** Neamen takes $\phi = 0$ at $-x_p$. Another choice of zero shifts $\phi(x)$ but not V_{bi} or W.

2.(d) Answer the following questions on Fermi level: i. At what energy level above Fermi level (in unit of kT) the probability of finding an electron is 10% ? ii. Find the electron concentration(n_0) for Si at room temperature if Fermi level is 0.5 eV below the conduction band. (at room temperature, $N_c = 2.8 \times 10^{19}$ cm^{-3}). Marks: 10

1. Theory

The basics

In a solid, electrons can have only certain allowed energies, called states. At absolute zero, they fill the lowest states. At a higher temperature, heat lifts some of them higher. So we ask for the probability that a state at energy E holds an electron. The energy at which this probability is one half is the **Fermi level**.

1. The Fermi-Dirac function (a short recap)

The full derivation of the Fermi-Dirac function and of n_0 is in 1(a). In short: the probability that an allowed state at energy E is occupied by an electron is

$$f_F(E) = \frac{1}{1 + \exp\left(\dfrac{E - E_F}{kT}\right)}$$

where

- E_F is the **Fermi energy** (Fermi level)
- $k = 8.62 \times 10^{-5}$ eV/K is Boltzmann's constant; $kT = 0.0259$ eV at $T = 300$ K

It is useful to measure energy from E_F in units of kT: write $u = (E - E_F)/kT$. Then $f = 1/(1 + e^u)$ depends only on u, for any material and any temperature.

2. What the curve tells us

- At $E = E_F$ ($u = 0$): $f = 1/2$. The Fermi level is the energy at which a state is **half likely** to be filled.
- Above E_F ($u > 0$): $f < 1/2$ and it falls quickly. A few kT above E_F states are nearly empty.
- Below E_F ($u < 0$): $f > 1/2$, nearly 1 a few kT below.
- **Symmetry:** $f(E_F + \Delta E) = 1 - f(E_F - \Delta E)$. The probability that a state is **empty** (a hole) at ΔE below E_F equals the probability that a state is **full** at ΔE above.

3. The Boltzmann approximation

When $E - E_F \gg kT$, the exponential in the denominator is much larger than 1, so

$$f_F(E) \approx \exp\left[\frac{-(E - E_F)}{kT}\right]$$

The error is small once $E - E_F > 3kT$: at $3kT$ the exact value is 0.0474 and the approximation gives 0.0498 (5 % high). At $2kT$ the error is 13.5 %, at $1kT$ it is 37 %.

4. Electron concentration in the conduction band

The number of electrons is the number of states times the probability that each is full, summed over the conduction band. When $E_c - E_F > 3kT$ (non-degenerate material), the Boltzmann form can be used and the sum gives (Neamen, 1(a)):

$$n_0 = N_c \exp\left[\frac{-(E_c - E_F)}{kT}\right]$$

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where

- N_c is the **effective density of states** in the conduction band: all the band's states behave as if N_c of them sat at E_c. For Si at 300 K, $N_c = 2.8 \times 10^{19}$ cm^{-3} (it grows as $T^{3/2}$)
- $\exp[-(E_c - E_F)/kT]$ is the probability that a state at E_c is occupied

Meaning: every kT (0.0259 eV) that E_F moves down from E_c divides n_0 by $e = 2.718$; every 0.0596 eV ($kT \ln 10$) divides it by 10.

2. Answer

(i) Energy at which the occupation probability is 10 %

Given: $f_F(E) = 0.10$.

$$\frac{1}{1 + \exp\left(\dfrac{E - E_F}{kT}\right)} = 0.10$$

$$1 + \exp\left(\frac{E - E_F}{kT}\right) = 10 \quad \Rightarrow \quad \exp\left(\frac{E - E_F}{kT}\right) = 9$$

$$\frac{E - E_F}{kT} = \ln 9 = 2.197$$

$$\boxed{E - E_F = 2.197\, kT \approx 2.2\, kT}$$

At $T = 300$ K this is $2.197 \times 0.0259 = 0.0569$ eV above E_F. So a state only about $2.2\,kT$ above the Fermi level is already filled just one time in ten.

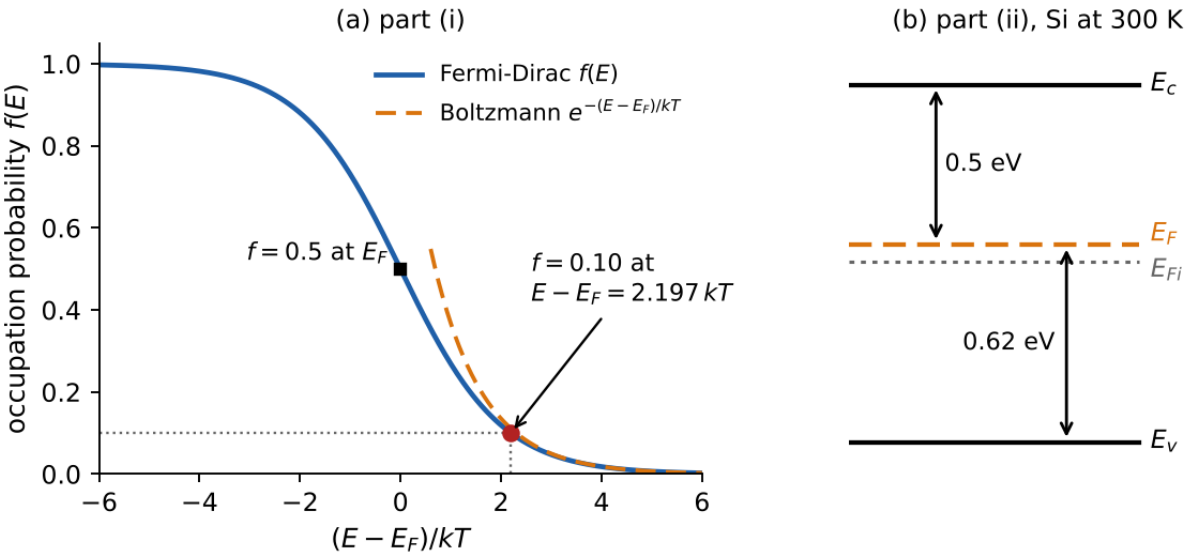


Figure 14: Fermi-Dirac function with the 10 percent point, and the band diagram of part (ii)

Reading the figure. Panel (a): the solid blue curve is the Fermi-Dirac function against $(E - E_F)/kT$; it passes 0.5 at E_F (black square) and 0.10 at $2.197\,kT$ (red dot, dotted grey guide lines). The dashed orange curve is the Boltzmann approximation; near $2.2\,kT$ it is still visibly above the exact curve, so the exact function must be used in part (i). Panel (b): the band diagram of part (ii) to scale, with E_F (dashed orange) 0.5 eV below E_c and just above E_{Fi} (dotted grey).

(ii) Electron concentration in Si

Given:

- $E_c - E_F = 0.5$ eV
- $N_c = 2.8 \times 10^{19}$ cm^{-3}

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- $T = 300$ K, $kT = 0.0259$ eV

Formula:

$$n_0 = N_c \exp\left[\frac{-(E_c - E_F)}{kT}\right]$$

Valid here because $E_c - E_F = 0.5$ eV $= 19.3\,kT \gg 3kT$ (Boltzmann approximation).

Step 1: the exponent.

$$\frac{E_c - E_F}{kT} = \frac{0.5}{0.0259} = 19.305, \qquad \exp(-19.305) = 4.130 \times 10^{-9}$$

Step 2: the concentration.

$$n_0 = (2.8 \times 10^{19})(4.130 \times 10^{-9})$$

$$\boxed{n_0 = 1.16 \times 10^{11}\ \text{cm}^{-3}}$$

This is larger than $n_i = 1.5 \times 10^{10}$ cm^{-3}, so the silicon is very lightly n-type. E_F sits a little above E_{Fi}:

$$E_F - E_{Fi} = 0.0259 \ln \frac{1.16 \times 10^{11}}{1.5 \times 10^{10}} = 0.053 \text{ eV}$$

Final Answer:

- (i) $E - E_F = kT \ln 9 = 2.197\,kT \approx 2.2\,kT$ (0.0569 eV at 300 K)
- (ii) $n_0 = 1.16 \times 10^{11}$ cm^{-3}

3. Keyword Explanation

1. Fermi level

The energy E_F at which the occupation probability is exactly one half; in equilibrium it is the same everywhere in a material.

2. Fermi-Dirac distribution

The probability $f_F(E) = 1/[1+\exp((E - E_F)/kT)]$ that an allowed state at energy E holds an electron.

3. Boltzmann approximation

$f_F(E) \approx \exp[-(E - E_F)/kT]$, valid when $E - E_F$ is more than about $3kT$.

4. Effective density of states

N_c (or N_v): the number of states per cm^3 that, placed at the band edge, would hold the same number of carriers as the whole band.

4. Notes and Remarks

- **Do not use Boltzmann in (i):** with $f \approx e^{-u} = 0.10$ one gets $u = \ln 10 = 2.303$, i.e. $2.3\,kT$, about 5 % too far. At only $2.2\,kT$ from E_F the approximation is not yet good.
- **Holes in (i):** by symmetry, a state $2.197\,kT$ **below** E_F is **empty** with probability 10 % (filled with probability 0.90).
- **Holes in (ii):** $p_0 = n_i^2/n_0 = (1.5\times10^{10})^2/(1.16\times10^{11}) = 1.95\times10^9$ cm^{-3}; E_F is $1.12-0.5 = 0.62$ eV above E_v.
- **In (ii) the exact and Boltzmann answers agree** to within 4×10^{-9} (relative), because $19.3\,kT \gg 3kT$.

5. Final Concise Answer

(a) A compensated semiconductor has both donors and acceptors in the same region. In its band diagram, donor states (E_d, just below E_c) are all ionized (+) and acceptor states (E_a, just above

E_v) are all filled (−), because donor electrons fall into the acceptors. For $N_d > N_a$ it is n-type with $n_0 \approx N_d - N_a$ and E_F above E_{Fi}; for $N_a > N_d$ it is p-type with $p_0 \approx N_a - N_d$ and E_F below E_{Fi}; for $N_d = N_a$ it is fully compensated, $n_0 = p_0 = n_i$ and $E_F = E_{Fi}$. Example: $N_d = 10^{16}$, $N_a = 3 \times 10^{15}$ cm^{-3} gives $n_0 = 7 \times 10^{15}$ cm^{-3} and $E_c - E_F = 0.215$ eV.

(b) In a graded n-type sample in equilibrium, $J_n = en\mu_n E_x + eD_n\, dn/dx = 0$ with $n = N_c \exp[-(E_c - E_F)/kT]$ and $dE_c/dx = eE_x$ gives $D_n/\mu_n = kT/e$; likewise $D_p/\mu_p = kT/e$ (0.0259 V at 300 K). Diffusion and drift come from the same scattered thermal motion, so knowing μ gives D: for Si, $D_n = 35.0$ cm^2/s and $D_p = 12.4$ cm^2/s.

(c) With $\rho = -eN_a$ $(-x_p < x < 0)$ and $+eN_d$ $(0 < x < x_n)$, Poisson's equation gives a triangular field with $|E_{max}| = eN_d x_n/\epsilon_s$ at $x = 0$, the charge balance $N_a x_p = N_d x_n$, and $V_{bi} = (e/2\epsilon_s)(N_d x_n^2 + N_a x_p^2)$. Solving, $W = x_n + x_p = \{(2\epsilon_s V_{bi}/e)[(N_a + N_d)/(N_a N_d)]\}^{1/2}$.

(d) (i) $f_F = 0.10$ at $E - E_F = kT \ln 9 = 2.197\, kT \approx 2.2\, kT$ (0.0569 eV at 300 K). (ii) $n_0 = 2.8 \times 10^{19} \exp(-0.5/0.0259) = 1.16 \times 10^{11}$ cm^{-3}.

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3. (a) Starting from charge neutrality condition $n_0 + N_a^- = p_0 + N_d^+$, derive the expressions for n_0 and p_0 in terms of N_a, N_d, and n_i. Marks: 5

(b) Sketch Energy-band diagram of a PN junction in thermal equilibrium and show P region, N region, E_C, E_V, E_F, E_{Fi}, $e\phi_{Fn}$, $e\phi_{Fp}$ and eV_{bi}. Marks: 5

(c) Define drift mobility. Explain the mechanism of carrier scattering in semiconductor materials and significance of τ. Marks: 5

(d) For the following questions: Marks: 10

i. Solve the ambipolar transport equation for a semiconductor with $N_a = 5 \times 10^{16}$ cm^{-3} for $x > 0$ at steady state with zero electric field where excess charges are created only at $x = 0$. ($\delta n_0 = 5 \times 10^{15}$ cm^{-3}).

ii. Find the distance where the excess charge will drop to 10×10^{13} cm^{-3}. (Assume $\tau_{n0} = 7 \times 10^{-7}$ s, $\tau_{p0} = 5 \times 10^{-7}$ s, $D_n = 30$ cm^2 s^{-1}, $D_p = 25$ cm^2 s^{-1})

iii. Plot n and p as function of x in log scale.

3.(a) Starting from charge neutrality condition $n_0 + N_a^- = p_0 + N_d^+$, derive the expressions for n_0 and p_0 in terms of N_a, N_d, and n_i. Marks: 5

1. Theory

1. What does charge neutrality mean? (Core Idea)

A piece of uniformly doped semiconductor, far from any junction or surface, has **no net electric charge**. It is built from neutral atoms, and adding a dopant atom adds a nucleus and an electron together. So, however the electrons rearrange themselves, every positive charge must be matched by a negative one.

Four kinds of charged particles can exist in the crystal:

- **free electrons** in the conduction band, concentration n_0, each with charge $-e$
- **free holes** in the valence band, concentration p_0, each with charge $+e$
- **ionized donors** N_d^+: a donor atom that has given away its extra electron is left as a fixed positive ion
- **ionized acceptors** N_a^-: an acceptor atom that has taken an electron is a fixed negative ion

The subscript 0 means **thermal equilibrium**: no light, no applied voltage, no current, a steady temperature. Total negative charge equals total positive charge:

$$n_0 + N_a^- = p_0 + N_d^+$$

This is the **charge neutrality condition** (Neamen, Section 4.5). All densities are in cm^{-3}; the common factor e has been cancelled.

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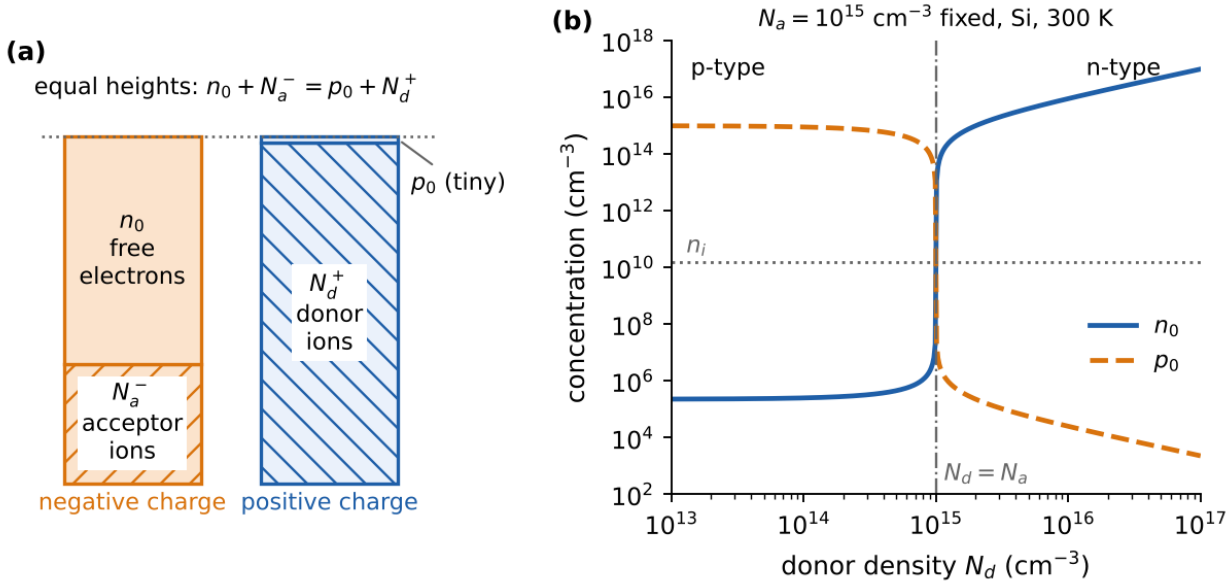


Figure 15: Charge balance in a compensated semiconductor and the electron and hole concentrations as the donor density is varied

Reading the figure. Panel (a) shows the neutrality condition as two stacks of equal height. The left stack (orange outline) is the negative charge: fixed acceptor ions (hatched) plus free electrons n_0. The right stack (blue outline) is the positive charge: fixed donor ions (hatched) plus a tiny number of free holes p_0. Here $N_d > N_a$, so the material is n-type. Panel (b) is computed from the formulas derived below for Si with $N_a = 10^{15}$ cm^{-3} held fixed while N_d is varied. The solid blue curve is n_0, the dashed orange curve p_0, the dotted grey line n_i. Left of the dash-dot grey line ($N_d < N_a$) the material is p-type and $p_0 \approx N_a - N_d$. At $N_d = N_a$ both curves pass through n_i: the material is fully compensated and behaves as intrinsic. Right of it, $n_0 \approx N_d - N_a$. On a log scale the change from p-type to n-type is very sharp.

2. A material with both donors and acceptors

A semiconductor that contains both donors and acceptors in the same region is called **compensated** (its band diagram is built in 2(a)). The electrons given away by the donors first fill the acceptors, so the two kinds of dopant partly cancel. Only the **difference** $N_d - N_a$ is left over to supply free carriers. If $N_d > N_a$ the material is n-type; if $N_a > N_d$ it is p-type.

3. Two more facts we need

Complete ionization. At room temperature in Si almost every donor has lost its electron and almost every acceptor has gained one (the proof is in 1(d) and 6(b)). So

$$N_d^+ = N_d, \qquad N_a^- = N_a$$

Mass action law. In equilibrium the product of electron and hole concentrations does not depend on doping (built in 1(a)):

$$n_0 p_0 = n_i^2 \quad \Rightarrow \quad p_0 = \frac{n_i^2}{n_0}$$

where n_i is the intrinsic carrier concentration (1.5×10^{10} cm^{-3} for Si at 300 K). Physically: more electrons means more chances for a hole to meet an electron and recombine, so holes become rarer in exact proportion.

4. Why a quadratic, and which root?

Putting both facts into the neutrality condition leaves one unknown, n_0, but it appears both as n_0 and as $1/n_0$. Multiplying through by n_0 gives a **quadratic** equation. A quadratic has two roots, $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. Here the square root is always larger than the first term, so the minus root is **negative**. A concentration cannot be negative, so only the plus root is physical.

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5. Limits that check the result

- **Strongly doped**, $|N_d - N_a| \gg n_i$: the square root is almost $|N_d - N_a|/2$, so the majority carrier concentration is simply the net doping, $n_0 \approx N_d - N_a$ (n-type), and the minority carrier follows from $p_0 = n_i^2/n_0$.
- **Intrinsic or fully compensated**, $N_d = N_a$: the formula gives $n_0 = p_0 = n_i$, as it must.
- **Lightly doped**, $|N_d - N_a|$ comparable to n_i (or a high temperature where n_i is large): both terms matter and the full formula must be used.

2. Answer

Assumptions: thermal equilibrium; complete ionization, so $N_a^- = N_a$ and $N_d^+ = N_d$; non-degenerate material, so the mass action law $n_0 p_0 = n_i^2$ holds.

Step 1: Charge neutrality

In thermal equilibrium a uniformly doped sample has no net charge. The negative charges are the free electrons n_0 and the ionized acceptors N_a^-; the positive charges are the holes p_0 and the ionized donors N_d^+. Their densities must balance (Neamen, Eq. 4.56):

$$n_0 + N_a^- = p_0 + N_d^+$$

With complete ionization this becomes $n_0 + N_a = p_0 + N_d$ (Neamen, Eq. 4.58).

Step 2: Remove the holes with the mass action law

Put $p_0 = n_i^2/n_0$ (Neamen, Eq. 4.59a):

$$n_0 + N_a = \frac{n_i^2}{n_0} + N_d$$

Step 3: Make it a quadratic in n_0

Multiply every term by n_0 and bring everything to the left:

$$n_0^2 + N_a n_0 - n_i^2 - N_d n_0 = 0 \quad \Rightarrow \quad n_0^2 - (N_d - N_a)\, n_0 - n_i^2 = 0$$

(Neamen, Eq. 4.59b).

Step 4 (Basic): The quadratic formula

For $ax^2 + bx + c = 0$, completing the square gives $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. Here $a = 1$, $b = -(N_d - N_a)$, $c = -n_i^2$:

$$n_0 = \frac{(N_d - N_a) \pm \sqrt{(N_d - N_a)^2 + 4n_i^2}}{2} = \frac{N_d - N_a}{2} \pm \sqrt{\left(\frac{N_d - N_a}{2}\right)^2 + n_i^2}$$

Step 5: Choose the root

The square root is larger than $|N_d - N_a|/2$, so the minus sign gives $n_0 < 0$, which is impossible. Keeping the plus sign:

$$\boxed{n_0 = \frac{N_d - N_a}{2} + \sqrt{\left(\frac{N_d - N_a}{2}\right)^2 + n_i^2}}$$

(Neamen, Eq. 4.60).

Step 6: The hole concentration

Go back to Step 1 and remove the electrons instead, with $n_0 = n_i^2/p_0$ (Neamen, Eq. 4.61a):

$$\frac{n_i^2}{p_0} + N_a = p_0 + N_d$$

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Multiply by p_0 and collect terms:

$$n_i^2 + N_a p_0 = p_0^2 + N_d p_0 \quad \Rightarrow \quad p_0^2 - (N_a - N_d)\, p_0 - n_i^2 = 0$$

The quadratic formula with $a = 1$, $b = -(N_a - N_d)$, $c = -n_i^2$, and the positive root for the same reason as in Step 5, give

$$\boxed{p_0 = \frac{N_a - N_d}{2} + \sqrt{\left(\frac{N_a - N_d}{2}\right)^2 + n_i^2}}$$

(Neamen, Eq. 4.62).

Steps to remember

Neutrality $n_0 + N_a = p_0 + N_d \to p_0 = n_i^2/n_0 \to$ multiply by n_0: quadratic $\to$ quadratic formula $\to$ keep the positive root $\to$ for p_0 put $n_0 = n_i^2/p_0$ instead.

Interpretation:

- n-type with $N_d - N_a \gg n_i$: $n_0 \approx N_d - N_a$, $p_0 = n_i^2/(N_d - N_a)$.
- p-type with $N_a - N_d \gg n_i$: $p_0 \approx N_a - N_d$, $n_0 = n_i^2/(N_a - N_d)$.
- $N_d = N_a$ (or no doping): $n_0 = p_0 = n_i$.

Example (Si, 300 K)

$N_d = 10^{16}$, $N_a = 3 \times 10^{15}$ cm^{-3} gives $n_0 = 7.0 \times 10^{15}$ cm^{-3} and $p_0 = n_i^2/n_0 = 3.2 \times 10^4$ cm^{-3}. The rejected minus root would give $n_0 = -3.2 \times 10^4$ cm^{-3}.

3. Keyword Explanation

1. Charge neutrality

In a uniform semiconductor in equilibrium the total positive charge (holes and ionized donors) equals the total negative charge (electrons and ionized acceptors).

2. Complete ionization

The state in which every donor has given up its electron and every acceptor has accepted one; true for Si at room temperature for normal doping.

3. Compensated semiconductor

A semiconductor that contains both donors and acceptors in the same region; its free carriers come from the net doping $|N_d - N_a|$.

4. Mass action law

$n_0 p_0 = n_i^2$ in thermal equilibrium, for a non-degenerate semiconductor, whatever the doping.

4. Notes and Remarks

- **When the result fails:** at very low temperature the dopants are not ionized (freeze-out, 1(d)), so $N_d^+ < N_d$; and in degenerate (very heavily doped) material $n_0 p_0 = n_i^2$ no longer holds.
- **Temperature:** n_i grows fast with temperature. When n_i approaches $|N_d - N_a|$, the full formula shows the material drifting towards intrinsic, which sets the upper working temperature of a device.
- **Example near intrinsic**

  $N_d = 5 \times 10^{10}$, $N_a = 3 \times 10^{10}$ cm^{-3} gives $n_0 = 2.80 \times 10^{10}$ and $p_0 = 8.03 \times 10^9$ cm^{-3}; here n_0 is clearly not equal to $N_d - N_a = 2 \times 10^{10}$, so the simple approximation would be wrong.

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3.(b) Sketch Energy-band diagram of a PN junction in thermal equilibrium and show P region, N region, E_C, E_V, E_F, E_{Fi}, $e\phi_{Fn}$, $e\phi_{Fp}$ and eV_{bi}. Marks: 5

1. Theory

1. What is a PN junction? (Core Idea)

A **PN junction** is one crystal of semiconductor that is p-type (doped with acceptors, N_a) on one side and n-type (doped with donors, N_d) on the other. The plane where the doping changes is the **metallurgical junction**, placed at $x = 0$. We take a **step junction**: uniform N_a for $x < 0$, uniform N_d for $x > 0$.

An **energy band diagram** plots the electron energy levels against position x:

- E_c: bottom of the conduction band (where free electrons live)
- E_v: top of the valence band (where holes live)
- E_{Fi}: the intrinsic Fermi level, near the middle of the gap
- E_F: the Fermi level, which shows how full the states are (built in 1(a))

Electron energy increases upward. Holes, having the opposite charge, have energy increasing downward.

2. The two sides before contact

In each neutral region the position of E_F relative to E_{Fi} is set by the doping (from 1(a), $n_0 = n_i e^{(E_F - E_{Fi})/kT}$ and $p_0 = n_i e^{(E_{Fi} - E_F)/kT}$). Neamen names these distances with potentials:

p side ($p_0 = N_a$): E_F is below E_{Fi} by $e\phi_{Fp}$:

$$e\phi_{Fp} = E_{Fi} - E_F, \qquad \phi_{Fp} = V_t \ln \frac{N_a}{n_i} \quad \text{(positive)}$$

n side ($n_0 = N_d$): E_F is above E_{Fi}. Neamen keeps the same definition, so the potential is negative:

$$e\phi_{Fn} = E_{Fi} - E_F, \qquad \phi_{Fn} = -V_t \ln \frac{N_d}{n_i} \quad \text{(negative)}, \qquad E_F - E_{Fi} = e|\phi_{Fn}|$$

where $V_t = kT/e = 0.0259$ V at 300 K is the **thermal voltage**.

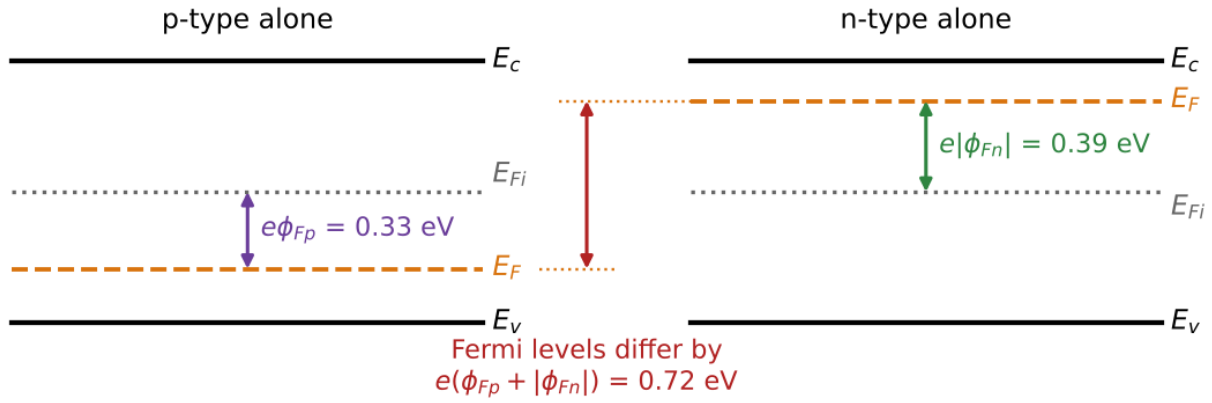


Figure 16: The p-type and n-type regions drawn separately before contact, for $N_a = 5 \times 10^{15}$ and $N_d = 5 \times 10^{16}$ cm^{-3} in Si

Reading the figure. Both pieces are the same material, so their bands (E_c, E_v, solid black) and intrinsic levels (dotted grey) are at the same heights. Only the Fermi level (dashed orange) differs: $e\phi_{Fp} = 0.33$ eV below E_{Fi} on the p side (purple double arrow) and $e|\phi_{Fn}| = 0.39$ eV above it on the n side (green double arrow). The red double arrow is the difference between the two Fermi levels, 0.72 eV. Numbers are for $N_a = 5 \times 10^{15}$ and $N_d = 5 \times 10^{16}$ cm^{-3}, the junction of 6(d).

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3. What happens at contact?

- The n side has many electrons, the p side many holes. Electrons **diffuse** from n to p, holes from p to n (diffusion: 1(b)).
- Electrons leaving the n side uncover fixed positive donor ions; holes leaving the p side uncover fixed negative acceptor ions. A layer around $x = 0$, from $-x_p$ to x_n, is emptied of free carriers. It is the **space charge region** (or depletion region).
- These ions create an **electric field** pointing from the positive (n) side to the negative (p) side. The field pushes electrons back to n and holes back to p (drift).
- Diffusion and drift soon balance exactly. In **thermal equilibrium** no net current flows, and this forces the Fermi level to be **flat** (constant) through the whole device. If E_F had a slope, a current would flow.

4. How the bands bend

Far from the junction each side keeps its own neutral picture: E_F is $e\phi_{Fp}$ below E_{Fi} on the left and $e|\phi_{Fn}|$ above E_{Fi} on the right. Since E_F is now one straight line, the bands of the p side must sit **higher** than those of the n side. Inside the space charge region they bend smoothly from one height to the other. E_c, E_v and E_{Fi} all bend by the same amount, because the band gap does not change.
The step in the bands is the **built-in potential barrier** eV_{bi}: the energy an electron in the n region must gain to climb into the p region's conduction band. The shape of the bending (two parabolas, from Poisson's equation) and the widths x_p, x_n are derived in 2(c).

5. Deriving the built-in potential

From the diagram, the total bend equals the two Fermi distances added:

$$eV_{bi} = e\phi_{Fp} + e|\phi_{Fn}| \quad \Rightarrow \quad V_{bi} = |\phi_{Fp}| + |\phi_{Fn}|$$

$$V_{bi} = V_t \ln \frac{N_a}{n_i} + V_t \ln \frac{N_d}{n_i} = V_t \ln \frac{N_a N_d}{n_i^2}$$

Check by carrier densities. The electron density on the n side is $n_{n0} = N_d$; on the p side it is $n_{p0} = n_i^2/N_a$. An electron must climb eV_{bi} to go from n to p, and in equilibrium (Boltzmann) the density falls by $e^{-eV_{bi}/kT}$ across the barrier:

$$n_{p0} = n_{n0}\, e^{-eV_{bi}/kT} \quad \Rightarrow \quad V_{bi} = V_t \ln \frac{n_{n0}}{n_{p0}} = V_t \ln \frac{N_a N_d}{n_i^2}$$

Typical number: $N_a = 5 \times 10^{15}$, $N_d = 5 \times 10^{16}$ cm^{-3} in Si at 300 K: $\phi_{Fp} = 0.0259 \times 12.717 = 0.329$ V, $|\phi_{Fn}| = 0.0259 \times 15.020 = 0.389$ V, $V_{bi} = 0.718$ V. A Si junction typically has V_{bi} of 0.6 to 0.9 V.

2. Answer

Sketch (thermal equilibrium, zero applied voltage):

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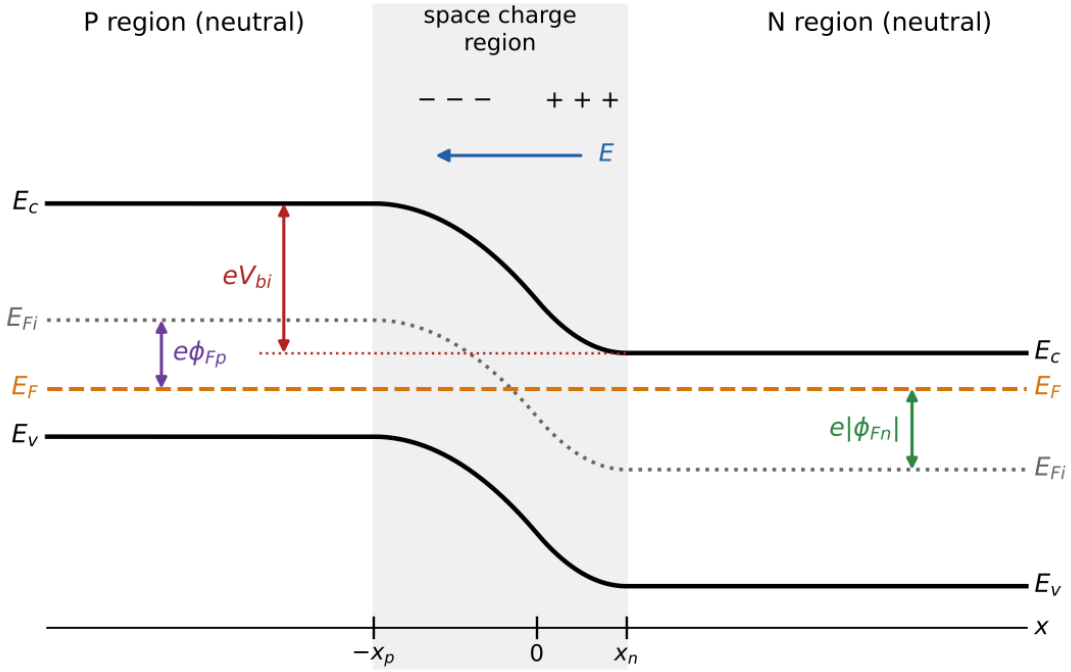


Figure 17: Energy band diagram of a PN junction in thermal equilibrium with the P and N regions, the space charge region and the energies e phi Fp, e phi Fn and eVbi

Reading the figure. The P region is on the left and the N region on the right; the shaded band from $-x_p$ to x_n is the space charge region (negative acceptor ions on the p side, positive donor ions on the n side, field E pointing from n to p, blue arrow). E_c and E_v (solid black) and E_{Fi} (dotted grey) are high on the p side and bend down across the space charge region. The Fermi level E_F (dashed orange) is a flat straight line.

Features of the diagram:

- E_F **is constant** through the whole device, because in equilibrium there is no net current.
- **P region:** E_F is below E_{Fi} by $e\phi_{Fp}$ (purple double arrow), with $\phi_{Fp} = V_t \ln(N_a/n_i)$.
- **N region:** E_F is above E_{Fi} by $e|\phi_{Fn}|$ (green double arrow), with $\phi_{Fn} = -V_t \ln(N_d/n_i)$ (negative in Neamen's sign convention, $e\phi_{Fn} = E_{Fi} - E_F$).
- **Space charge region:** only here are the bands bent; outside it both regions are neutral and the bands are flat.
- **Built-in barrier:** the bands of the p side are higher by eV_{bi} (red double arrow), the potential energy barrier that stops the majority electrons of the n side from diffusing into the p side (and holes the other way).

The built-in potential is the sum of the two Fermi distances:

$$eV_{bi} = e\phi_{Fp} + e|\phi_{Fn}| \quad \Rightarrow \quad V_{bi} = |\phi_{Fp}| + |\phi_{Fn}| = V_t \ln \frac{N_a N_d}{n_i^2}$$

3. Keyword Explanation

1. Built-in potential barrier

The potential difference V_{bi} across the space charge region in equilibrium. It keeps diffusion and drift in balance; it cannot be measured with a voltmeter, because contact potentials cancel it.

2. Space charge region

The region around the junction, from $-x_p$ to x_n, that has lost its free carriers and holds the charge of the fixed dopant ions.

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3. Fermi potentials ϕ_{Fp} and ϕ_{Fn}

The distance between E_{Fi} and E_F in each neutral region, divided by e: $\phi_{Fp} = (E_{Fi} - E_F)/e > 0$ on the p side, $\phi_{Fn} = (E_{Fi} - E_F)/e < 0$ on the n side.

4. Thermal equilibrium

No external voltage, light or current and a uniform temperature; then E_F is the same everywhere.

4. Notes and Remarks

- **Sign convention:** Neamen defines both Fermi potentials as $(E_{Fi} - E_F)/e$, so ϕ_{Fn} is negative and the diagram labels the n-side distance $e|\phi_{Fn}|$. Some books (for example Streetman) write $q\phi_n$ as a positive number for the same distance. The sum $V_{bi} = |\phi_{Fp}| + |\phi_{Fn}|$ is the same in all books.
- **Drawing to scale:** in the figure the energies are to scale for the example, but the widths are not. The lighter-doped side holds most of the space charge region ($N_a x_p = N_d x_n$), so for this example x_p is ten times x_n (6(d)), and most of the band bending is on the p side.
- **With applied voltage** the Fermi level is no longer flat: a forward bias V_a lowers the barrier to $e(V_{bi} - V_a)$, a reverse bias raises it to $e(V_{bi} + V_R)$ (4(d)).

3.(c) Define drift mobility. Explain the mechanism of carrier scattering in semiconductor materials and significance of τ. Marks: 5

1. Theory

1. Thermal motion and drift (Core Idea)

Free electrons and holes in a crystal are never at rest. At room temperature each carrier moves with a high **thermal velocity**, about 10^7 cm/s, in a random direction. It travels a short distance, hits something, flies off in a new random direction, and so on. Because all directions are equally likely, the carriers go nowhere on average, and no current flows.

When an electric field E is applied, every carrier feels a force ($+eE$ on a hole, $-eE$ on an electron). Between collisions this force bends each flight a little in the same direction. The random motion is not removed, but a slow average motion is added on top of it. This average motion is called **drift**, and its average velocity is the **drift velocity** v_d.

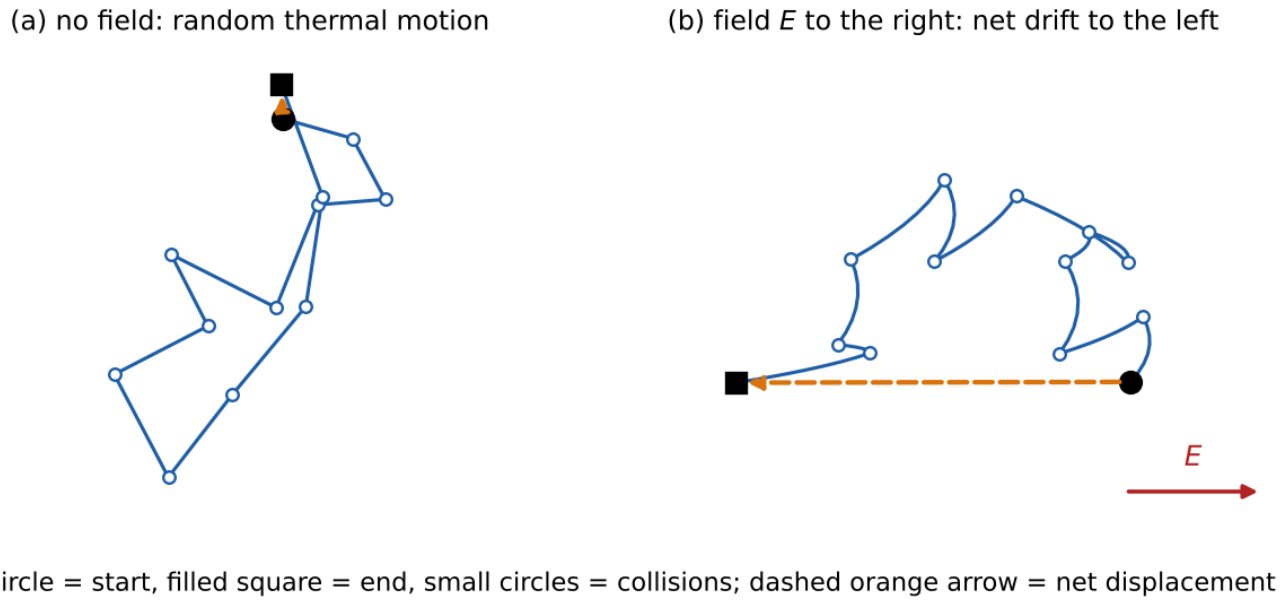


Figure 18: Path of an electron with no field and with an applied field

Reading the figure. The solid blue line is the path of one electron; each small open circle is a collision, the filled circle the start and the filled square the end. In panel (a) there is no field: after many flights

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the dashed orange arrow (net displacement) is very short and in no particular direction. In panel (b) a field E points to the right (red arrow). Each flight curves to the left because the force on an electron is $-eE$, and the dashed orange arrow is long and points to the left: a net drift opposite to the field.

2. Where does the drift velocity come from?

Take a hole of effective mass m_p^* (the effective mass is the mass that makes Newton's law work inside the crystal). Newton's law during a free flight is

$$m_p^* \frac{dv}{dt} = eE$$

After a collision its velocity is random, with zero average. It then gains extra velocity eEt/m_p^* during a flight of time t. Averaged over all flights, the time between collisions is τ, so the average extra velocity is

$$v_d = \frac{e\tau}{m_p^*} E = \mu_p E$$

where

- τ is the **mean free time**: the average time between two collisions (s)
- $\mu_p = e\tau/m_p^*$ is the **hole mobility** ($\text{cm}^2/\text{V s}$)

For electrons the same steps give $v_d = -\mu_n E$ (opposite to the field) with $\mu_n = e\tau/m_n^*$. Neamen writes the mean free times as τ_{cp} and τ_{cn}.

3. What does a collision mean? (Scattering)

A perfect, motionless crystal lattice would not scatter a carrier at all: the carrier moves through the periodic potential as a wave, and that is already included in its effective mass. A carrier is **scattered** only by things that break the perfect periodicity. In a semiconductor there are two main ones.

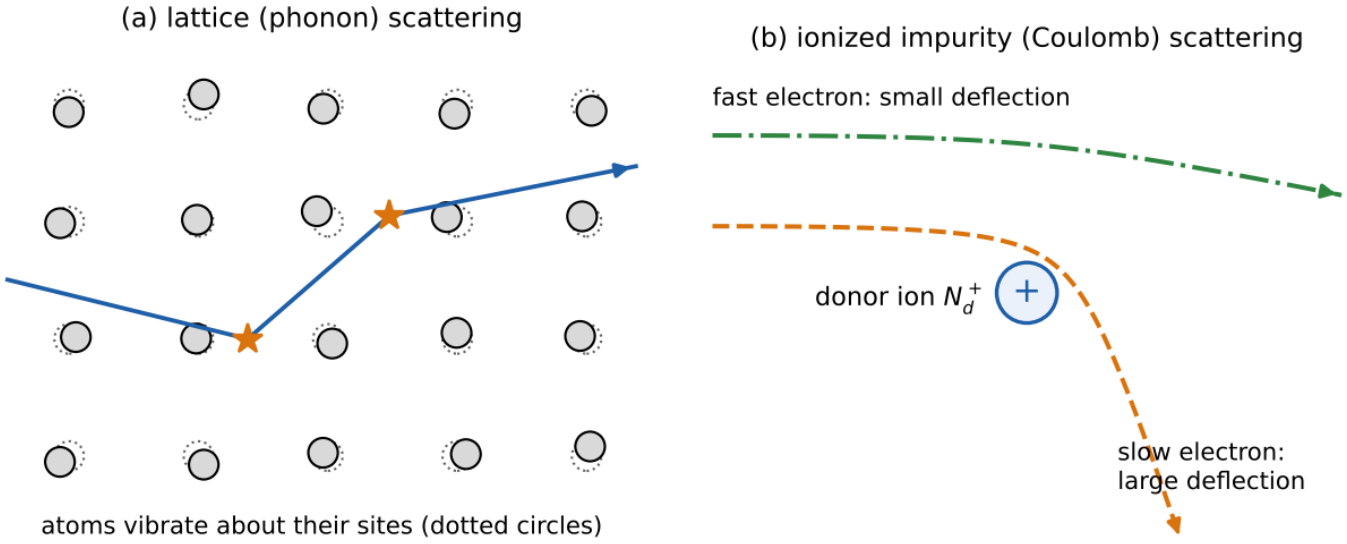


Figure 19: Lattice scattering by vibrating atoms and ionized impurity scattering by a donor ion

Reading the figure. In panel (a) the atoms (grey circles) are displaced from their ideal sites (dotted circles) because they vibrate with heat. The electron (solid blue line) changes direction at each collision (orange stars). In panel (b) a positive donor ion attracts a passing electron. The fast electron (dash-dot green) is bent only a little; the slow electron (dashed orange) spends longer near the ion and is bent much more.

Lattice (phonon) scattering. Above 0 K the atoms vibrate about their positions. The vibrations disturb the periodic potential, and a carrier exchanges energy and momentum with them. A quantum of lattice vibration is called a **phonon**, so this is also called **phonon scattering**. More heat means bigger vibrations and more scattering, so the lattice-limited mobility falls as temperature rises: $\mu_L \propto T^{-3/2}$.

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Ionized impurity scattering. Each ionized donor (+) or acceptor (−) has a Coulomb field that bends the path of a passing carrier. The effect is larger for *more* ions and for **slower** carriers (a slow carrier spends more time near the ion). So the impurity-limited mobility falls with doping and rises with temperature (hotter carriers are faster): $\mu_I \propto T^{3/2}/N_I$, where $N_I = N_d^+ + N_a^-$ is the total density of ionized impurities.

The two mechanisms act together. Their scattering rates add, so the inverse mobilities add (**Matthiessen's rule**): $1/\mu = 1/\mu_L + 1/\mu_I$. The temperature behaviour is built in 4(a). Other, smaller mechanisms are neutral impurities, defects and other carriers.

4. Typical numbers for Si

Electron mobility of lightly doped Si is $\mu_n = 1350$ cm^2/V s. With the conductivity effective mass $m^* = 0.26\, m_0$:

$$\tau = \frac{\mu_n m^*}{e} = \frac{(0.135 \text{ m}^2/\text{V s})(0.26 \times 9.11 \times 10^{-31} \text{ kg})}{1.602 \times 10^{-19} \text{ C}} = 2.0 \times 10^{-13} \text{ s} = 0.2 \text{ ps}$$

The thermal velocity $\sqrt{3kT/m^*}$ is 2.3×10^7 cm/s, so a carrier travels only about $v_{th}\tau \approx 46$ nm between collisions (the **mean free path**). At $E = 1000$ V/cm the drift velocity is $\mu_n E = 1.35 \times 10^6$ cm/s, much less than the thermal velocity.

5. From mobility to current

A drift velocity of charged carriers is a current. Electrons and holes drift in opposite directions but, having opposite charges, both carry current along the field:

$$J_{drf} = e(\mu_n n + \mu_p p)E = \sigma E, \qquad \sigma = e(\mu_n n + \mu_p p)$$

where J_{drf} is the drift current density (A/cm^2), σ the conductivity (S/cm) and its inverse $\rho = 1/\sigma$ the resistivity (Ω cm). Example: n-type Si with $n = 10^{16}$ cm^{-3} has $\sigma = (1.602 \times 10^{-19})(10^{16})(1350) = 2.16$ S/cm, $\rho = 0.46$ Ω cm.

2. Answer

Definition. The **drift mobility** μ is the average drift velocity of a carrier per unit applied electric field:

$$\mu = \frac{|v_d|}{E}, \qquad v_{dp} = \mu_p E, \qquad v_{dn} = -\mu_n E$$

Unit: cm^2/V s. For Si at 300 K, $\mu_n = 1350$ and $\mu_p = 480$ cm^2/V s. It tells how easily a carrier moves through the crystal under a field. From Newton's law for a carrier accelerated by the field between collisions,

$$\mu = \frac{e\tau}{m^*}$$

where e is the electron charge, m^* the effective mass and τ the mean free time between collisions.

Mechanism of carrier scattering. A carrier moving in a field is accelerated, but it soon collides with something that breaks the perfect periodicity of the crystal. Its velocity is then randomised and the drift gained is lost. This repeated collision is **scattering**; it limits the average (drift) velocity. The two main mechanisms are:

1. **Lattice (phonon) scattering:** above 0 K the atoms vibrate (phonons). The vibrations disturb the periodic potential, and carriers exchange energy and momentum with them. It increases with temperature, so $\mu_L \propto T^{-3/2}$.
2. **Ionized impurity scattering:** the Coulomb field of each ionized donor or acceptor deflects passing carriers. It increases with the density of ionized impurities N_I and is stronger for slow carriers, so $\mu_I \propto T^{3/2}/N_I$.

The scattering rates add, so $\dfrac{1}{\mu} = \dfrac{1}{\mu_L} + \dfrac{1}{\mu_I}$ (Matthiessen's rule). Lattice scattering dominates in pure material at high temperature; impurity scattering in heavily doped material at low temperature.

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Significance of τ. τ is the **mean free time**, the average time between two collisions (about 10^{-13} s in Si):

- It sets the mobility, $\mu = e\tau/m^*$: a longer time between collisions means a larger drift velocity for the same field.
- It sets the conductivity, $\sigma = e(n\mu_n + p\mu_p)$, and so the resistance and current of a device.
- It sets the speed of the device: high τ (high μ) lets carriers cross the device quickly, so it switches faster.
- It measures the scattering: $1/\tau$ is the collision rate, and the rates of the mechanisms add, $1/\tau = 1/\tau_L + 1/\tau_I$.

3. Keyword Explanation

1. Drift velocity

The average velocity of carriers caused by an electric field, added on top of their random thermal motion.

2. Phonon

A quantum of lattice vibration. Carriers scatter by absorbing or emitting phonons.

3. Mean free time and mean free path

The mean free time τ is the average time between collisions; the mean free path is the average distance travelled between collisions, about $v_{th}\tau$.

4. Effective mass

The mass m^* that, put in Newton's law, describes a carrier's response to a force inside the crystal.

4. Notes and Remarks

- **Two different "tau"s:** the mean free time between collisions (τ, τ_{cn}, about 10^{-13} s) is *not* the minority carrier lifetime (τ_{n0}, τ_{p0}, about 10^{-7} s) used in 3(d). The lifetime is the time before a carrier recombines and disappears; a carrier suffers about half a million collisions ($7 \times 10^{-7}/2 \times 10^{-13}$, using τ_{n0} of 3(d)) in its life.
- **High fields:** $v_d = \mu E$ holds only at low fields. At about 10^4 to 10^5 V/cm in Si the carriers lose energy to optical phonons and the drift velocity saturates near 10^7 cm/s.
- **Mobility from a measurement:** the Hall effect gives n and μ directly (5(d)); mobility is linked to the diffusion coefficient by the Einstein relation $D/\mu = kT/e$ (2(b)).
- The mass in $\mu = e\tau/m^*$ is the **conductivity** effective mass ($0.26\,m_0$ for electrons in Si), not the density-of-states mass $1.08\,m_0$.

3.(d) For the following questions: i. Solve the ambipolar transport equation for a semiconductor with $N_a = 5 \times 10^{16}$ cm^{-3} for $x > 0$ at steady state with zero electric field where excess charges are created only at $x = 0$. ($\delta n_0 = 5 \times 10^{15}$ cm^{-3}). ii. Find the distance where the excess charge will drop to 10×10^{13} cm^{-3}. (Assume $\tau_{n0} = 7 \times 10^{-7}$ s, $\tau_{p0} = 5 \times 10^{-7}$ s, $D_n = 30$ cm^2 s^{-1}, $D_p = 25$ cm^2 s^{-1}) iii. Plot n and p as function of x in log scale. Marks: 10

1. Theory

1. What are excess carriers? (Core Idea)

In thermal equilibrium a semiconductor has n_0 electrons and p_0 holes per cm^3, with $n_0 p_0 = n_i^2$. Light, a forward-biased junction or high-energy particles can add extra electron-hole pairs. The added carriers are the **excess carriers**:

$$n = n_0 + \delta n, \qquad p = p_0 + \delta p$$

where δn and δp are the excess electron and hole concentrations (cm^{-3}). When light breaks a bond, it makes one electron and one hole together, so $\delta n = \delta p$.

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2. Generation and recombination

- **Generation** creates electron-hole pairs. Heat does it all the time (thermal generation $G_{n0} = G_{p0}$); an outside source adds an **excess generation rate** g' ($\mathrm{cm^{-3}\ s^{-1}}$).
- **Recombination** destroys pairs: a conduction electron falls into a hole. In equilibrium thermal generation and recombination exactly balance.

When there are excess carriers, recombination becomes faster than thermal generation, and the excess decays. In a p-type material the excess **electrons** are the **minority carriers**. Each excess electron is surrounded by many holes, so its chance of recombining in a short time is a constant. The net recombination rate is therefore proportional to the excess:

$$R' = \frac{\delta n}{\tau_{n0}}$$

where τ_{n0} is the **excess minority carrier (electron) lifetime**: the average time an excess electron lives before it recombines (typically 10^{-7} s in Si). In n-type material the same role is played by τ_{p0} for excess holes.

Low injection. This simple law holds when the excess is small compared with the majority concentration, $\delta n \ll p_0$, so that the hole density hardly changes. This is called **low-level injection**.

3. The continuity equation: counting carriers in a slab

Take a thin slab of the semiconductor from x to $x+dx$, of unit area. Let $F_n(x)$ be the **flux** of electrons: the number crossing unit area per second in the $+x$ direction. The number of electrons in the slab changes for three reasons:

- electrons flow in at x and out at $x+dx$: net gain $F_n(x) - F_n(x+dx) = -\dfrac{\partial F_n}{\partial x}dx$
- generation adds $g_n\,dx$
- recombination removes $(n/\tau_{nt})\,dx$, where τ_{nt} includes thermal recombination

Dividing by dx gives the **continuity equation** for electrons (the same for holes with p and F_p):

$$\frac{\partial n}{\partial t} = -\frac{\partial F_n}{\partial x} + g_n - \frac{n}{\tau_{nt}}$$

It says only that carriers are not lost: whatever is not carried away or recombined stays.

4. The time-dependent diffusion equation

Electrons move by two mechanisms: **drift** in a field (3(c)) and **diffusion** from high to low concentration (1(b)). Their flux is

$$F_n = -\mu_n n E - D_n \frac{\partial n}{\partial x}$$

where D_n is the electron diffusion coefficient ($\mathrm{cm^2/s}$) and μ_n the mobility. Putting this into the continuity equation, with $n = n_0 + \delta n$ (n_0 uniform and constant), $g_n - n/\tau_{nt} = g' - \delta n/\tau_{n0}$ (thermal generation cancels thermal recombination), gives (Neamen)

$$D_n \frac{\partial^2(\delta n)}{\partial x^2} + \mu_n\left(E\frac{\partial(\delta n)}{\partial x} + n\frac{\partial E}{\partial x}\right) + g' - \frac{\delta n}{\tau_{n0}} = \frac{\partial(\delta n)}{\partial t}$$

and a similar equation for holes with D_p, μ_p, τ_{p0}.

5. Ambipolar transport: why the electrons and holes move together

Electrons diffuse faster than holes ($D_n > D_p$). If the excess electrons ran ahead of the excess holes, the two would separate and create a charge. That charge makes an **internal electric field** which pulls the electrons back and the holes forward. So the excess electrons and holes move **together**, as one cloud, keeping $\delta n \approx \delta p$ (**charge neutrality**). This joint motion is called **ambipolar transport**.

Eliminating the internal field between the electron and hole equations gives one equation, the **ambipolar transport equation** (Neamen):

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$$D'\frac{\partial^2(\delta n)}{\partial x^2} + \mu' E\frac{\partial(\delta n)}{\partial x} + g' - R' = \frac{\partial(\delta n)}{\partial t}$$

with the **ambipolar diffusion coefficient** and **ambipolar mobility**

$$D' = \frac{D_n D_p(n+p)}{D_n n + D_p p}, \qquad \mu' = \frac{\mu_n \mu_p (p-n)}{\mu_n n + \mu_p p}$$

The p-type, low-injection case. If $p \gg n$, then $D' \to D_n$ and $\mu' \to \mu_n$: the cloud moves with the properties of the **minority carrier** (electrons). Also $R' = \delta n/\tau_{n0}$. The equation becomes

$$D_n\frac{\partial^2(\delta n)}{\partial x^2} + \mu_n E\frac{\partial(\delta n)}{\partial x} + g' - \frac{\delta n}{\tau_{n0}} = \frac{\partial(\delta n)}{\partial t}$$

This is why, in a p-type sample, only D_n and τ_{n0} matter. The hole values D_p and τ_{p0} would be used for an n-type sample. A pulse that drifts in a field is treated with this equation in 5(a).

6. Steady state with no field: the diffusion length

With **steady state** ($\partial/\partial t = 0$), **no field** ($E = 0$) and **no generation** where we look ($g' = 0$), the equation keeps only diffusion and recombination:

$$D_n\frac{d^2(\delta n)}{dx^2} - \frac{\delta n}{\tau_{n0}} = 0 \quad \Rightarrow \quad \frac{d^2(\delta n)}{dx^2} - \frac{\delta n}{L_n^2} = 0, \qquad L_n = \sqrt{D_n \tau_{n0}}$$

It is a second-order linear equation with constant coefficients. Trying $\delta n = e^{sx}$ gives $s^2 = 1/L_n^2$, $s = \pm 1/L_n$, so

$$\delta n(x) = A\, e^{-x/L_n} + B\, e^{+x/L_n}$$

A and B come from the two boundary conditions. The physical picture: carriers injected at one place diffuse away, and on the way they recombine. Diffusion spreads them out; recombination removes them. The balance is an exponential decay.

L_n is the **minority carrier diffusion length**: the distance over which the excess falls by a factor $e = 2.718$. It is also the **average distance** an excess electron diffuses before it recombines. Typical values in Si are tens to hundreds of μm.

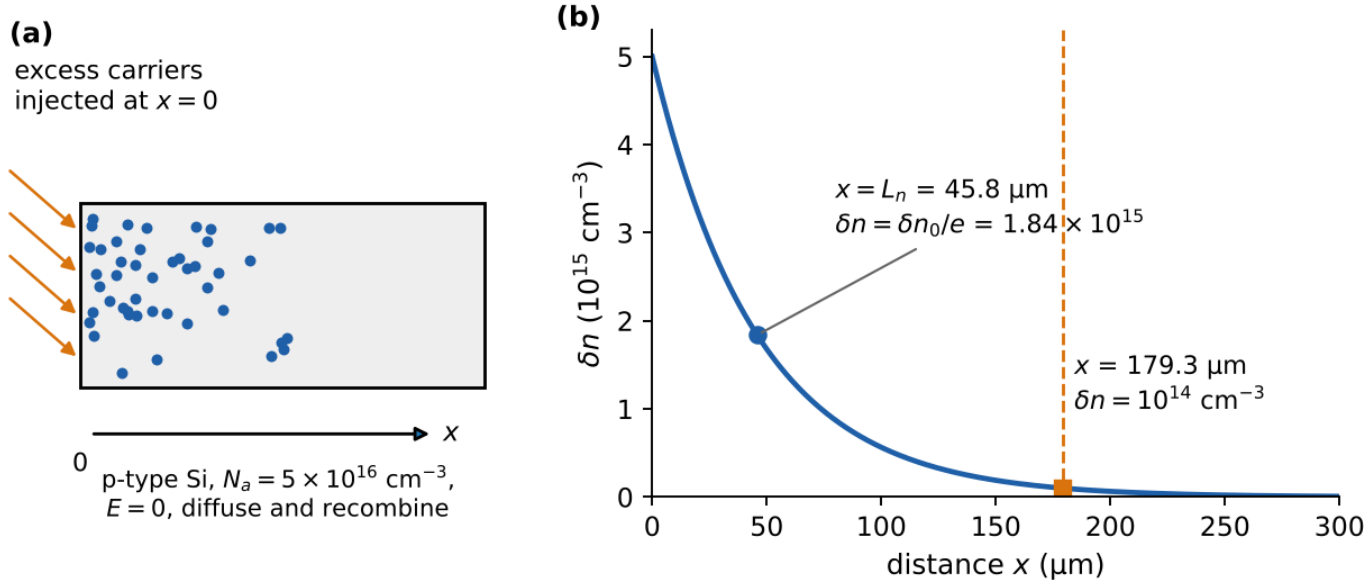


Figure 20: Excess electrons injected at x = 0 diffuse into the p-type bar and decay exponentially

Reading the figure. Panel (a): carriers are injected at the surface $x = 0$ (orange arrows) and spread into the bar; the dots (blue) thin out with distance because the carriers recombine as they diffuse. Panel (b) is the solution of this question on a linear scale: the solid blue curve $\delta n(x)$ starts at 5×10^{15}

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cm^{-3}, falls to $1/e$ of that (1.84×10^{15}) at $x = L_n = 45.8$ μm (blue dot), and reaches 10^{14} cm^{-3} at $x = 179.3$ μm (dashed orange line, orange square).

2. Answer

Given:

- p-type Si, $N_a = 5 \times 10^{16}$ cm^{-3}, so $p_0 = N_a = 5 \times 10^{16}$ cm^{-3}
- excess at $x = 0$: $\delta n_0 = 5 \times 10^{15}$ cm^{-3}; excess generated only at $x = 0$
- steady state, $E = 0$, region $x > 0$
- $\tau_{n0} = 7 \times 10^{-7}$ s, $D_n = 30$ cm^2/s ($\tau_{p0} = 5 \times 10^{-7}$ s, $D_p = 25$ cm^2/s)
- $n_i = 1.5 \times 10^{10}$ cm^{-3} (Si, 300 K)

Step 1: (i) Reduce the ambipolar transport equation

The ambipolar transport equation is

$$D' \frac{\partial^2(\delta n)}{\partial x^2} + \mu' E \frac{\partial(\delta n)}{\partial x} + g' - R' = \frac{\partial(\delta n)}{\partial t}$$

where D' and μ' are the ambipolar diffusion coefficient and mobility, g' the excess generation rate and R' the excess recombination rate.

Low injection check: $\delta n_0/p_0 = 5 \times 10^{15}/5 \times 10^{16} = 0.1$, so $\delta n \ll p_0$ (low injection) everywhere. In p-type material with low injection, $D' = D_n$, $\mu' = \mu_n$ and $R' = \delta n/\tau_{n0}$: the excess carriers behave as minority electrons.

Conditions: steady state, $\partial(\delta n)/\partial t = 0$; zero field, $E = 0$; no generation for $x > 0$, $g' = 0$. The equation becomes

$$D_n \frac{d^2(\delta n)}{dx^2} - \frac{\delta n}{\tau_{n0}} = 0 \quad \Rightarrow \quad \frac{d^2(\delta n)}{dx^2} - \frac{\delta n}{L_n^2} = 0$$

where $L_n = \sqrt{D_n \tau_{n0}}$ is the minority carrier diffusion length.

Step 2: (i) Solve it with the boundary conditions

General solution:

$$\delta n(x) = A\, e^{-x/L_n} + B\, e^{+x/L_n}$$

- As $x \to \infty$ the excess must die out (carriers recombine; δn cannot grow without limit), so $B = 0$.
- At $x = 0$, $\delta n(0) = \delta n_0$, so $A = \delta n_0$.

$$\boxed{\delta n(x) = \delta n_0\, e^{-x/L_n} = 5 \times 10^{15}\, e^{-x/L_n} \text{ cm}^{-3}, \quad x \geq 0}$$

and, by charge neutrality, $\delta p(x) = \delta n(x)$.

Step 3: (i) Diffusion length

$$L_n = \sqrt{D_n \tau_{n0}} = \sqrt{(30)(7 \times 10^{-7})} = \sqrt{2.1 \times 10^{-5}} \text{ cm} = 4.583 \times 10^{-3} \text{ cm}$$

$$\boxed{L_n = 45.8\ \mu\text{m}}$$

So $\delta n(x) = 5 \times 10^{15} \exp(-x/45.8\ \mu\text{m})$ cm^{-3}: the excess falls by a factor e every 45.8 μm.

Step 4: (ii) Distance where the excess drops to the given value

$10 \times 10^{13} = 1 \times 10^{14}$ cm^{-3}. Set $\delta n(x) = 10^{14}$:

$$10^{14} = 5 \times 10^{15}\, e^{-x/L_n} \quad \Rightarrow \quad x = L_n \ln \frac{\delta n_0}{\delta n} = L_n \ln \frac{5 \times 10^{15}}{10^{14}} = (45.83\ \mu\text{m}) \ln 50$$

$$x = 45.83 \times 3.912 = 179.3\ \mu\text{m}$$

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$$\boxed{x = 179.3\ \mu\text{m} = 1.79 \times 10^{-2}\ \text{cm}}$$

The excess has fallen fifty times in about four diffusion lengths.

Step 5: (iii) Total electron and hole concentrations

Equilibrium values:

$$p_0 = N_a = 5 \times 10^{16}\ \text{cm}^{-3}, \qquad n_0 = \frac{n_i^2}{N_a} = \frac{(1.5 \times 10^{10})^2}{5 \times 10^{16}} = 4.5 \times 10^{3}\ \text{cm}^{-3}$$

Totals, with $\delta p = \delta n$:

$$n(x) = n_0 + \delta n_0\, e^{-x/L_n} = 4.5 \times 10^3 + 5 \times 10^{15}\, e^{-x/L_n}$$

$$p(x) = p_0 + \delta n_0\, e^{-x/L_n} = 5 \times 10^{16} + 5 \times 10^{15}\, e^{-x/L_n}$$

x (μm)	δn (cm^{-3})	n (cm^{-3})	p (cm^{-3})
0	5.00×10^{15}	5.00×10^{15}	5.50×10^{16}
45.8 (L_n)	1.84×10^{15}	1.84×10^{15}	5.18×10^{16}
91.7 ($2L_n$)	6.77×10^{14}	6.77×10^{14}	5.07×10^{16}
179.3	1.00×10^{14}	1.00×10^{14}	5.01×10^{16}
458.3 ($10L_n$)	2.27×10^{11}	2.27×10^{11}	5.00×10^{16}
916.5 ($20L_n$)	1.03×10^{7}	1.03×10^{7}	5.00×10^{16}
1374.8 ($30L_n$)	4.68×10^{2}	4.97×10^{3}	5.00×10^{16}

Table 2: Excess and total carrier concentrations along the bar

On a log scale $\log n$ falls as a **straight line** (an exponential), one decade every $L_n \ln 10 = 105.5\ \mu$m, until δn reaches n_0 at $x = L_n \ln(\delta n_0/n_0) = 1271\ \mu$m; beyond that n levels off at $n_0 = 4.5 \times 10^3$ cm^{-3}. The hole concentration is almost flat: it falls only from 5.5×10^{16} to 5.0×10^{16} cm^{-3} within a few diffusion lengths.

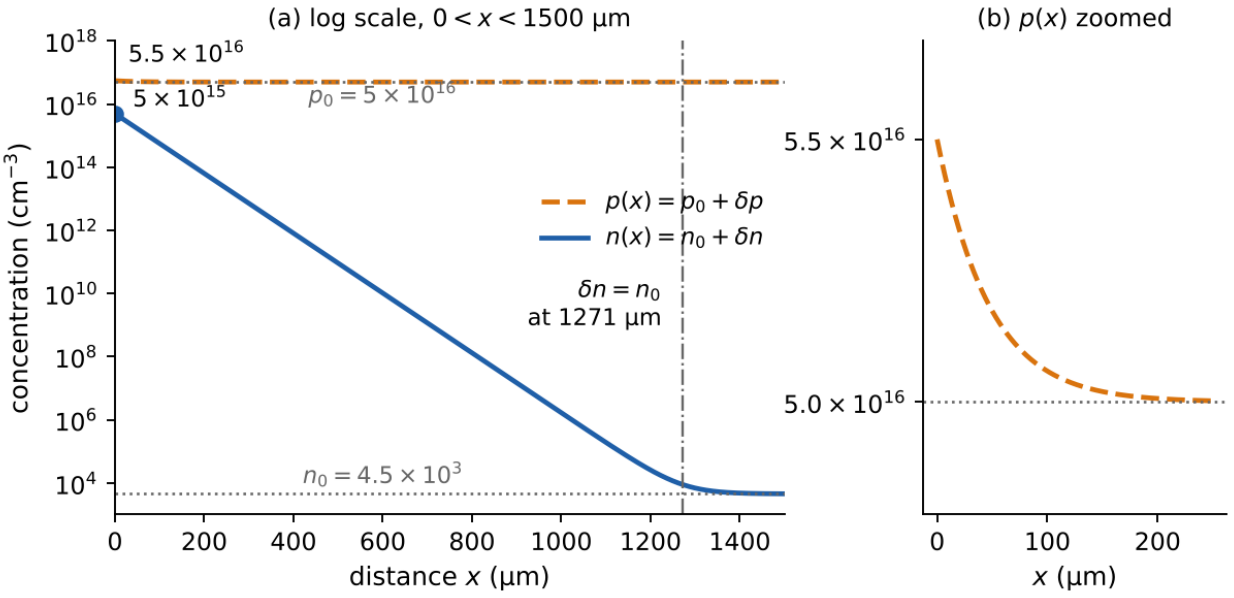


Figure 21: Total electron and hole concentrations versus distance on a log scale, with the hole curve zoomed

Reading the figure. In panel (a) the solid blue curve is $n(x)$: a straight line on the log scale from 5×10^{15} cm^{-3} at $x = 0$ down to the dotted grey level $n_0 = 4.5 \times 10^3$ cm^{-3}, which it joins near 1271

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μm (dash-dot grey line). The dashed orange curve is $p(x)$, almost on top of the dotted grey level $p_0 = 5 \times 10^{16}$ cm^{-3}. Panel (b) zooms on $p(x)$ (dashed orange) for the first 250 μm: it falls from 5.5×10^{16} to 5.0×10^{16} cm^{-3} with the same decay length L_n.

Final Answer:

- (i) $\delta n(x) = \delta p(x) = 5 \times 10^{15}\, e^{-x/L_n}$ cm^{-3} for $x \geq 0$, with $L_n = \sqrt{D_n \tau_{n0}} = 45.8$ μm
- (ii) $\delta n = 10^{14}$ cm^{-3} at $x = L_n \ln 50 = 179.3$ μm
- (iii) $n(x) = 4.5 \times 10^3 + 5 \times 10^{15} e^{-x/L_n}$ (straight line on a log scale down to n_0 near 1271 μm); $p(x) = 5 \times 10^{16} + 5 \times 10^{15} e^{-x/L_n}$ (nearly constant at 5×10^{16} cm^{-3})

3. Keyword Explanation

1. Excess carriers

Electrons and holes above their equilibrium values, $\delta n = n - n_0$ and $\delta p = p - p_0$, created by light, injection or radiation.

2. Minority carrier lifetime

The average time τ_{n0} (or τ_{p0}) an excess minority carrier exists before it recombines; it sets the recombination rate $\delta n/\tau_{n0}$.

3. Low-level injection

The excess carrier concentration is much smaller than the majority carrier concentration, so the majority concentration is almost unchanged.

4. Ambipolar transport

The joint diffusion and drift of excess electrons and holes as one neutral cloud, held together by their internal field, with the properties of the minority carrier under low injection.

5. Diffusion length

$L_n = \sqrt{D_n \tau_{n0}}$, the average distance an excess minority electron diffuses before it recombines; the decay length of $\delta n(x)$.

4. Notes and Remarks

- **Unused data:** $\tau_{p0} = 5 \times 10^{-7}$ s and $D_p = 25$ cm^2/s are not needed. The material is p-type, so holes are the majority carriers and the excess cloud moves and recombines as minority electrons. They would give $L_p = \sqrt{D_p \tau_{p0}} = 35.4$ μm, which matters only for an n-type sample.
- **How good is low injection here?** At $x = 0$ the exact ambipolar coefficient is $D' = D_n D_p (n + p)/(D_n n + D_p p) = 29.5$ cm^2/s instead of $D_n = 30$ cm^2/s, a 1.6 % difference that shrinks quickly with x. So using D_n and τ_{n0} is justified.
- **Reading 10×10^{13}:** the paper's value is taken literally as 1×10^{14} cm^{-3}. If 10^{13} cm^{-3} were meant, the distance would be $L_n \ln 500 = 284.8$ μm.
- **Why $B = 0$:** a solution with e^{+x/L_n} grows without limit into the bar, which no real sample can have. For a short bar of length W (comparable to L_n) both terms are kept and a second boundary condition at $x = W$ fixes them.
- **Not equilibrium:** with excess carriers $np \neq n_i^2$. At $x = 0$, $np = 2.75 \times 10^{32}$ cm^{-6}, far above $n_i^2 = 2.25 \times 10^{20}$ cm^{-6}; the quasi-Fermi levels split.

5. Final Concise Answer

(a) With complete ionization ($N_a^- = N_a$, $N_d^+ = N_d$) and $n_0 p_0 = n_i^2$, the neutrality condition becomes $n_0^2 - (N_d - N_a) n_0 - n_i^2 = 0$. Taking the positive root: $n_0 = \frac{N_d - N_a}{2} + \sqrt{\left(\frac{N_d - N_a}{2}\right)^2 + n_i^2}$ and $p_0 = \frac{N_a - N_d}{2} + \sqrt{\left(\frac{N_a - N_d}{2}\right)^2 + n_i^2}$. For $|N_d - N_a| \gg n_i$ the majority concentration is the net doping; for $N_d = N_a$, $n_0 = p_0 = n_i$.

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(b) In equilibrium E_F is flat through the device. On the P side E_F lies $e\phi_{Fp} = kT\ln(N_a/n_i)$ below E_{Fi}; on the N side it lies $e|\phi_{Fn}| = kT\ln(N_d/n_i)$ above E_{Fi}. So E_c, E_v and E_{Fi} are higher on the P side by eV_{bi} and bend only inside the space charge region $-x_p < x < x_n$, with $V_{bi} = |\phi_{Fp}| + |\phi_{Fn}| = V_t\ln(N_aN_d/n_i^2)$.

(c) Drift mobility is the drift velocity per unit field, $\mu = |v_d|/E = e\tau/m^*$ ($\text{cm}^2/\text{V s}$). Carriers are scattered by lattice vibrations (phonons, $\mu_L \propto T^{-3/2}$) and by the Coulomb field of ionized impurities ($\mu_I \propto T^{3/2}/N_I$), combined by $1/\mu = 1/\mu_L + 1/\mu_I$. The mean free time τ (about 10^{-13} s) between collisions sets the mobility, and through it the conductivity $\sigma = e(n\mu_n + p\mu_p)$ and the device speed.

(d) For p-type Si under low injection ($\delta n_0/N_a = 0.1$) the ambipolar equation reduces to $D_n\, d^2(\delta n)/dx^2 - \delta n/\tau_{n0} = 0$, giving $\delta n(x) = \delta p(x) = 5\times10^{15}e^{-x/L_n}$ cm^{-3} with $L_n = 45.8$ μm. The excess falls to 10^{14} cm^{-3} at $x = 179.3$ μm. $n(x) = 4.5\times10^3 + \delta n$ is a straight line on a log scale down to n_0 near 1271 μm; $p(x) = 5\times10^{16} + \delta n$ stays almost constant.

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4. (a) Explain the effects of temperature on carrier mobility inside a semiconductor. Marks: 5
   (b) Draw the energy band diagram of a Metal-Oxide-Semiconductor structure for the three modes of operation of MOSFET and thus explain how the channel is formed. Marks: 5
   (c) Show the four operating modes of a bipolar transistor as a function of V_{BE} and V_{CB}. Draw the minority carrier distribution of the four modes of a NPN transistor with proper labelling of all the quantities. Marks: 5
   (d) Consider a PN junction at room temperature with $n_i = 1.5 \times 10^{10}$ cm^{-3}, assume doping in N side is 1×10^6 cm^{-3} [*sic*: should be 1×10^{16} cm^{-3}]. What will be the minority carrier hole concentrations at the edge in N side if forward bias and reverse bias of 0.7 V is applied? Do the same for majority carrier electron concentration in N side. Marks: 10

4.(a) Explain the effects of temperature on carrier mobility inside a semiconductor. Marks: 5

1. Theory

1. What is mobility? (Core Idea)

Mobility μ tells us how fast a carrier drifts for each volt per centimetre of electric field: $v_d = \mu E$, in cm^2/V s. A carrier is accelerated by the field, but every short while it hits something (it is **scattered**) and loses its extra velocity. The average time between two collisions is the mean free time τ, and

$$\mu = \frac{e\,\tau}{m^*}$$

where e is the electron charge and m^* the conductivity effective mass. This is built in full in 3(c). Here we only need one fact: **mobility is large when collisions are rare** (long τ). So to see how temperature changes mobility, we must see how temperature changes each kind of collision.
The two main kinds of collision in a semiconductor are:

- **Lattice (phonon) scattering:** collisions with the vibrating atoms of the crystal.
- **Ionized impurity scattering:** collisions with the charged dopant ions (the N_d^+ and N_a^- ions).

Temperature acts on these two in **opposite** directions.

2. Lattice scattering: more heat, more collisions

Above 0 K the atoms of the crystal vibrate about their places. The hotter the crystal, the larger the vibrations. A vibrating lattice is no longer perfectly periodic, and these disturbances (their quanta are called **phonons**) deflect the carriers.

- More heat means more phonons and larger vibrations.
- So collisions become more frequent, τ falls, and the mobility falls.

Theory gives (Neamen):

$$\mu_L \propto T^{-3/2}$$

where μ_L is the mobility we would have if only lattice scattering existed (the **lattice-limited mobility**) and T is the absolute temperature in K.

Example

going from 200 K to 400 K (twice the temperature), μ_L falls by $2^{3/2} = 2.83$ times. In real silicon the measured fall is a little steeper than $T^{-3/2}$ (about $T^{-2.4}$ for electrons), because other phonon processes also act, but the trend is the same.

3. Ionized impurity scattering: more heat, fewer collisions

A donor ion N_d^+ has a positive charge. Its Coulomb force bends the path of a passing electron (it pulls a passing electron, pushes a passing hole). How much the path is bent depends on how long the carrier

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stays near the ion:

- A **slow** carrier passes the ion slowly, feels the force for a long time and is strongly deflected.
- A **fast** carrier rushes past and is hardly deflected.

The average thermal speed of carriers grows with temperature ($\frac{1}{2}m^*v_{th}^2 = \frac{3}{2}kT$). So at higher T carriers pass the ions faster, are deflected less, and the mobility **rises**. The more ions there are, the more collisions, so the mobility falls with the total ionized impurity concentration. Theory gives (Neamen):

$$\mu_I \propto \frac{T^{3/2}}{N_I}$$

where

- μ_I is the **impurity-limited mobility** (only ionized impurity scattering)
- $N_I = N_d^+ + N_a^-$ is the total ionized impurity concentration (cm^{-3}); in a compensated material both kinds of ion count

4. Adding the two: Matthiessen's rule

The two collision processes are independent. In a time dt the probability of a lattice collision is dt/τ_L and of an impurity collision dt/τ_I; the total probability is the sum:

$$\frac{1}{\tau} = \frac{1}{\tau_L} + \frac{1}{\tau_I}$$

Dividing by e/m^* (since $\mu = e\tau/m^*$) gives **Matthiessen's rule**:

$$\frac{1}{\mu} = \frac{1}{\mu_L} + \frac{1}{\mu_I}$$

So the total mobility is always **smaller than the smaller of the two**. The mechanism with the lower mobility (the more frequent collision) controls the total. This is like two resistors in series: the larger resistance dominates.

5. The total mobility against temperature

Putting the two laws into Matthiessen's rule:

$$\frac{1}{\mu} = a\,T^{3/2} + b\,\frac{N_I}{T^{3/2}}$$

with a and b constants of the material.

- At **high** T the first term wins: the lattice term dominates, and $\mu \propto T^{-3/2}$ falls with temperature.
- At **low** T the second term wins: impurity scattering dominates, and $\mu \propto T^{3/2}$ rises with temperature.
- In between, μ has a **maximum**. Setting $d(1/\mu)/dT = 0$ gives $aT^{3/2} = bN_I/T^{3/2}$, that is, the peak is where $\mu_L = \mu_I$. Since $T_{peak}^3 \propto N_I$, a heavier doping moves the peak to a higher temperature, and lowers the whole curve.

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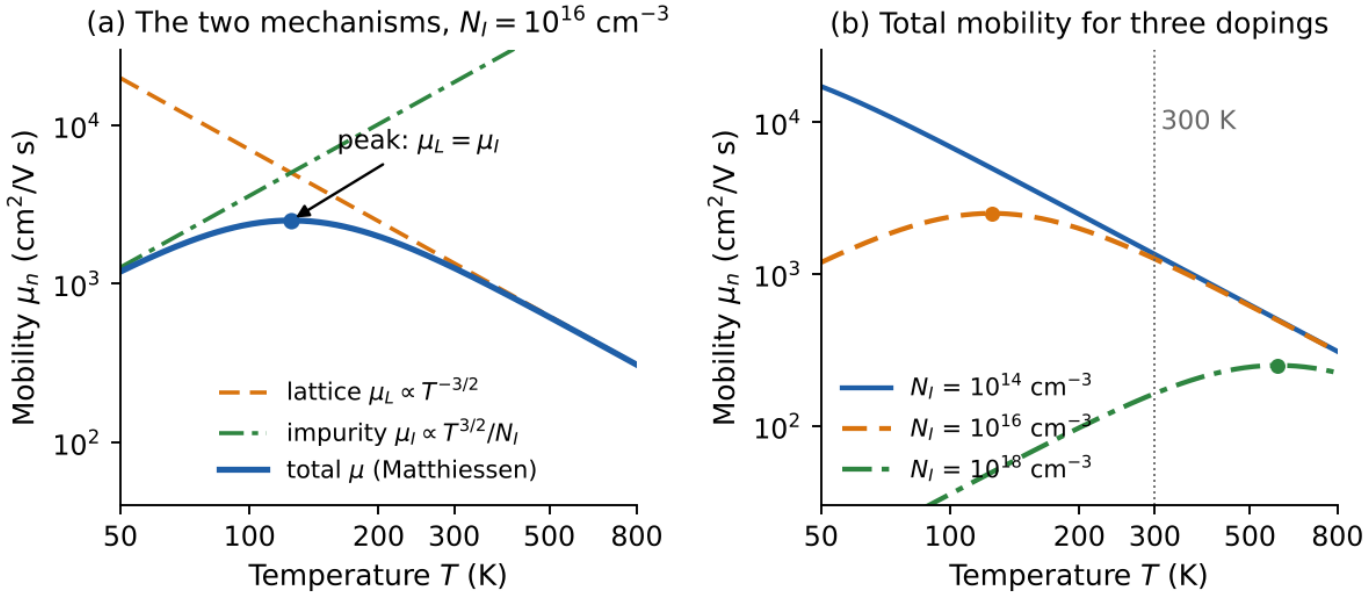


Figure 22: Electron mobility against temperature on log-log axes from a simple model scaled so that the lightly doped mobility at 300 K is 1350 cm^2/V s: (a) the two mechanisms and their Matthiessen sum, (b) the total mobility for three dopings

Reading the figure. The curves come from an illustrative model, $\mu_L = 1350\,(T/300)^{-3/2}$ and $\mu_I = (1.867 \times 10^{20}/N_I)\,(T/300)^{3/2}$ cm^2/V s, chosen so that lightly doped Si has about 1350 cm^2/V s at 300 K. They show the shape, not measured values. In panel (a), for $N_I = 10^{16}$ cm^{-3}, the dashed orange line is the lattice-limited μ_L (slope $-3/2$ on log-log axes), the dash-dot green line is the impurity-limited μ_I (slope $+3/2$), and the thick solid blue curve is their Matthiessen sum. The blue curve always lies below both lines; it follows the green line at low T and the orange line at high T, with its peak (dot, about 2500 cm^2/V s near 125 K) where the two lines cross. In panel (b) the solid blue curve (10^{14} cm^{-3}) is almost pure lattice scattering above 50 K and is 1349 cm^2/V s at 300 K. The dashed orange curve (10^{16}) is a little lower at 300 K (1259) and peaks near 125 K. The dash-dot green curve (10^{18}) is much lower (164 at 300 K) and its peak has moved up to about 580 K, so it still rises with T at room temperature.

6. Typical numbers

Real data for Si at 300 K (Neamen, Chapter 5): electron mobility about 1350 cm^2/V s in lightly doped material, falling to a few hundred cm^2/V s near 10^{18} cm^{-3}. For lightly doped Si the mobility at 400 K is about half of that at 300 K. Holes follow the same laws with smaller values ($\mu_p = 480$ cm^2/V s, lightly doped, 300 K).

2. Answer

Mobility is $\mu = e\tau/m^*$, so it depends on how often carriers are scattered (the mean free time τ). Temperature changes the two main scattering mechanisms in opposite ways.

1. Lattice (phonon) scattering. The atoms vibrate more as T rises, so carriers collide with phonons more often and τ falls:

$$\mu_L \propto T^{-3/2}$$

The lattice-limited mobility **decreases** with temperature.

2. Ionized impurity scattering. A charged donor or acceptor ion deflects a passing carrier by its Coulomb force. At higher T carriers move faster (thermal speed grows with T), spend less time near each ion and are deflected less:

$$\mu_I \propto \frac{T^{3/2}}{N_I}$$

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where $N_I = N_d^+ + N_a^-$ is the total ionized impurity concentration. The impurity-limited mobility **increases** with temperature and **decreases** with doping.

3. Total mobility (Matthiessen's rule). The collision rates add, so

$$\frac{1}{\mu} = \frac{1}{\mu_L} + \frac{1}{\mu_I}$$

The smaller of the two controls the total.

4. Result (see the figure in the Theory):

- **Low temperature:** impurity scattering dominates, μ rises with T.
- **High temperature (including room temperature for light doping):** lattice scattering dominates, μ falls with T roughly as $T^{-3/2}$.
- In between there is a **peak** where $\mu_L = \mu_I$.
- **Heavier doping** lowers the whole curve and moves the peak to a higher temperature; for heavy doping ($\sim 10^{18}$ cm^{-3}) the mobility can still be rising at 300 K.

Practical note: in lightly doped Si the electron mobility falls from about 1350 cm^2/V s at 300 K to roughly half of that at 400 K, so device currents (and the conductivity, when the carrier density is fixed) drop as a chip heats up.

3. Keyword Explanation

1. Phonon

A quantum of lattice vibration. Its number grows with temperature, so phonon (lattice) scattering grows with temperature.

2. Ionized impurity concentration

$N_I = N_d^+ + N_a^-$, the total density of charged dopant ions. Every ion scatters carriers, whatever its sign, so in a compensated material both kinds count.

3. Matthiessen's rule

The rule $1/\mu = \sum_i 1/\mu_i$: the inverse mobilities (scattering rates) of independent scattering mechanisms add.

4. Notes and Remarks

- **Exponents are ideal:** $T^{-3/2}$ and $T^{3/2}$ come from simple models. Measured Si data fall faster with T (about $T^{-2.4}$ for electrons and $T^{-2.2}$ for holes, Sze), but the shapes and the peak are the same.
- **Freeze-out:** at very low T the donors capture electrons back and become neutral (1(d)), so N_I itself falls. Neutral impurities scatter much more weakly.
- **Conductivity vs temperature:** $\sigma = en\mu$. In the extrinsic range $n \approx N_d$ is constant, so σ follows μ and falls as T rises. At high T the intrinsic carriers grow exponentially and σ rises again.
- **Mobility at high field** also stops following $v_d = \mu E$: the drift velocity saturates near 10^7 cm/s in Si. That is a field effect, not a temperature effect.

4.(b) Draw the energy band diagram of a Metal-Oxide-Semiconductor structure for the three modes of operation of MOSFET and thus explain how the channel is formed. Marks: 5

1. Theory

1. What is a MOS structure? (Core Idea)

A **MOS** (metal-oxide-semiconductor) structure is a sandwich: a metal (or heavily doped polysilicon) **gate** on top, a thin insulating layer of silicon dioxide (SiO_2, thickness t_{ox}, a few nm) in the middle, and

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a semiconductor **substrate** (also called the body) below. With a contact at the back of the substrate, it is a **MOS capacitor**: two conductors separated by an insulator. No DC current flows through the oxide.
The MOSFET is a MOS capacitor with two extra heavily doped regions, the **source** and the **drain**, at the two ends of the gate. For an **n-channel** MOSFET the substrate is **p-type** and source and drain are n^+. Between source and drain there are two back-to-back PN junctions, so no current can flow, unless the gate creates a thin layer of electrons at the surface that connects them. That layer is the **channel**. This sub-part shows how the gate creates it.

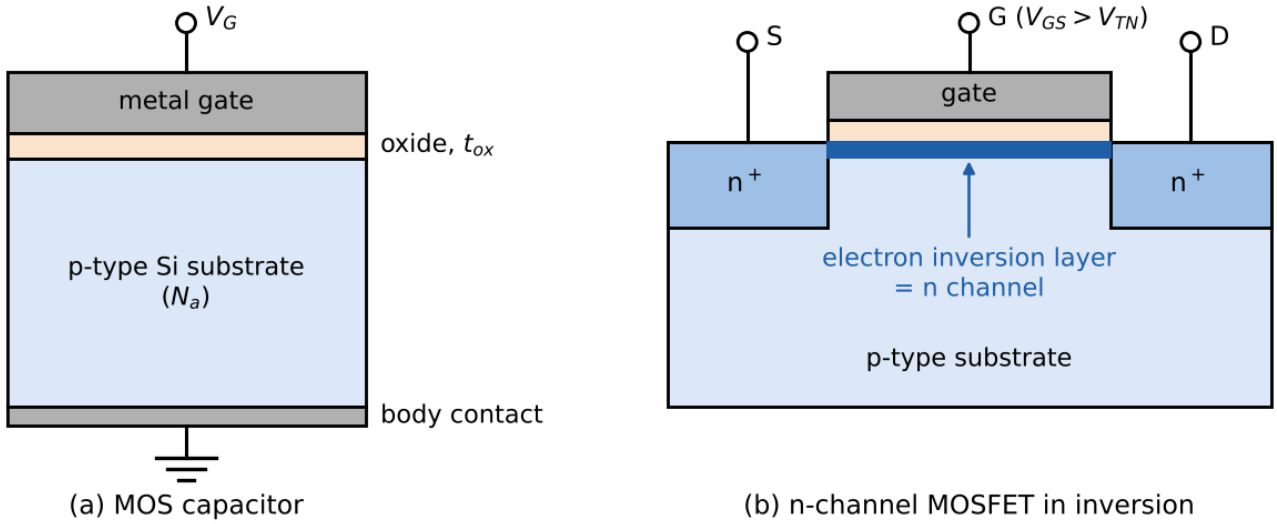


Figure 23: Cross-sections: (a) a MOS capacitor on a p-type substrate, (b) an n-channel MOSFET whose gate has created an electron inversion layer between the n^+ source and drain

Reading the figure. In panel (a) the gate (grey), the thin oxide (light orange) and the p-type substrate (light blue) are stacked, with the gate voltage V_G applied against the grounded body. In panel (b) the same stack sits between two n^+ regions; the thin dark blue strip under the oxide is the electron inversion layer, which joins source S and drain D when V_{GS} exceeds the threshold voltage V_{TN}.

2. The words we need for band diagrams

- E_c and E_v are the conduction and valence band edges; E_{Fi} is the intrinsic Fermi level, near mid-gap; E_F is the Fermi level. Their meaning is built in 1(a).
- In p-type material E_F lies **below** E_{Fi}. The distance is written $e\phi_{fp}$ (Neamen):

$$\phi_{fp} = \frac{E_{Fi} - E_F}{e} = V_t \ln\left(\frac{N_a}{n_i}\right)$$

where N_a is the acceptor concentration, n_i the intrinsic concentration and $V_t = kT/e = 0.0259$ V at 300 K. **Example:** $N_a = 10^{16}$ cm^{-3} gives $\phi_{fp} = 0.0259 \ln(10^{16}/1.5 \times 10^{10}) = 0.347$ V.

- The key rule: the electron density at any point is $n = n_i \exp[(E_F - E_{Fi})/kT]$ and the hole density $p = n_i \exp[(E_{Fi} - E_F)/kT]$. So wherever E_{Fi} is **above** E_F the material is p-type, and wherever E_{Fi} is **below** E_F it is n-type.
- **Band bending:** if the electric potential at the surface changes by ϕ_s relative to the bulk, all the energy levels at the surface move by $-e\phi_s$ (an electron's energy is $-e$ times the potential). ϕ_s is the **surface potential**. A positive ϕ_s bends the bands *down* at the surface.
- In equilibrium (no current flows through the oxide) E_F stays flat inside the semiconductor. A gate voltage V_G moves the metal's Fermi level by $-eV_G$ relative to the semiconductor's.

We use the **ideal** MOS capacitor: no charge in the oxide and equal work functions of metal and semiconductor, so with $V_G = 0$ the bands are flat (the flat-band case).

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3. Accumulation (negative gate voltage)

A negative gate voltage puts negative charge on the gate. To balance it, positive charge gathers in the semiconductor right at the surface: the **majority holes** are attracted up to the oxide. The bands bend *up* at the surface ($\phi_s < 0$), E_v comes closer to E_F, and the hole density at the surface is larger than in the bulk. This is **accumulation** of holes. There is no channel of electrons.

4. Depletion (small positive gate voltage)

A small positive gate voltage puts positive charge on the gate. The holes near the surface are pushed down into the substrate, leaving behind the fixed **negative acceptor ions** in a **space charge region** of width x_d. This region balances the gate charge. The bands bend *down* ($\phi_s > 0$), but E_{Fi} at the surface is still above E_F: the surface is still p-type, only weakly. The width grows with ϕ_s, by the same Poisson-equation argument as in a one-sided PN junction (2(c)):

$$x_d = \left(\frac{2\epsilon_s \phi_s}{eN_a} \right)^{1/2}$$

where ϵ_s is the permittivity of silicon.

5. Inversion (gate voltage above threshold)

As V_G grows, the bands bend down more. When E_{Fi} at the surface goes **below** E_F, the surface has more electrons than holes: it has been **inverted** from p-type to n-type. These electrons are minority carriers of the substrate. They come from thermal generation in the space charge region (and, in a MOSFET, from the n^+ source), and the field pulls them to the surface.

The usual definition of the **threshold inversion point** (Neamen): the electron density at the surface equals the hole density in the bulk, $n_s = N_a$. Since $E_F - E_{Fi}$ at the surface must then equal $E_{Fi} - E_F$ in the bulk, E_{Fi} has dropped by $2e\phi_{fp}$:

$$\phi_s = 2\phi_{fp}$$

Example

for $N_a = 10^{16}$ cm^{-3}, $2\phi_{fp} = 0.695$ V, and then $n_s = n_i \exp(\phi_{fp}/V_t) = 10^{16}$ cm^{-3}.

Beyond this point a small increase of ϕ_s multiplies the electron density by a large factor, $n_s \propto \exp(e\phi_s/kT)$. So the electrons now supply all the extra charge, and the space charge region **stops growing** at its maximum width:

$$x_{dT} = \left(\frac{4\epsilon_s \phi_{fp}}{eN_a} \right)^{1/2}$$

Example

$N_a = 10^{16}$ cm^{-3} gives $x_{dT} = 0.300$ μm.

The band diagrams of the three modes are drawn in the Answer below, with a guide to reading them.

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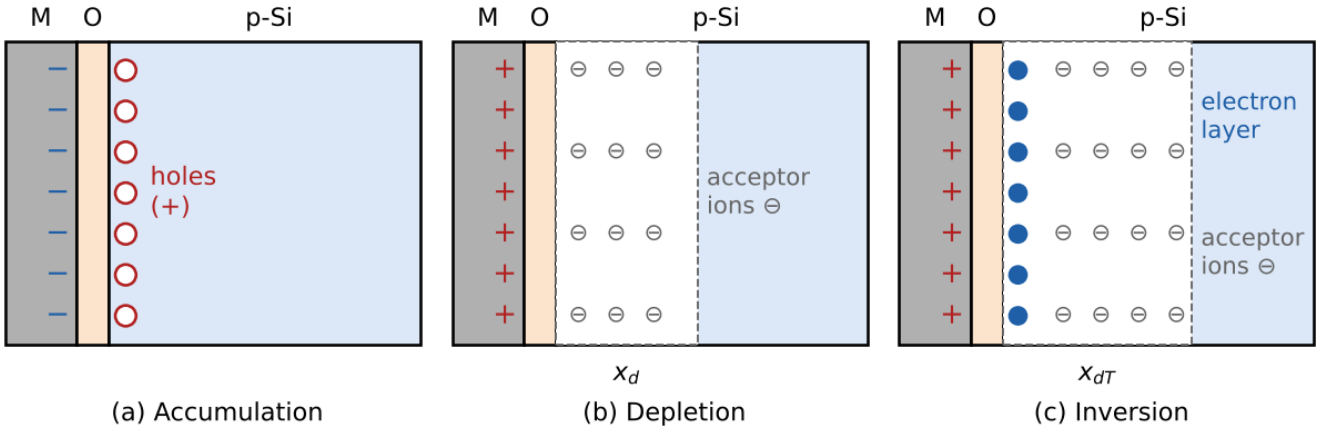


Figure 24: Charge picture of the MOS capacitor in accumulation, depletion and inversion

Reading the figure. In (a) negative charge on the gate is balanced by a thin sheet of holes (open circles) at the surface. In (b) positive charge on the gate is balanced by negative acceptor ions (circled minus signs, $\ominus$) in the region x_d that the holes have left. In (c) a larger positive gate charge is balanced by the ions in the maximum width x_{dT} plus a thin sheet of electrons (filled dots) at the surface: this sheet is the channel.

6. The threshold voltage

The gate voltage needed to reach $\phi_s = 2\phi_{fp}$ is the **threshold voltage** V_{TN}. At threshold the gate voltage is shared between the oxide and the silicon. The oxide is a capacitor of capacitance per area $C_{ox} = \epsilon_{ox}/t_{ox}$ that carries the depletion charge $|Q'_{SD}(max)| = eN_a x_{dT}$, so its voltage is $|Q'_{SD}(max)|/C_{ox}$. Adding the flat-band voltage V_{FB} (zero for the ideal structure; it comes from the work-function difference ϕ_{ms} and the oxide charge Q'_{ss}) gives Neamen's result:

$$V_{TN} = \frac{|Q'_{SD}(max)|}{C_{ox}} + V_{FB} + 2\phi_{fp}, \qquad V_{FB} = \phi_{ms} - \frac{Q'_{ss}}{C_{ox}}$$

where

- $C_{ox} = \epsilon_{ox}/t_{ox}$ is the oxide capacitance per unit area (F/cm^2), $\epsilon_{ox} = 3.9\,\epsilon_0$
- $Q'_{SD}(max) = -eN_a x_{dT}$ is the depletion charge per unit area at threshold (C/cm^2)

Example

$N_a = 10^{16}$ cm^{-3}, $t_{ox} = 10$ nm, ideal ($V_{FB} = 0$): $C_{ox} = 3.45\times10^{-7}$ F/cm^2, $|Q'_{SD}(max)| = 4.80\times10^{-8}$ C/cm^2, so $V_{TN} = 0.139 + 0.695 = 0.834$ V.

2. Answer

MOS structure: metal gate / thin SiO_2 (t_{ox}) / p-type Si substrate; in a MOSFET, n^+ source and drain sit at the two ends of the gate (structure figure in the Theory). The gate voltage V_G is applied against the substrate. Ideal structure: flat bands at $V_G = 0$. In the p-type bulk E_F is $e\phi_{fp}$ below E_{Fi}, with $\phi_{fp} = V_t \ln(N_a/n_i)$.

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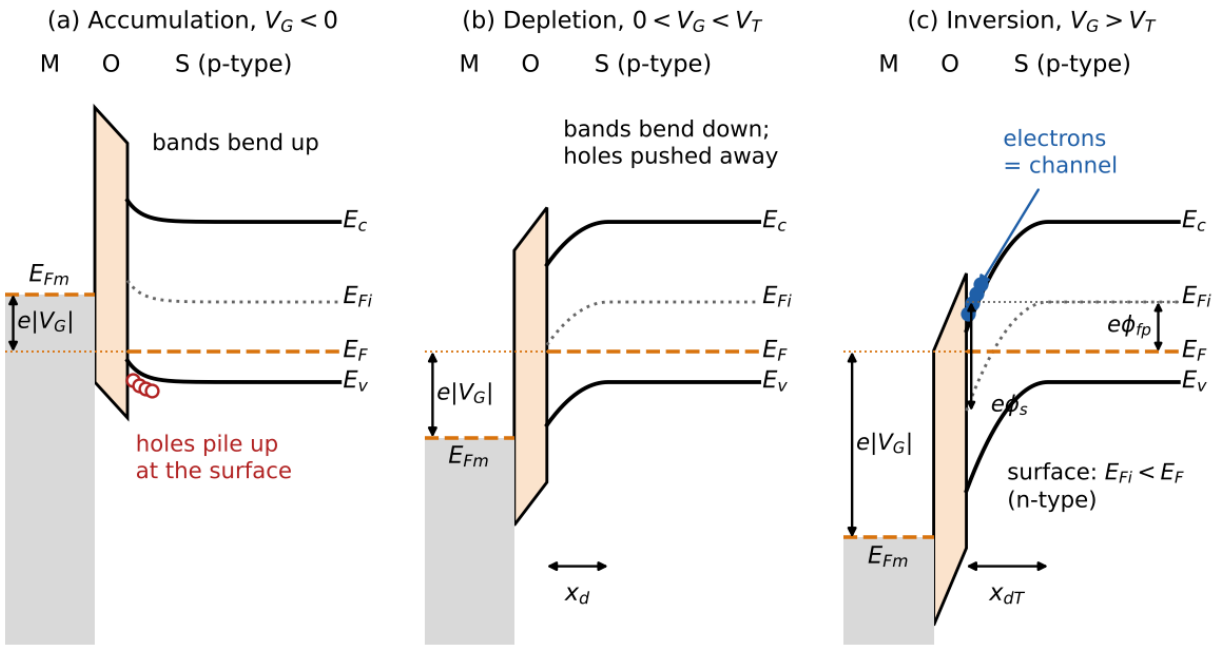


Figure 25: Energy band diagrams of an ideal MOS capacitor on p-type Si in accumulation, depletion and inversion (oxide band edges schematic, not to scale)

Reading the figure. Each panel shows the metal (M, grey, with its Fermi level E_{Fm} as a dashed orange line), the oxide (O, light orange) and the p-type semiconductor (S). In the semiconductor the solid black lines are E_c and E_v, the dotted grey curve is E_{Fi} and the dashed orange line is E_F, flat everywhere. The double arrow $e|V_G|$ shows how far the metal Fermi level has moved from E_F. In (a) $V_G < 0$: E_{Fm} is raised, the bands bend up, E_v approaches E_F and holes (open red circles) pile up at the surface. In (b) $V_G > 0$ but small: the bands bend down over the depletion width x_d, but E_{Fi} stays above E_F. In (c) $V_G > V_T$: the bands bend down by $e\phi_s \geq 2e\phi_{fp}$, so near the surface E_{Fi} is below E_F (the surface is n-type); E_c comes close to E_F and electrons (filled blue dots) form the inversion layer; the depletion width has reached x_{dT}. The tilt of the oxide band edges shows the field in the oxide.

(a) Accumulation, $V_G < 0$. Negative charge on the gate attracts the majority holes to the surface. The bands bend *up* at the surface, E_v moves closer to E_F, and a thin layer of excess holes forms. No channel exists.

(b) Depletion, $0 < V_G < V_T$. Positive charge on the gate repels the holes and leaves uncovered negative acceptor ions in a depletion region of width $x_d = (2\epsilon_s\phi_s/eN_a)^{1/2}$. The bands bend *down*, but E_{Fi} at the surface is still above E_F, so the surface is still p-type (weakly). No channel exists.

(c) Inversion, $V_G > V_T$. The bands bend down so far that E_{Fi} at the surface crosses **below** E_F. The surface now has more electrons than holes: it is n-type, an **inversion layer**. Threshold is defined by

$$\phi_s = 2\phi_{fp} \quad (n_s = N_a)$$

and the depletion width stops at its maximum $x_{dT} = (4\epsilon_s\phi_{fp}/eN_a)^{1/2}$. The threshold voltage is

$$V_{TN} = \frac{|Q'_{SD}(max)|}{C_{ox}} + V_{FB} + 2\phi_{fp}$$

where $|Q'_{SD}(max)| = eN_a x_{dT}$, $C_{ox} = \epsilon_{ox}/t_{ox}$ and V_{FB} is the flat-band voltage.

How the channel forms. As V_G rises above V_T, the extra positive gate charge is balanced almost entirely by electrons at the surface, because their density grows as $\exp(e\phi_s/kT)$. These electrons (from thermal generation and from the n^+ source) form a thin n-type sheet under the oxide. In a MOSFET this sheet connects the n^+ source to the n^+ drain: it is the **n-channel**. Its charge is about $Q'_n = -C_{ox}(V_{GS} - V_{TN})$, so a higher gate voltage gives more electrons and more drain current when V_{DS} is applied. Below V_T the two back-to-back PN junctions block current, so the gate turns the transistor on and off.

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3. Keyword Explanation

1. Surface potential

ϕ_s, the change of electric potential from the bulk to the semiconductor surface. The bands at the surface move by $-e\phi_s$.

2. Inversion layer

A thin surface layer whose type is opposite to the substrate (electrons on p-type), created by the gate field. Strong inversion starts at $\phi_s = 2\phi_{fp}$.

3. Threshold voltage

The gate voltage that brings the surface to the threshold inversion point; below it the MOSFET is off, above it a channel exists.

4. Flat-band voltage

$V_{FB} = \phi_{ms} - Q'_{ss}/C_{ox}$: the gate voltage that makes the bands flat in a real (non-ideal) MOS structure. It is zero for the ideal structure drawn here.

4. Notes and Remarks

- **Real structures:** with a work-function difference ϕ_{ms} (for example an n^+ polysilicon gate on p-Si, $\phi_{ms} \approx -1$ V) the bands are already bent at $V_G = 0$, and every voltage in the description is shifted by V_{FB}. The three modes and the diagrams are the same about $V_G = V_{FB}$ instead of $V_G = 0$.
- **n-type substrate (p-channel):** everything reverses. Accumulation is at $V_G > 0$, inversion at $V_G < 0$ with a hole channel, and the threshold is $\phi_s = -2\phi_{fn}$.
- **Where do the inversion electrons come from?** In a MOS capacitor alone, only thermal generation supplies them, which takes time (this is why a high-frequency C–V curve does not show the inversion capacitance). In a MOSFET the source supplies them at once.
- **Inversion with band diagrams** is also asked in 6(c), and the double-gate version in 6(a).
- The example numbers ($N_a = 10^{16}$ cm^{-3}, $t_{ox} = 10$ nm) are only an illustration; the paper gives no values.

4.(c) Show the four operating modes of a bipolar transistor as a function of V_{BE} and V_{CB}. Draw the minority carrier distribution of the four modes of a NPN transistor with proper labelling of all the quantities. Marks: 5

1. Theory

1. What is a bipolar transistor? (Core Idea)

A **bipolar junction transistor (BJT)** is three doped regions in a row, so it has two PN junctions back to back. In an **NPN** transistor:

- the **emitter** (E) is n-type and very heavily doped (N_E);
- the **base** (B) is p-type, moderately doped (N_B) and **very thin** (width x_B much smaller than the diffusion length of electrons in it);
- the **collector** (C) is n-type and lightly doped (N_C), so $N_E > N_B > N_C$.

The **B–E junction** lies between emitter and base, the **B–C junction** between base and collector. The idea of the device: the B–E junction **injects** electrons into the thin base; they diffuse across it, and the B–C junction **collects** them. Because the base is thin, almost none of them are lost on the way, so a small base current controls a large collector current.

The structure and the four modes in the V_{BE}–V_{CB} plane are drawn in the Answer below.

2. When is each junction forward or reverse biased?

A PN junction is forward biased when its p side is at a higher potential than its n side.

- **B–E junction** (base is p, emitter is n): forward when $V_{BE} > 0$, reverse when $V_{BE} < 0$.

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- **B–C junction** (base is p, collector is n): forward when $V_{BC} = V_B - V_C > 0$. With the axis $V_{CB} = -V_{BC}$, this means: **reverse when** $V_{CB} > 0$**, forward when** $V_{CB} < 0$.

Two junctions, each forward or reverse, give four combinations: the four **operating modes**.

Mode	B–E junction	B–C junction	V_{BE}	V_{CB}	Use
Forward active	forward	reverse	> 0	> 0	amplifier
Saturation	forward	forward	> 0	< 0	switch ON
Cutoff	reverse	reverse	< 0	> 0	switch OFF
Inverse active	reverse	forward	< 0	< 0	rarely used

Table 3: Summary

3. Minority carriers in each region

In each region of an NPN transistor the **minority** carriers are: holes in the emitter (concentration p_E), electrons in the base (n_B), holes in the collector (p_C). In equilibrium their values are (Neamen):

$$p_{E0} = \frac{n_i^2}{N_E}, \qquad n_{B0} = \frac{n_i^2}{N_B}, \qquad p_{C0} = \frac{n_i^2}{N_C}$$

Since $N_E > N_B > N_C$, we have $p_{E0} < n_{B0} < p_{C0}$.
The minority carriers move by **diffusion** (as in 3(d)), and their values at the edge of each depletion region are set by the voltage across that junction, the **law of the junction** (4(d)):

$$n_B(0) = n_{B0} \exp\left(\frac{V_{BE}}{V_t}\right), \qquad p_E(0) = p_{E0} \exp\left(\frac{V_{BE}}{V_t}\right)$$

$$n_B(x_B) = n_{B0} \exp\left(\frac{V_{BC}}{V_t}\right), \qquad p_C(0) = p_{C0} \exp\left(\frac{V_{BC}}{V_t}\right)$$

where $V_t = kT/e$, x is measured in the base from the B–E edge, and in the emitter and the collector the distance is measured from each junction outwards.

- A **forward**-biased junction ($V > 0$) multiplies the equilibrium value by a large factor: many minority carriers are **injected**.
- A **reverse**-biased junction (V a few V_t negative) makes the factor almost zero: the edge is **emptied** of minority carriers.

Far from the junctions (more than a few diffusion lengths) each concentration returns to its equilibrium value. Inside the thin base the profile between the two edge values is almost a **straight line**, because hardly any carriers recombine in a base much thinner than its diffusion length L_B. The forward-active case with its exact sinh solution is worked out in 5(b).

4. What each mode looks like

- **Forward active:** n_B falls from a large value $n_{B0}e^{V_{BE}/V_t}$ at the emitter edge to about 0 at the collector edge. The steep slope means a large electron diffusion current across the base into the collector: I_C is large and controlled by V_{BE}.
- **Cutoff:** both edges are emptied; there are almost no minority carriers anywhere near the junctions, so only tiny leakage currents flow. The transistor is OFF.
- **Saturation:** both edges are raised. The base fills with electrons from both sides; the slope (and so I_C) is smaller than the B–E voltage alone would give, and the collector is flooded with injected holes. The transistor is a closed switch with a small V_{CE} (about 0.1–0.2 V), but much stored charge must be removed before it can turn off.
- **Inverse active:** the roles of emitter and collector are swapped: the B–C junction injects, the B–E junction collects. Because the collector is lightly doped and large, the current gain is poor.

The four distributions are drawn in the Answer below, with a guide to reading them.

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2. Answer

NPN structure: n^{++} emitter, thin p^{+} base, n collector ($N_E > N_B > N_C$), with the B–E and B–C junctions. The mode is set by V_{BE} (B–E forward if $V_{BE} > 0$) and V_{CB} (B–C **reverse** if $V_{CB} > 0$, forward if $V_{CB} < 0$, since $V_{BC} = -V_{CB}$).

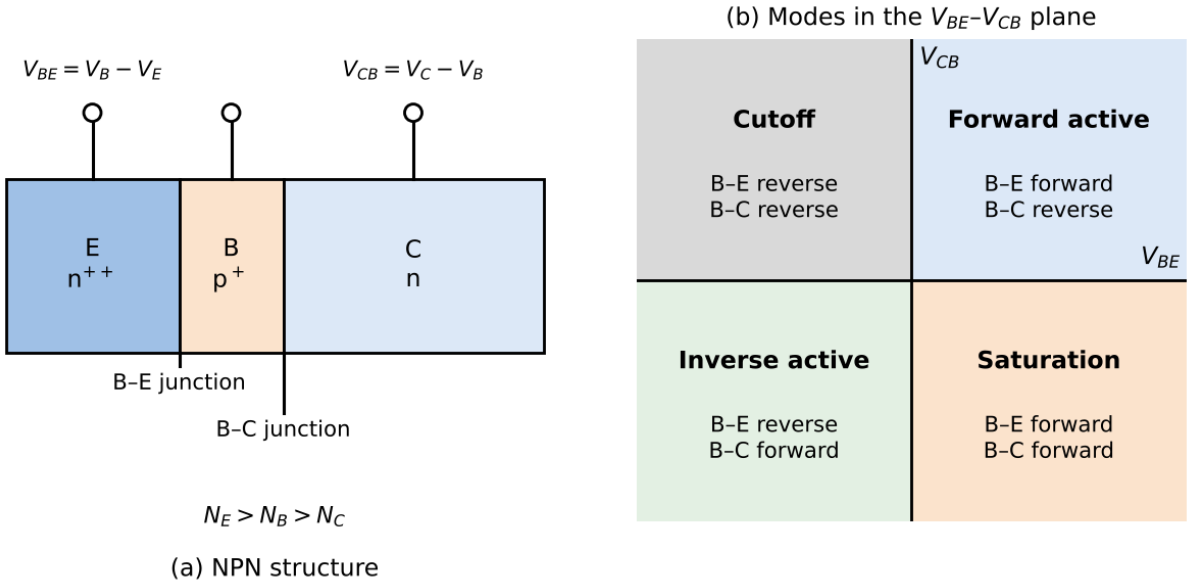


Figure 26: (a) Structure of an NPN transistor; (b) the four operating modes in the plane of V_{BE} and V_{CB}

Reading the figure. Panel (a) shows the three regions with their doping and the two junctions; $V_{BE} = V_B - V_E$ and $V_{CB} = V_C - V_B$ are the two voltages that set the mode. Panel (b) has V_{BE} on the horizontal axis and V_{CB} on the vertical axis; each quadrant (shaded differently) is one mode, with the bias of the two junctions written in it.

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Quadrant	Mode	B–E	B–C
$V_{BE} > 0$, $V_{CB} > 0$	Forward active	forward	reverse
$V_{BE} > 0$, $V_{CB} < 0$	Saturation	forward	forward
$V_{BE} < 0$, $V_{CB} > 0$	Cutoff	reverse	reverse
$V_{BE} < 0$, $V_{CB} < 0$	Inverse active	reverse	forward

Table 4: Summary

Minority carriers: holes p_E in the emitter, electrons n_B in the base, holes p_C in the collector, with equilibrium values $p_{E0} = n_i^2/N_E$, $n_{B0} = n_i^2/N_B$, $p_{C0} = n_i^2/N_C$. At each depletion edge the law of the junction applies:

$$p_E(0) = p_{E0}e^{V_{BE}/V_t}, \quad n_B(0) = n_{B0}e^{V_{BE}/V_t}, \quad n_B(x_B) = n_{B0}e^{V_{BC}/V_t}, \quad p_C(0) = p_{C0}e^{V_{BC}/V_t}$$

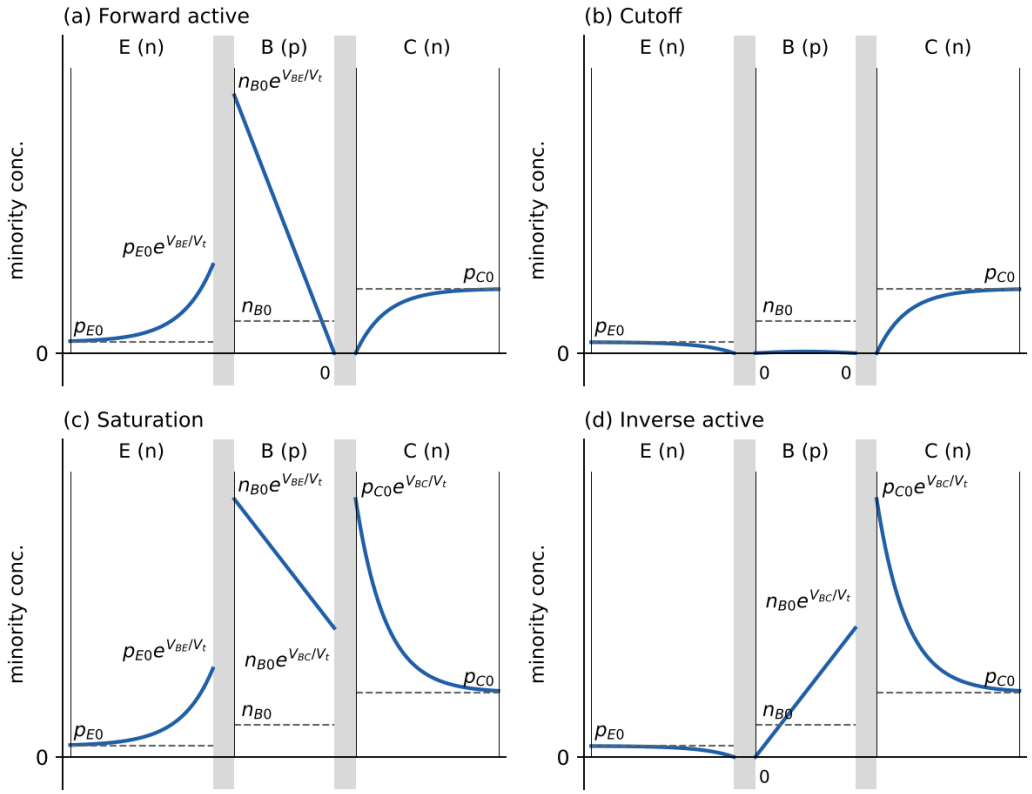


Figure 27: Minority carrier distributions in the emitter, base and collector of an NPN transistor in the four modes (schematic, not to scale)

Reading the figure. In each panel the regions are, from left to right, the emitter E (n), the base B (p) and the collector C (n); the grey bands between them are the two depletion regions. The dashed grey lines are the equilibrium values p_{E0}, n_{B0} and p_{C0}; the solid blue curves are the actual minority carrier concentrations p_E, n_B and p_C. Each edge value is labelled. A forward junction lifts its two edge values by e^{V/V_t}; a reverse junction pulls them to 0, marked "0".

(a) Forward active: p_E rises from p_{E0} to $p_{E0}e^{V_{BE}/V_t}$ at the B–E edge; n_B falls almost linearly from $n_{B0}e^{V_{BE}/V_t}$ to ≈ 0 at the B–C edge; p_C is ≈ 0 at the B–C edge and rises to p_{C0}. The large gradient of n_B gives the collector current.

(b) Cutoff: both junctions reverse: p_E, n_B and p_C are ≈ 0 at every junction edge and return to p_{E0} and p_{C0} away from them; the base is almost empty. Only leakage currents flow.

(c) Saturation: both junctions forward: $p_E(0) = p_{E0}e^{V_{BE}/V_t}$; n_B goes from $n_{B0}e^{V_{BE}/V_t}$ to $n_{B0}e^{V_{BC}/V_t}$ (both large, so a smaller slope); $p_C(0) = p_{C0}e^{V_{BC}/V_t}$, decaying to p_{C0}. Much excess charge is stored.

(d) Inverse active: B–E reverse, B–C forward: $p_E(0) \approx 0$ and $n_B(0) \approx 0$; n_B rises to $n_{B0}e^{V_{BC}/V_t}$ at the collector side; $p_C(0) = p_{C0}e^{V_{BC}/V_t}$, decaying to p_{C0}. Electrons flow from collector to emitter: the

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transistor works backwards with a low gain.

3. Keyword Explanation

1. Law of the junction

The minority carrier density at the edge of a depletion region equals its equilibrium value times $\exp(V/V_t)$, where V is the voltage across that junction (positive for forward bias).

2. Base width

x_B, the width of the neutral base between the two depletion regions. It is made much smaller than the electron diffusion length so that injected electrons cross it without recombining.

3. Saturation (BJT)

The mode with both junctions forward biased: the transistor acts as a closed switch with a small $V_{CE(sat)}$ and stores excess minority charge.

4. Notes and Remarks

- **Sign of the B–C bias:** the axis is V_{CB}, not V_{BC}. A positive V_{CB} reverse biases the B–C junction (forward active), a negative V_{CB} forward biases it (saturation). Mixing the two is the most common error in this question.
- **Not to scale:** the factor e^{V/V_t} is huge in a real transistor (for $V_{BE} = 0.6$ V it is about 10^{10}), so on a true linear scale the equilibrium values would be invisible. Textbooks (Neamen, Chapter 12) draw these profiles schematically, as here.
- **Straight line in the base:** the exact base solution is a sum of sinh terms; for $x_B \ll L_B$ it becomes linear (5(b)).
- **Emitter and collector widths:** if the emitter is shorter than its hole diffusion length (common in modern devices), p_E is a straight line to p_{E0} at the contact instead of an exponential tail. The labels and edge values are unchanged.

4.(d) Consider a PN junction at room temperature with $n_i = 1.5 \times 10^{10}$ cm^{-3}, assume doping in N side is 1×10^6 cm^{-3} [*sic:* should be 1×10^{16} cm^{-3}]. What will be the minority carrier hole concentrations at the edge in N side if forward bias and reverse bias of 0.7 V is applied? Do the same for majority carrier electron concentration in N side. Marks: 10

1. Theory

Note on the paper

The paper prints the N-side doping as 1×10^6 cm^{-3}. That is far **below** $n_i = 1.5 \times 10^{10}$ cm^{-3}. A doping that small changes almost nothing: solving charge neutrality gives $n_0 = 1.5001 \times 10^{10}$ and $p_0 = 1.5000 \times 10^{10}$ cm^{-3}, so the region would be **intrinsic**, not an N side, and "minority" and "majority" carriers would have no meaning. A missing digit is surely the misprint: the value meant is $N_d = 1 \times 10^{16}$ cm^{-3}, a common textbook doping. This solution uses $N_d = 1 \times 10^{16}$ cm^{-3}.

1. What happens at a PN junction in equilibrium? (Core Idea)

At a PN junction the electrons of the n side and the holes of the p side diffuse across and leave a **space charge region** (depletion region) of uncovered ions, from $-x_p$ on the p side to $+x_n$ on the n side. Its field creates a **built-in potential barrier** V_{bi} that stops further net flow. The equilibrium band diagram and V_{bi} are built in 3(b), and the width of the region in 2(c).

Outside the space charge region the material is **neutral**. On the n side the **majority** carriers are electrons, $n_{n0} = N_d$; the **minority** carriers are holes, $p_{n0} = n_i^2/N_d$ (the mass-action law $n_0 p_0 = n_i^2$, 1(b)).

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2. The barrier controls how many carriers cross

The holes of the p side ($p_{p0} \approx N_a$) must climb the barrier to reach the n side. Only those with enough energy succeed, and Boltzmann statistics says the fraction is $\exp(-eV_{bi}/kT)$. So in equilibrium

$$p_{n0} = p_{p0} \exp\left(-\frac{eV_{bi}}{kT}\right)$$

This is exactly consistent with $V_{bi} = V_t \ln(N_a N_d / n_i^2)$.

3. Applying a voltage: the law of the junction

Apply a voltage V_a with the p side positive ($V_a > 0$ is **forward** bias, $V_a < 0$ is **reverse** bias). Almost all of it appears across the high-resistance space charge region, so the barrier becomes $V_{bi} - V_a$: lower in forward bias, higher in reverse bias.

The same Boltzmann argument now gives the hole concentration at the edge $x = x_n$ of the n side:

$$p_n(x_n) = p_{p0} \exp\left(-\frac{e(V_{bi} - V_a)}{kT}\right) = p_{p0} \exp\left(-\frac{eV_{bi}}{kT}\right) \exp\left(\frac{eV_a}{kT}\right)$$

Using step 2, the first part is p_{n0}, so (Neamen):

$$p_n(x_n) = p_{n0} \exp\left(\frac{eV_a}{kT}\right)$$

and in the same way for electrons on the p side, $n_p(-x_p) = n_{p0} \exp(eV_a/kT)$. These are the **boundary conditions** of the PN junction, also called the **law of the junction**.

- **Forward bias:** the barrier is lower, holes pour into the n side, and the edge concentration rises far above p_{n0}. This is **minority carrier injection**.
- **Reverse bias:** the barrier is higher; for $|V_a|$ more than about $4V_t \approx 0.1$ V the factor is almost zero, so the edge is **emptied** of minority carriers. The field sweeps away any hole that reaches the edge.

The excess holes at the edge are $\delta p_n(x_n) = p_n(x_n) - p_{n0} = p_{n0}[\exp(eV_a/kT) - 1]$. Away from the edge they diffuse into the n side and recombine, so the excess decays as $\exp[-(x - x_n)/L_p]$, where L_p is the hole diffusion length (3(d)).

4. The assumption behind the law: low injection

The derivation used the equilibrium Boltzmann relation inside the space charge region and assumed that the majority carrier density is unchanged. Both need **low injection**: the injected minority carriers must be much fewer than the majority carriers,

$$\delta p_n \ll n_{n0} = N_d$$

When this holds, the electron density on the n side stays $\approx N_d$ and the law of the junction is accurate.

5. What happens to the majority carriers?

The n side must stay **neutral** (quasi-neutrality). Every extra hole (+) brought in at the edge is matched by an extra electron (−) drawn in from the n contact, so

$$\delta n_n = \delta p_n, \qquad n_n(x_n) = N_d + \delta p_n(x_n)$$

In low injection $\delta p_n \ll N_d$, so $n_n \approx N_d$: the majority carrier density hardly changes, while the minority density changes by many orders of magnitude. In reverse bias $\delta p_n = -p_{n0}$ (a tiny deficit), so $n_n = N_d - p_{n0} \approx N_d$.

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6. The band diagram under bias

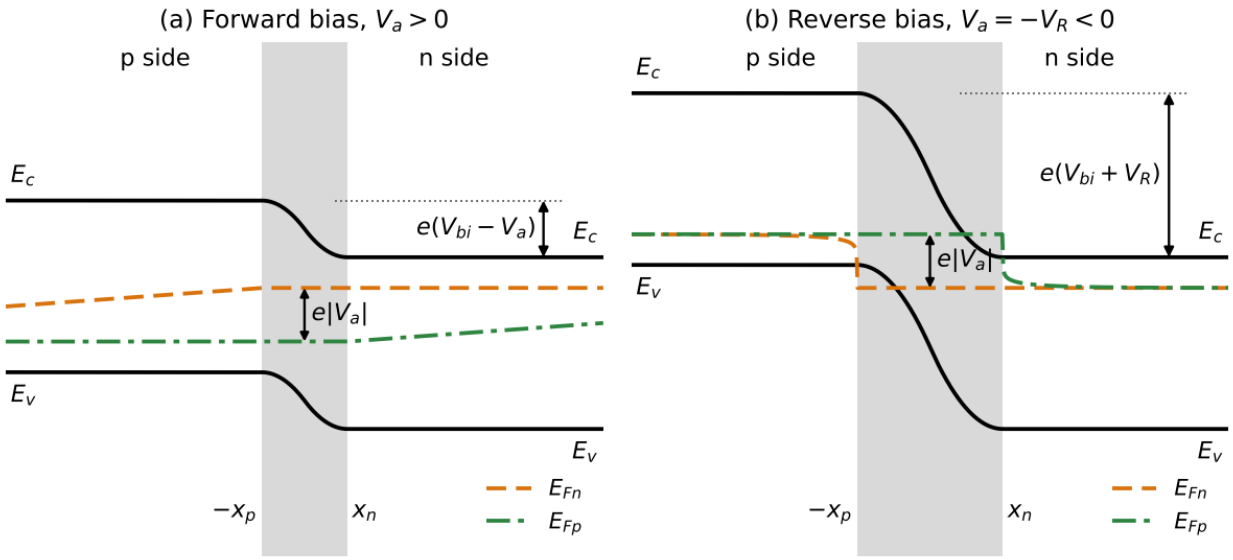


Figure 28: Energy band diagrams of a PN junction under (a) forward and (b) reverse bias, with the quasi-Fermi levels (schematic)

Reading the figure. The solid black lines are E_c and E_v; the grey band from $-x_p$ to x_n is the space charge region. Under bias there is no single Fermi level; electrons and holes each have a **quasi-Fermi level**, E_{Fn} (dashed orange) and E_{Fp} (dash-dot green), and they are flat through the space charge region and split by $e|V_a|$. In (a) forward bias the barrier on E_c falls to $e(V_{bi} - V_a)$ and E_{Fn} lies above E_{Fp}; the gradual bending of E_{Fp} on the n side shows the injected holes decaying over a few diffusion lengths. In (b) reverse bias the barrier grows to $e(V_{bi} + V_R)$ and E_{Fp} lies above E_{Fn}; the quasi-Fermi levels meet again within a very short distance because the edge is simply emptied. The split $E_{Fn} - E_{Fp} = eV_a$ at the edge is just another way of writing $p_n n_n = n_i^2 \exp(eV_a/kT)$.

2. Answer

Given:

- Si at room temperature, $T = 300$ K, $V_t = kT/e = 0.0259$ V
- $n_i = 1.5 \times 10^{10}$ cm^{-3}
- N side: $N_d = 1 \times 10^{16}$ cm^{-3} (the paper's 1×10^6 is a misprint, see the note above)
- Forward bias $V_a = +0.7$ V; reverse bias $V_a = -0.7$ V

Relationships:

$$p_{n0} = \frac{n_i^2}{N_d}, \qquad p_n(x_n) = p_{n0} \exp\left(\frac{eV_a}{kT}\right), \qquad n_n(x_n) = N_d + \delta p_n(x_n)$$

where

- p_{n0} is the equilibrium hole (minority) concentration in the n region (cm^{-3})
- $p_n(x_n)$ is the hole concentration at the edge x_n of the space charge region
- $\delta p_n = p_n(x_n) - p_{n0}$ is the excess hole concentration, and n_n is the electron (majority) concentration, from charge neutrality $\delta n_n = \delta p_n$

Step 1: Equilibrium carrier concentrations

$$n_{n0} = N_d = 1 \times 10^{16} \text{ cm}^{-3}, \qquad p_{n0} = \frac{n_i^2}{N_d} = \frac{(1.5 \times 10^{10})^2}{1 \times 10^{16}} = \frac{2.25 \times 10^{20}}{1 \times 10^{16}}$$

$$\boxed{p_{n0} = 2.25 \times 10^4 \text{ cm}^{-3}}$$

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Step 2: The bias factor

$$\frac{eV_a}{kT} = \frac{V_a}{V_t} = \frac{0.7}{0.0259} = 27.027, \qquad \exp(27.027) = 5.466 \times 10^{11}$$

For reverse bias the factor is $\exp(-27.027) = 1/(5.466 \times 10^{11}) = 1.829 \times 10^{-12}$.

Step 3: Minority holes at the edge, forward bias of 0.7 V

$$p_n(x_n) = p_{n0} \exp\left(\frac{V_a}{V_t}\right) = (2.25 \times 10^4)(5.466 \times 10^{11})$$

$$\boxed{p_n(x_n) = 1.23 \times 10^{16} \text{ cm}^{-3} \quad \text{(forward)}}$$

The hole density at the edge has risen by a factor of 5.5×10^{11}; the excess is $\delta p_n = 1.23 \times 10^{16}$ cm^{-3}.

Step 4: Minority holes at the edge, reverse bias of 0.7 V

$$p_n(x_n) = p_{n0} \exp\left(-\frac{0.7}{V_t}\right) = \frac{2.25 \times 10^4}{5.466 \times 10^{11}}$$

$$\boxed{p_n(x_n) = 4.1 \times 10^{-8} \text{ cm}^{-3} \approx 0 \quad \text{(reverse)}}$$

This is less than one hole in 10^7 cm^3: the edge is completely emptied of minority holes, $\delta p_n \approx -p_{n0}$.

Step 5: Majority electrons at the edge

From charge neutrality, $\delta n_n = \delta p_n$, so $n_n(x_n) = N_d + \delta p_n$.
Forward bias:

$$n_n(x_n) = 1 \times 10^{16} + 1.23 \times 10^{16}$$

$$\boxed{n_n(x_n) = 2.23 \times 10^{16} \text{ cm}^{-3} \quad \text{(forward)}}$$

Reverse bias:

$$n_n(x_n) = N_d - p_{n0} = 1 \times 10^{16} - 2.25 \times 10^4$$

$$\boxed{n_n(x_n) \approx 1 \times 10^{16} \text{ cm}^{-3} = N_d \quad \text{(reverse)}}$$

In reverse bias the majority electrons are unchanged; in forward bias they would be more than doubled.

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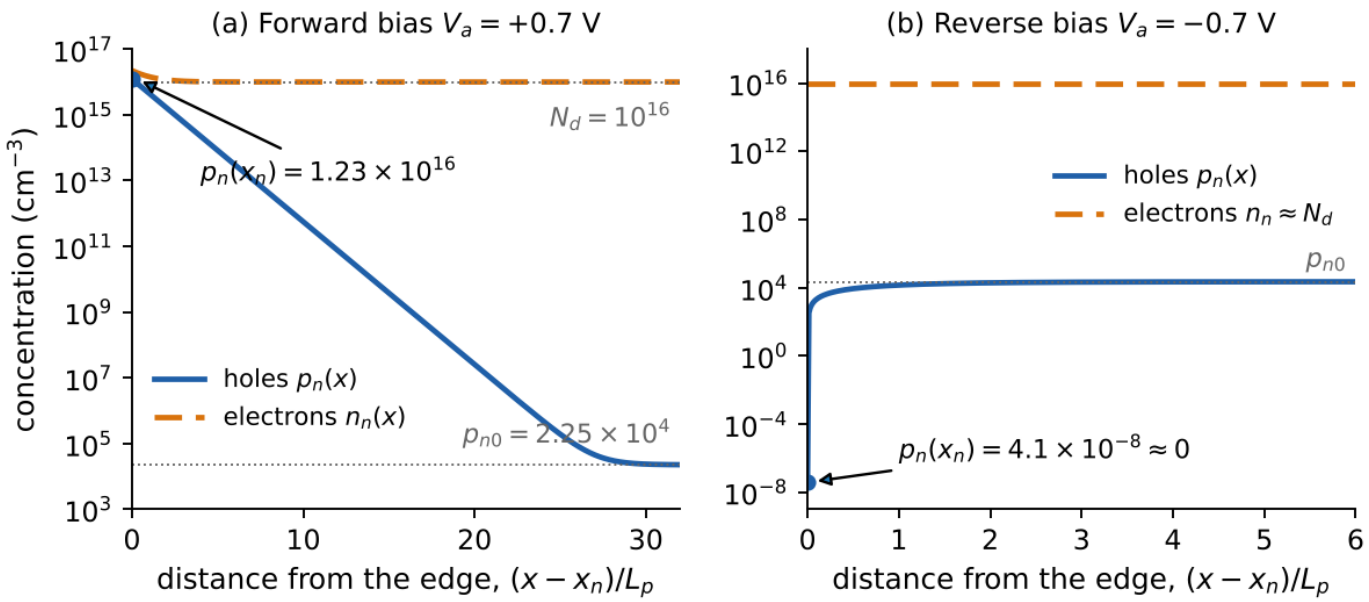


Figure 29: Hole and electron concentrations on the n side against distance from the space charge edge for forward and reverse bias of 0.7 V

Reading the figure. The vertical axis is logarithmic. In (a), forward bias, the solid blue line is the hole density: 1.23×10^{16} cm^{-3} at the edge (dot), falling as $\exp[-(x - x_n)/L_p]$ (a straight line on the log axis) and reaching $p_{n0} = 2.25 \times 10^4$ cm^{-3} after about $27L_p$. The dashed orange line is the electron density, 2.23×10^{16} at the edge, settling to N_d after a few L_p. In (b), reverse bias, the hole density is $4.1 \times 10^{-8} \approx 0$ at the edge and rises to p_{n0} within a few L_p; the electron density stays at N_d.

Step 6: Check of low injection

Forward bias gives $\delta p_n/N_d = 1.23$, which is *not* much smaller than 1. So **low injection is violated**: 0.7 V is too large a forward bias for this doping, and the forward numbers above are only the first estimate that the law of the junction gives. (Low injection holds up to about $V_a = V_t \ln(0.1N_d/p_{n0}) = 0.635$ V, where $p_n = 0.1N_d$.) The corrected high-injection values are $p_n \approx 7.2 \times 10^{15}$ cm^{-3} and $n_n \approx 1.7 \times 10^{16}$ cm^{-3} (Notes and Remarks). The reverse-bias results are exact.

Final Answer:

- $p_{n0} = 2.25 \times 10^4$ cm^{-3}, $n_{n0} = 1 \times 10^{16}$ cm^{-3} (using $N_d = 10^{16}$ cm^{-3})
- Forward 0.7 V: minority holes $p_n(x_n) = 1.23 \times 10^{16}$ cm^{-3}; majority electrons $n_n(x_n) = 2.23 \times 10^{16}$ cm^{-3} (low injection violated; high-injection estimate $p_n = 7.2 \times 10^{15}$, $n_n = 1.7 \times 10^{16}$ cm^{-3})
- Reverse 0.7 V: minority holes $p_n(x_n) = 4.1 \times 10^{-8} \approx 0$ cm^{-3}; majority electrons $n_n(x_n) \approx 1 \times 10^{16}$ cm^{-3}

3. Keyword Explanation

1. Law of the junction

$p_n(x_n) = p_{n0} \exp(eV_a/kT)$: the minority carrier density at the edge of the space charge region in terms of the applied voltage. It holds for low injection.

2. Low injection

The condition that the excess carrier density is much smaller than the majority carrier density, $\delta p_n \ll N_d$; then the majority density is unchanged.

3. Quasi-Fermi levels

E_{Fn} and E_{Fp}, separate Fermi levels for electrons and holes out of equilibrium:

$$n = n_i \exp\left[\frac{E_{Fn} - E_{Fi}}{kT}\right], \qquad p = n_i \exp\left[\frac{E_{Fi} - E_{Fp}}{kT}\right]$$

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4. Charge neutrality (quasi-neutrality)

In a neutral region the excess electrons and holes are equal, $\delta n = \delta p$, so no large field builds up.

4. Notes and Remarks

A more exact forward result. The general relation behind the law of the junction is that the quasi-Fermi levels are flat through the space charge region, so the product of the two densities at the edge is

$$p_n(x_n)\, n_n(x_n) = n_i^2 \exp\left(\frac{eV_a}{kT}\right)$$

With $n_n = N_d + p_n$ (charge neutrality, p_{n0} negligible) this is a quadratic, $p_n^2 + N_d p_n - n_i^2 e^{V_a/V_t} = 0$. With $n_i^2 e^{V_a/V_t} = 2.25 \times 10^{20} \times 5.466 \times 10^{11} = 1.23 \times 10^{32}$ cm^{-6}:

$$p_n = \frac{-N_d + \sqrt{N_d^2 + 4n_i^2 e^{V_a/V_t}}}{2} = 7.17 \times 10^{15} \text{ cm}^{-3}, \qquad n_n = N_d + p_n = 1.72 \times 10^{16} \text{ cm}^{-3}$$

This is the better answer at 0.7 V. The simple law of the junction overestimates p_n because it ignores the rise of n_n. In low injection ($n_n \approx N_d$) the quadratic gives back $p_n = p_{n0} e^{V_a/V_t}$.

Other remarks:

- **Why the answer matters:** at high injection the diode current grows as $\exp(eV_a/2kT)$ instead of $\exp(eV_a/kT)$, and the ideal diode equation fails. Real Si diodes at 0.7 V usually have heavier doping or a series resistance that takes part of the voltage.
- **The literal paper value.** With $N_d = 10^6$ cm^{-3} the region is intrinsic ($p_0 \approx n_i$), and the law of the junction would give $1.5 \times 10^{10} \times 5.466 \times 10^{11} = 8.2 \times 10^{21}$ cm^{-3}, far above $N_c = 2.8 \times 10^{19}$ cm^{-3}, which is impossible for a non-degenerate semiconductor. This is another sign that 10^{16} is meant.
- **Reverse saturation:** the empty edge in reverse bias is why the reverse current is tiny and constant: it is set only by the minority carriers that diffuse to the edge from within about L_p, not by the size of the reverse voltage.

5. Final Concise Answer

(a) Mobility $\mu = e\tau/m^*$ depends on how often carriers are scattered. Lattice (phonon) scattering grows with temperature, so $\mu_L \propto T^{-3/2}$ falls; ionized impurity scattering weakens as carriers become faster, so $\mu_I \propto T^{3/2}/N_I$ rises. By Matthiessen's rule $1/\mu = 1/\mu_L + 1/\mu_I$: impurity scattering dominates at low T (mobility rises), lattice scattering at high T (mobility falls), with a peak where $\mu_L = \mu_I$; heavier doping lowers the curve and moves the peak to higher T.

(b) For a MOS capacitor on p-type Si: accumulation ($V_G < 0$) bends the bands up and gathers holes at the surface; depletion ($0 < V_G < V_T$) bends them down and leaves negative acceptor ions over x_d; inversion ($V_G > V_T$) bends them so far that E_{Fi} at the surface falls below E_F, with $\phi_s = 2\phi_{fp}$ at threshold and the depletion width fixed at x_{dT}. The electrons at the surface form an n-type inversion layer that connects the n$^+$ source and drain: the channel. $V_{TN} = |Q'_{SD}(max)|/C_{ox} + V_{FB} + 2\phi_{fp}$.

(c) NPN modes: forward active ($V_{BE} > 0$, $V_{CB} > 0$; B–E forward, B–C reverse), saturation ($V_{BE} > 0$, $V_{CB} < 0$; both forward), cutoff ($V_{BE} < 0$, $V_{CB} > 0$; both reverse), inverse active ($V_{BE} < 0$, $V_{CB} < 0$; B–E reverse, B–C forward). Edge values: forward junctions give $p_{E0}e^{V_{BE}/V_t}$, $n_{B0}e^{V_{BE}/V_t}$, $n_{B0}e^{V_{BC}/V_t}$, $p_{C0}e^{V_{BC}/V_t}$; reverse junctions give 0. Between the edges n_B is nearly linear, and p_E, p_C return to p_{E0}, p_{C0}.

(d) With $N_d = 10^{16}$ cm^{-3} (the printed 10^6 is below n_i, a misprint): $p_{n0} = 2.25 \times 10^4$ cm^{-3}. Forward 0.7 V: $p_n(x_n) = 1.23 \times 10^{16}$ cm^{-3} and $n_n = 2.23 \times 10^{16}$ cm^{-3} (low injection violated; the high-injection relation gives 7.2×10^{15} and 1.7×10^{16} cm^{-3}). Reverse 0.7 V: $p_n(x_n) = 4.1\times10^{-8} \approx 0$ and $n_n \approx N_d = 10^{16}$ cm^{-3}.

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5. (a) Write the ambipolar equation for $g' = 0$ for n type semiconductor with constant electric field E_0. At time $t = 0$, a number of electron-hole pair is generated at $x = 0$. Sketch excess electron concentration as a function of distance at different time. Marks: 5

   (b) For NPN transistor, plot minority carrier concentration in base, emitter, and collector as a function of space. What are the minority carrier concentrations at the boundaries of base, emitter, and collector? Marks: 5

   (c) Explain the working principle of Tunnel diode. Draw its characteristics curve. Marks: 5

   (d) Current is flowing through an extrinsic semiconductor placed in a magnetic field as shown in Fig. 1. Marks: 10

   i. What is the Hall voltage?
   ii. What is type of carrier?
   iii. What is the carrier concentration?
   iv. What is the mobility?

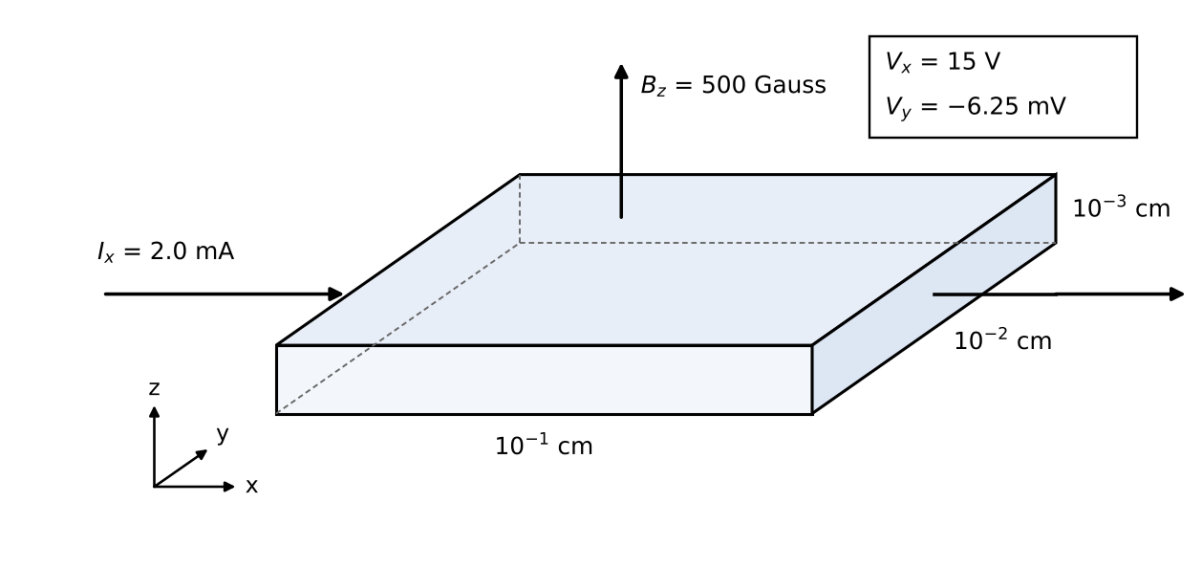


Figure 30: Fig. 1: A bar of an extrinsic semiconductor in a magnetic field (redrawn from the paper)

5.(a) Write the ambipolar equation for $g' = 0$ for n type semiconductor with constant electric field E_0. At time $t = 0$, a number of electron-hole pair is generated at $x = 0$. Sketch excess electron concentration as a function of distance at different time. Marks: 5

1. Theory

1. What are excess carriers, and what is the ambipolar equation? (Recap)

Light or a short pulse of radiation can break bonds and make extra electron-hole pairs. These extra carriers are **excess carriers**: δn extra electrons and δp extra holes above the equilibrium values n_0 and p_0. They are made in pairs, so $\delta n = \delta p$ at the start. Once the source is off, they **recombine** with an average lifetime, and they spread out by **diffusion** and move by **drift** in an electric field.

Generation, recombination, lifetime, the continuity equations and the full derivation of the **ambipolar transport equation** are built in 3(d). Its result (Neamen) is:

$$D' \frac{\partial^2(\delta n)}{\partial x^2} + \mu' E \frac{\partial(\delta n)}{\partial x} + g - R = \frac{\partial(\delta n)}{\partial t}$$

where

- D' is the ambipolar diffusion coefficient (cm^2/s) and μ' the ambipolar mobility ($\text{cm}^2/\text{V s}$)

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- E is the applied electric field (V/cm)
- g and R are the generation and recombination rates ($\text{cm}^{-3}\ \text{s}^{-1}$)

The word **ambipolar** means "both polarities": the equation describes the electrons and the holes *together*, as one cloud. Diffusion and drift (x-derivatives), generation and recombination, and change with time (t-derivative) all appear in one line.

2. Why must the electrons and holes move together?

The excess electrons and holes have opposite charges. Suppose they tried to move apart: the holes drifting with the field and the electrons against it (panel (a) of the figure). At once there would be a positive region and a negative region, and a strong **internal electric field** would appear between them. That field pulls the two clouds back together. Only a tiny separation is needed to make a field large enough to stop them, so in practice **the two clouds stay on top of each other**: $\delta n \approx \delta p$ everywhere (quasi-neutrality).

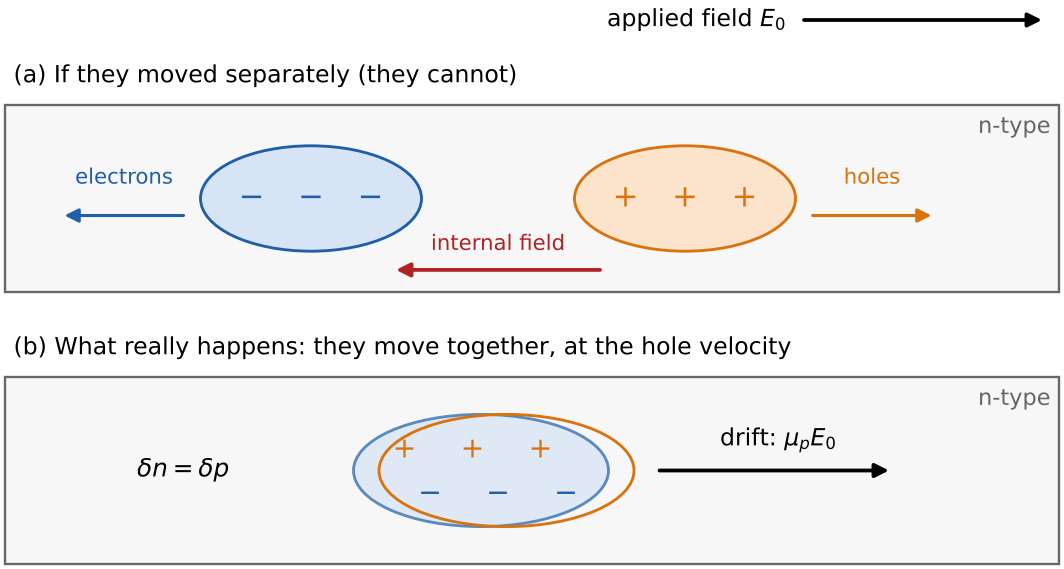


Figure 31: Excess electrons and holes in n-type material: if they moved apart an internal field would pull them back, so they drift together

Reading the figure. Panel (a) shows the impossible case: the electron cloud (solid blue outline, minus signs) drifting left and the hole cloud (orange outline, plus signs) drifting right with the applied field E_0 (black arrow at the top). The red arrow is the internal field this separation would create; it points from the holes to the electrons and pulls them together. Panel (b) is what happens: one neutral cloud with $\delta n = \delta p$, drifting to the right with the field at the hole speed $\mu_p E_0$.

3. Which carrier sets the motion? The n-type case

The ambipolar coefficients are (Neamen)

$$D' = \frac{D_n D_p (n+p)}{D_n n + D_p p}, \qquad \mu' = \frac{\mu_n \mu_p (p-n)}{\mu_n n + \mu_p p}$$

In an **extrinsic n-type** semiconductor under **low injection** ($\delta n \ll n_0$), $n \approx n_0$ is much larger than p. Putting $n \gg p$ into the two formulas:

$$D' \to \frac{D_n D_p\, n}{D_n n} = D_p, \qquad \mu' \to \frac{\mu_n \mu_p (-n)}{\mu_n n} = -\mu_p$$

The recombination rate becomes $R = \delta p / \tau_{p0}$, with τ_{p0} the **minority carrier (hole) lifetime**. So in n-type material the whole cloud diffuses, drifts and dies like the **minority carriers, the holes**. The reason is simple: there are so many majority electrons that a few extra ones hardly change anything, and the electrons simply follow the holes to keep the charge neutral.

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4. What does the pulse do? Solving the equation

At $t = 0$ a short flash makes N_0 electron-hole pairs (per cm^2 of cross-section) at $x = 0$; for $t > 0$ there is no generation, so $g' = 0$. The equation for n-type is

$$D_p \frac{\partial^2(\delta p)}{\partial x^2} - \mu_p E_0 \frac{\partial(\delta p)}{\partial x} - \frac{\delta p}{\tau_{p0}} = \frac{\partial(\delta p)}{\partial t}$$

Its solution for this impulse (Neamen) is

$$\delta n(x,t) = \delta p(x,t) = \frac{N_0}{\sqrt{4\pi D_p t}} \exp\left(-\frac{t}{\tau_{p0}}\right) \exp\left[-\frac{(x - \mu_p E_0 t)^2}{4 D_p t}\right]$$

Each factor has its own meaning:

- $\exp[-(x - \mu_p E_0 t)^2/4D_p t]$ is a **bell curve (Gaussian)** whose centre is at $x = \mu_p E_0 t$: the pulse **drifts** in the $+x$ direction (the direction of E_0) at the hole drift velocity $\mu_p E_0$.
- The width of the bell is $\sqrt{2D_p t}$: the pulse **spreads** by diffusion; the width grows as $\sqrt{t}$.
- $1/\sqrt{4\pi D_p t}$ lowers the peak as the pulse spreads, so the area stays the same.
- $\exp(-t/\tau_{p0})$ makes the whole pulse **decay** by recombination: the area under the curve, the number of pairs left, is $N_0 e^{-t/\tau_{p0}}$.

Typical numbers (used for the figure): n-type Si with $\mu_p = 480$ cm^2/V s, $D_p = \mu_p kT/e = 12.4$ cm^2/s, $\tau_{p0} = 2$ μs and $E_0 = 20$ V/cm. Then the drift velocity is $\mu_p E_0 = 9600$ cm/s $= 96$ μm/μs. At $t = 0.5$, 1 and 2 μs the centre is at 48, 96 and 192 μm and the width $\sqrt{2D_p t}$ is 35, 50 and 71 μm.
The pulse at three times is sketched in the Answer below, for $E_0 = 0$ and for $E_0 > 0$.

5. Where is this used?

This is the **Haynes-Shockley experiment**: a light flash makes the pulse, and a probe a distance d away records when it arrives. The arrival time gives the minority carrier mobility ($\mu_p = d/E_0 t_0$), the width gives D_p, and the fall of the area with time gives τ_{p0}.

2. Answer

Ambipolar equation. For an extrinsic n-type semiconductor at low injection ($n_0 \gg p_0$, $\delta p \ll n_0$) the ambipolar parameters reduce to those of the minority holes: $D' = D_p$, $\mu' = -\mu_p$, $\tau = \tau_{p0}$ (3(d)). With a constant field E_0 and no generation ($g' = 0$):

$$D_p \frac{\partial^2(\delta p)}{\partial x^2} - \mu_p E_0 \frac{\partial(\delta p)}{\partial x} - \frac{\delta p}{\tau_{p0}} = \frac{\partial(\delta p)}{\partial t}$$

where δp is the excess hole concentration, D_p the hole diffusion coefficient and τ_{p0} the minority carrier lifetime. By charge neutrality $\delta n = \delta p$, so the same equation holds for δn.
Solution for a pulse. N_0 pairs generated at $x = 0$, $t = 0$ give

$$\delta n(x,t) = \delta p(x,t) = \frac{N_0}{\sqrt{4\pi D_p t}} \exp\left(-\frac{t}{\tau_{p0}}\right) \exp\left[-\frac{(x - \mu_p E_0 t)^2}{4 D_p t}\right]$$

Behaviour with time ($t_1 < t_2 < t_3$):

- **Drift:** the peak moves in the direction of E_0 ($+x$) with the **hole** velocity $\mu_p E_0$; its position is $x = \mu_p E_0 t$.
- **Diffusion:** the pulse spreads; its width $\sqrt{2D_p t}$ grows as $\sqrt{t}$, and the peak falls.
- **Recombination:** the area under the pulse decays as $N_0 e^{-t/\tau_{p0}}$.

Sketch ($t_1 < t_2 < t_3$; typical n-Si numbers from the Theory, Section 4):

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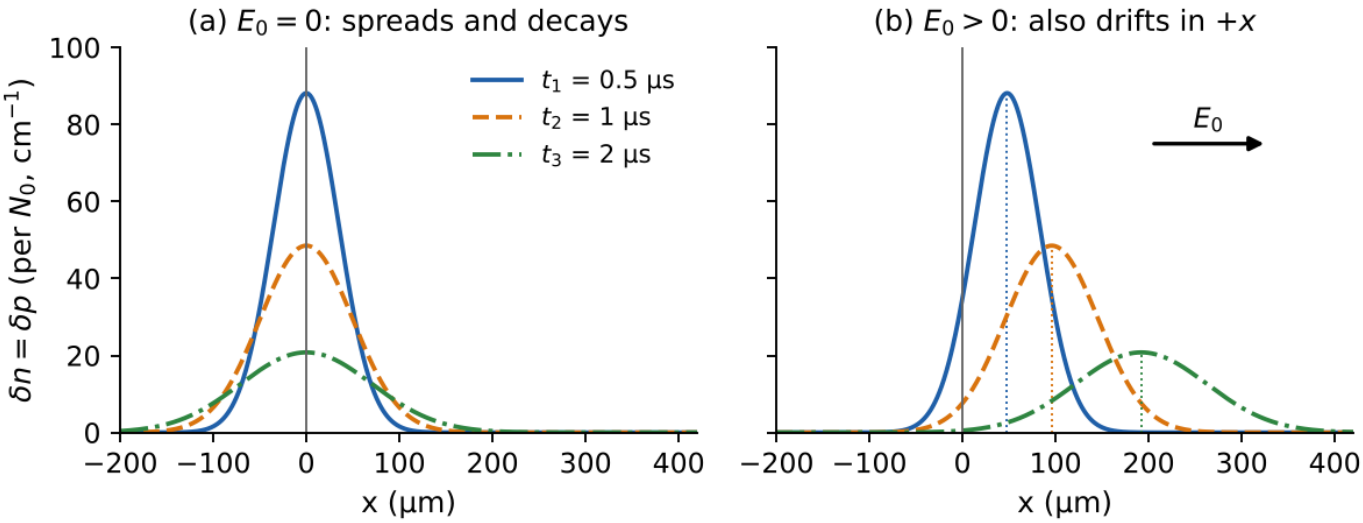


Figure 32: Excess carrier pulse at three times: with no field and with a field E0 in the +x direction

Reading the figure. The curves are $\delta n = \delta p$ per unit N_0, at $t_1 = 0.5$ μs (solid blue), $t_2 = 1$ μs (dashed orange) and $t_3 = 2$ μs (dash-dot green). In panel (a) there is no field: all three stay centred at $x = 0$; they get wider and lower. In panel (b) the field points in $+x$: the same shapes now also move to the right; the thin dotted lines mark the centres at $\mu_p E_0 t = 48$, 96 and 192 μm. The later pulses are wider and much lower, because of diffusion and recombination together.

Key point: the excess **electrons** move with the holes, in the direction of the field, although free electrons alone would drift against it. The internal field from any separation holds the two clouds together, and in n-type material the minority holes set the speed.

3. Keyword Explanation

1. Ambipolar transport

The joint drift and diffusion of excess electrons and holes as one neutral cloud, held together by the internal field between them.

2. Low injection

The excess concentration is much smaller than the majority carrier concentration ($\delta n \ll n_0$ in n-type), so the majority concentration hardly changes.

3. Minority carrier lifetime τ_{p0}

The average time an excess hole lives in n-type material before it recombines.

4. Haynes-Shockley experiment

The measurement of μ_p, D_p and τ_{p0} from the arrival time, width and size of a drifting carrier pulse.

4. Notes and Remarks

- **Check of the Einstein relation:** the measured D_p and μ_p obey $D_p/\mu_p = kT/e$ (2(b)).
- **p-type material** is the mirror case: $D' = D_n$, $\mu' = +\mu_n$, $\tau = \tau_{n0}$, and the pulse drifts *against* the field, with the minority electrons, at speed $\mu_n E_0$.
- **Neamen's form:** Neamen writes the solution without N_0 (it is proportional to it). The units of N_0 are cm^{-2}, so δp is in cm^{-3}.
- **Intrinsic material** ($n = p$) gives $\mu' = 0$: the pulse does not drift at all, it only spreads and decays.

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5.(b) For NPN transistor, plot minority carrier concentration in base, emitter, and collector as a function of space. What are the minority carrier concentrations at the boundaries of base, emitter, and collector? Marks: 5

1. Theory

1. What is an NPN transistor? (Core Idea)

An **NPN bipolar transistor** is three layers: a heavily doped n^+ **emitter**, a thin p-type **base**, and an n-type **collector**. It has two PN junctions: base-emitter (B-E) and base-collector (B-C). The structure, the four operating modes and the minority carrier pictures of each mode are built in 4(c). Here we need only the **forward-active mode**, the mode of an amplifier:

- B-E junction **forward biased** ($V_{BE} > 0$, about 0.6 to 0.7 V)
- B-C junction **reverse biased** ($V_{BC} < 0$, or $V_{CB} > 0$)

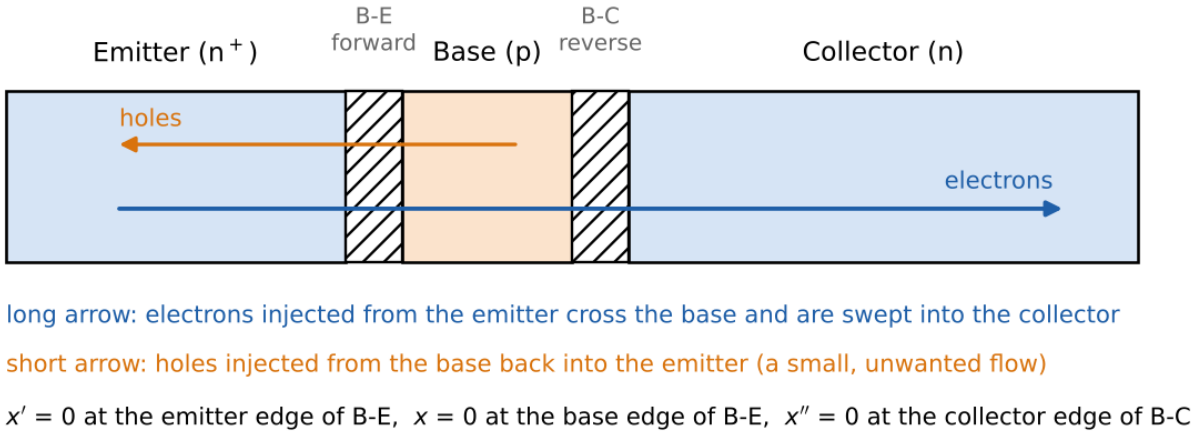


Figure 33: NPN transistor in forward-active mode: regions, depletion layers and carrier flows

Reading the figure. The three regions are drawn as blocks; the hatched strips are the two depletion layers. The long arrow is the main electron flow: electrons injected from the emitter into the base, crossing it, and swept into the collector. The short arrow is the hole flow from the base into the emitter. The bottom line gives the three distance coordinates used below.

2. Which carriers are "minority" in each region?

In n-type material electrons are majority and holes are minority; in p-type it is the other way round. So the minority carriers are:

- **Emitter (n^+):** holes, $p_E(x')$; equilibrium value $p_{E0} = n_i^2/N_E$
- **Base (p):** electrons, $n_B(x)$; equilibrium value $n_{B0} = n_i^2/N_B$
- **Collector (n):** holes, $p_C(x'')$; equilibrium value $p_{C0} = n_i^2/N_C$

Since the emitter is the most heavily doped, p_{E0} is the smallest of the three.

3. The law of the junction: the boundary values

At the edge of a biased depletion layer the minority carrier concentration is set by the voltage across the junction (Neamen; built in 4(d)):

$$n_p(\text{edge}) = n_{p0} \exp\left(\frac{eV_a}{kT}\right) = n_{p0} \exp\left(\frac{V_a}{V_t}\right)$$

where V_a is the junction voltage (> 0 forward) and $V_t = kT/e = 0.0259$ V at 300 K. A forward bias multiplies the minority concentration by a huge factor ($e^{0.65/0.0259} \approx 10^{11}$); a reverse bias of more than about 0.1 V makes the exponential nearly zero.

Applying it on both sides of both junctions:

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- **B-E (forward):** base side $n_B(0) = n_{B0}e^{V_{BE}/V_t}$; emitter side $p_E(0) = p_{E0}e^{V_{BE}/V_t}$
- **B-C (reverse):** base side $n_B(x_B) = n_{B0}e^{V_{BC}/V_t} \approx 0$; collector side $p_C(0) = p_{C0}e^{V_{BC}/V_t} \approx 0$

4. The shape between the boundaries

Inside each neutral region there is no field, so the excess minority carriers just **diffuse and recombine**. Their steady-state equation is the diffusion equation of 3(d), $D\,d^2(\delta n)/dx^2 = \delta n/\tau$, whose solutions are exponentials with the diffusion length $L = \sqrt{D\tau}$.

- **Base** (width x_B): Neamen's exact solution is a sum of sinh terms. The base is made much thinner than L_B ($x_B \ll L_B$), so few electrons recombine, and the curve becomes almost a **straight line** from $n_{B0}e^{V_{BE}/V_t}$ down to about 0:

$$n_B(x) \approx n_{B0} \exp\left(\frac{V_{BE}}{V_t}\right)\left(1 - \frac{x}{x_B}\right)$$

The steep, constant slope means a large diffusion current of electrons towards the collector. This is the collector current.

- **Emitter:** holes injected from the base fall from $p_{E0}e^{V_{BE}/V_t}$ at the junction back to p_{E0}, exponentially with length L_E (or linearly, for a short emitter with an ohmic contact).
- **Collector:** the reverse-biased junction pulls out the holes near it, so p_C is about zero at the edge and rises back to p_{C0} over a few L_C:

$$p_C(x'') = p_{C0}\left[1 - \exp\left(-\frac{x''}{L_C}\right)\right]$$

2. Answer

In forward-active mode the B-E junction is forward biased ($V_{BE} > 0$) and the B-C junction is reverse biased ($V_{BC} < 0$). The minority carriers are holes in the emitter, electrons in the base and holes in the collector. Their boundary values follow the law of the junction (4(d)); the modes themselves are in 4(c).

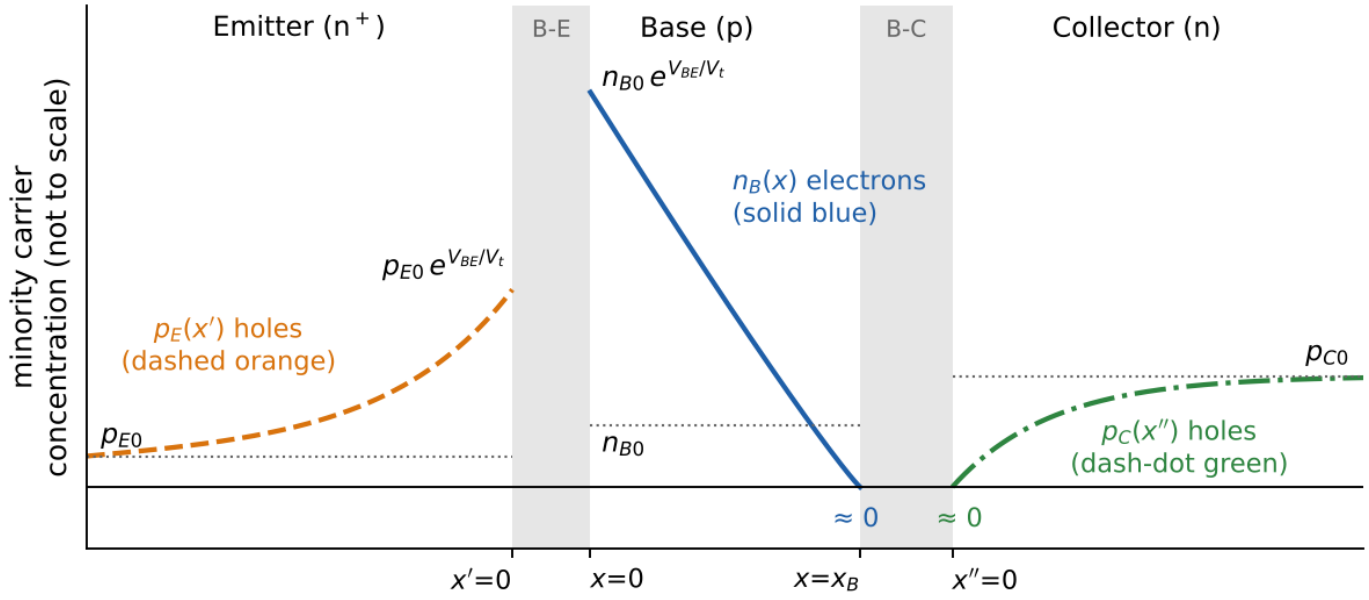


Figure 34: Minority carrier concentrations in an NPN transistor in forward-active mode (not to scale)

Reading the figure. The grey bands are the B-E and B-C depletion layers. The dashed orange curve is the hole concentration in the emitter: it rises from p_{E0} far from the junction to $p_{E0}e^{V_{BE}/V_t}$ at $x' = 0$.

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The solid blue line is the electron concentration in the base: it falls almost linearly from $n_{B0}e^{V_{BE}/V_t}$ at $x = 0$ to nearly zero at $x = x_B$; the dotted line n_{B0} is the equilibrium value. The dash-dot green curve is the hole concentration in the collector: nearly zero at $x'' = 0$, rising to p_{C0}.

Boundary values:

Region (minority carrier)	At the B-E junction	At the B-C junction	Far from the junction
Emitter (holes)	$p_E(0) = p_{E0}e^{V_{BE}/V_t}$	—	$p_E = p_{E0}$
Base (electrons)	$n_B(0) = n_{B0}e^{V_{BE}/V_t}$	$n_B(x_B) = n_{B0}e^{V_{BC}/V_t} \approx 0$	—
Collector (holes)	—	$p_C(0) = p_{C0}e^{V_{BC}/V_t} \approx 0$	$p_C = p_{C0}$

Table 5: Minority carrier concentrations at the boundaries in forward-active mode

Shapes:

- **Base:** nearly linear (narrow base, $x_B \ll L_B$), $n_B(x) \approx n_{B0}e^{V_{BE}/V_t}(1 - x/x_B)$. Its slope gives the electron diffusion current, i.e. the collector current.
- **Emitter:** excess holes decay from the junction towards the contact (length L_E).
- **Collector:** $p_C(x'') = p_{C0}[1 - e^{-x''/L_C}]$; holes are extracted by the reverse-biased junction.

Note: $p_{E0} = n_i^2/N_E$ is small because the emitter is heavily doped, so the hole injection into the emitter (a loss) is small. This gives a high emitter injection efficiency.

3. Keyword Explanation

1. Forward-active mode

B-E forward biased, B-C reverse biased; the transistor amplifies, $I_C = \beta I_B$.

2. Law of the junction

The minority carrier concentration at the edge of a depletion layer equals its equilibrium value times e^{V_a/V_t}.

3. Diffusion length

$L = \sqrt{D\tau}$, the average distance a minority carrier diffuses before it recombines.

4. Emitter injection efficiency

The fraction of the emitter current carried by electrons injected into the base; it is near 1 when $N_E \gg N_B$.

4. Notes and Remarks

- **Neamen's sign convention:** Neamen writes the B-C value as $n_B(x_B) = n_{B0}\exp(-V_{CB}/V_t)$ with $V_{CB} > 0$ for reverse bias. This is the same as $\exp(V_{BC}/V_t)$ above.
- **Excess form:** many books plot the *excess* concentrations: $\delta n_B(0) = n_{B0}[e^{V_{BE}/V_t} - 1]$, $\delta n_B(x_B) = -n_{B0}$, $\delta p_C(0) = -p_{C0}$.
- **Why not exactly linear?** A little recombination in the base makes $n_B(x)$ bend slightly downwards (the sinh solution). The difference in slope between $x = 0$ and $x = x_B$ is the base recombination current.
- **Scale:** the figure is not to scale. In a real device $n_{B0}e^{V_{BE}/V_t}$ is many orders of magnitude above n_{B0} and p_{C0}.

5.(c) Explain the working principle of Tunnel diode. Draw its characteristics curve. Marks: 5

1. Theory

1. What is tunnelling? (Core Idea)

In classical physics a ball that has less energy than the top of a hill cannot get to the other side. In quantum mechanics an electron is described by a **wave**. When the wave meets an energy barrier higher than its energy, it does not stop dead at the wall: it **decays exponentially** inside the barrier. If the barrier is thin, some of the wave is still left at the far side, and the electron can appear there. This is **quantum-mechanical tunnelling**.

The chance of getting through, the **transmission coefficient**, is roughly

$$T \approx \exp(-2\kappa a), \qquad \kappa = \frac{\sqrt{2m^*(V_0 - E)}}{\hbar}$$

where

- a is the barrier width (m)
- $V_0 - E$ is how far the barrier top is above the electron energy (J)
- m^* is the effective mass and $\hbar$ the reduced Planck constant

Because a is in the exponent, T is tiny for a thick barrier and becomes large only when the barrier is a few nanometres thin.

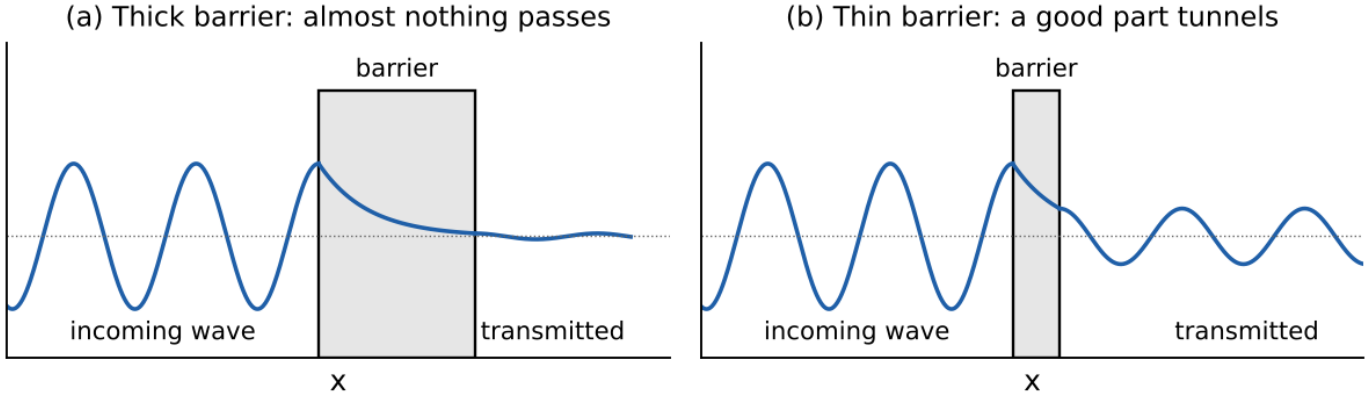


Figure 35: An electron wave meeting a thick and a thin energy barrier

Reading the figure. The solid blue curve is the electron wave. In panel (a) it decays inside the wide grey barrier to almost nothing, so the wave on the right is very small. In panel (b) the barrier is narrow; the wave has not decayed much when it reaches the far side, so a clear wave continues: the electron tunnels.

2. How do we make a junction so thin? Degenerate doping

In a PN junction the barrier for electrons is the **depletion region** (2(c)). Its width is

$$W = \sqrt{\frac{2\epsilon_s V_{bi}}{e}\left(\frac{N_a + N_d}{N_a N_d}\right)}$$

For $N_a = N_d = 10^{16}$ cm^{-3} in Si, $W \approx 0.42$ μm: far too thick to tunnel through. The tunnel diode (Esaki diode) dopes **both sides very heavily**, about 10^{19} to 10^{20} cm^{-3}. Then W falls to about 17 nm at 10^{19} and 6 nm at 10^{20} cm^{-3} (from the same formula with $V_{bi} \approx 1.1$ V): thin enough for tunnelling.

Such heavy doping also makes the semiconductor **degenerate**: the Fermi level moves **inside** the band (1(a) shows how E_F moves with doping):

- **n$^+$ side:** E_F is above E_c; the conduction band is filled with electrons up to E_F.
- **p$^+$ side:** E_F is below E_v; the valence band is empty of electrons (full of holes) down to E_F.

3. The condition for a tunnelling current

An electron tunnels horizontally in the band diagram (its energy does not change). A current flows only if, at the same energy,

- on one side there are **filled** states (electrons ready to go), and

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- on the other side there are **empty** allowed states (places to arrive).

If the opposite side has a filled state, or a forbidden gap, at that energy, no tunnelling current flows.

4. Band diagrams under bias (the working principle)

The equilibrium PN junction band diagram is built in 3(b). A forward bias V raises the n-side bands by eV relative to the p side.

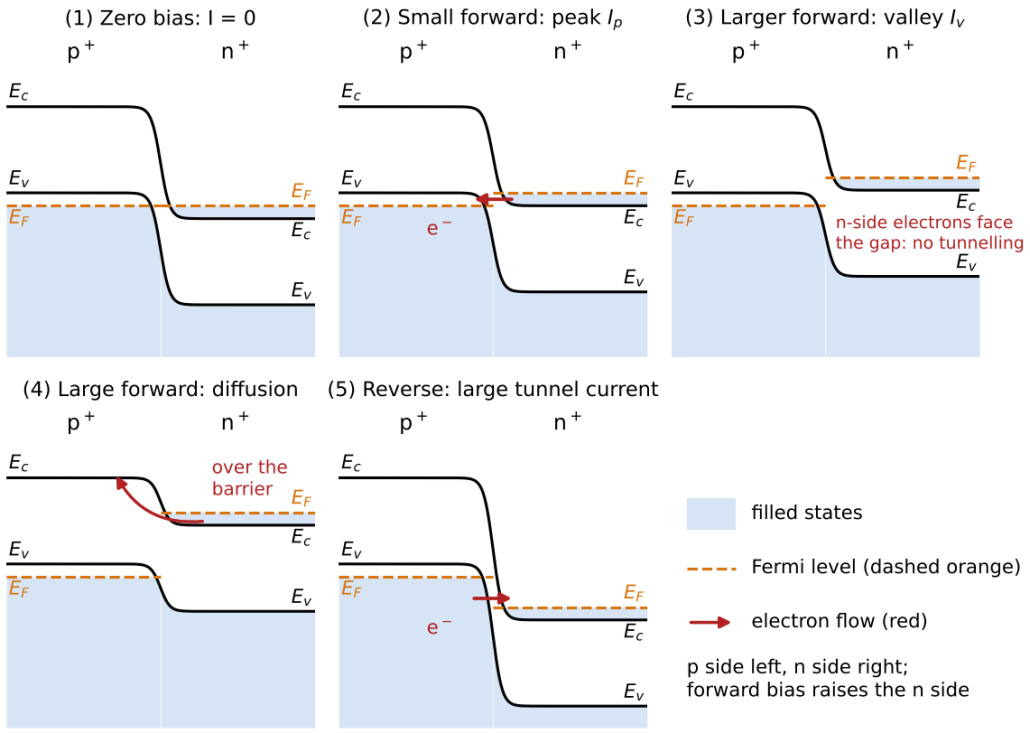


Figure 36: Band diagrams of a tunnel diode at zero bias, small forward bias, larger forward bias, large forward bias and reverse bias

Reading the figure. The p^+ side is on the left and the n^+ side on the right; solid black lines are E_c and E_v, the dashed orange lines the Fermi levels, and the shaded areas the filled states. Red arrows show where electrons go.

1. **Zero bias:** E_F is flat. Filled conduction-band states on the n side face *filled* valence-band states on the p side at the same energy. Tunnelling in both directions is equal: $I = 0$.
2. **Small forward bias:** the n side rises. Its filled conduction-band states now face the **empty** valence-band states of the p side (between E_F and E_v). Electrons tunnel from n to p. When the overlap is largest, the current is the **peak current** I_p at V_p.
3. **Larger forward bias:** the n-side E_c rises above the p-side E_v. The filled n-side states now face the p-side **forbidden gap**: there is nothing to tunnel into. The current *falls* as V rises, down to the **valley current** I_v at V_v. A falling current with rising voltage is a **negative differential resistance** ($dI/dV < 0$).
4. **Large forward bias:** the barrier is now low enough for the normal **diffusion current** of an ordinary diode: electrons go *over* the barrier, and the current rises exponentially again.
5. **Reverse bias:** the p side rises. Its filled valence-band states face the **empty** conduction-band states of the n side. Electrons tunnel from p to n, and a large reverse current flows even at a tiny reverse voltage. (The diode has no reverse blocking.)

The small current left at the valley, the **excess current**, comes from tunnelling through energy states inside the gap (impurities, band tails).

5. Why is it fast?

Tunnelling happens at almost the speed of the electron wave; no minority carriers have to diffuse or be stored. So a tunnel diode switches in picoseconds and works at microwave frequencies. Its drawback is its small voltage swing (a few hundred mV) and low power.

2. Answer

Principle. A tunnel diode is a PN junction with **both sides degenerately doped** ($N_a, N_d \approx 10^{19}$ to 10^{20} cm^{-3}). Two things follow:

- The Fermi level lies **inside the bands**: above E_c on the n$^+$ side and below E_v on the p$^+$ side.
- The depletion layer is **very thin**, about 5 to 17 nm, so electrons can pass through it by **quantum-mechanical tunnelling**, with no change of energy.

A tunnelling current flows when **filled** states on one side face **empty** allowed states on the other side at the same energy. The band diagrams (Theory, Section 4) show how bias changes this:

- **Zero bias:** filled states face filled states; net $I = 0$.
- **Small forward bias:** filled n-side conduction-band states face empty p-side valence-band states; electrons tunnel n $\rightarrow$ p. The overlap is largest at V_p: **peak current** I_p.
- **Larger forward bias:** the n-side conduction band faces the p-side forbidden gap; the overlap shrinks, so the current **falls** to the **valley current** I_v at V_v. This is the **negative differential resistance** region, $dI/dV < 0$.
- **Large forward bias:** the normal **diffusion current** takes over and rises exponentially, as in an ordinary diode.
- **Reverse bias:** filled p-side valence-band states face empty n-side conduction-band states; electrons tunnel p $\rightarrow$ n and a **large reverse current** flows.

Characteristic curve:

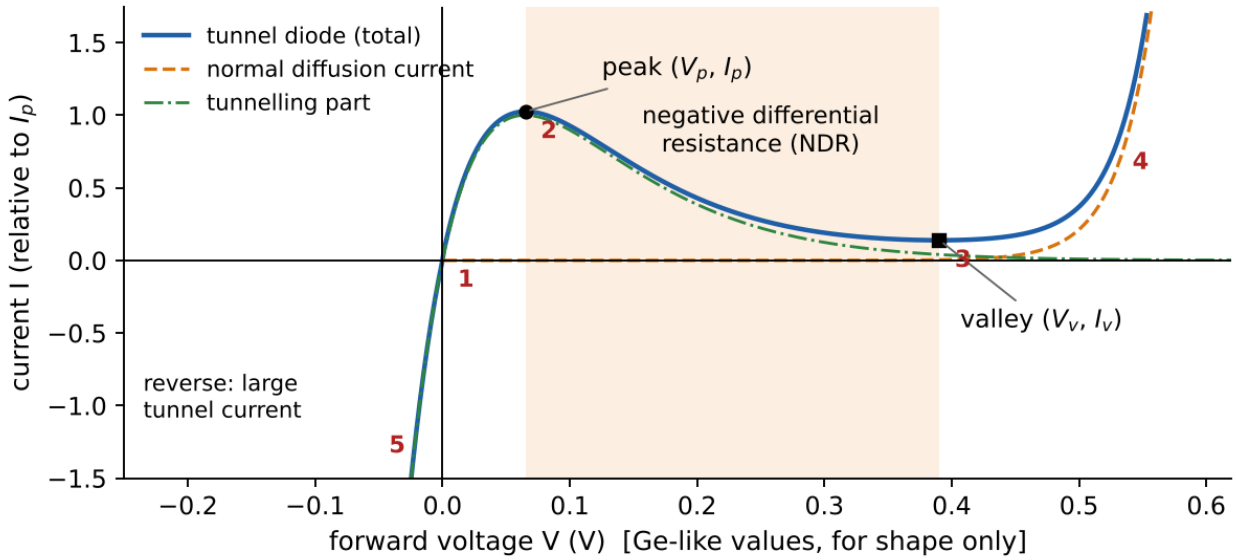


Figure 37: I-V characteristic of a tunnel diode, with the peak, the valley, the negative-resistance region and the diffusion current

Reading the figure. The solid blue curve is the tunnel diode current. It rises fast from 0 to the peak (filled circle, V_p, I_p), falls through the shaded negative-resistance region to the valley (filled square, V_v, I_v), then rises again along the dashed orange diffusion current of an ordinary diode. The dash-dot green curve is the tunnelling part alone. In reverse the current is large. The red numbers 1 to 5 match the five band diagrams. For a Ge tunnel diode, V_p is about 0.05 to 0.1 V, V_v about 0.3 V and $I_p/I_v \approx 8$.

Applications: microwave oscillators and amplifiers (the negative resistance cancels circuit losses), very fast switching and pulse circuits, and two-state memory circuits.

3. Keyword Explanation

1. Degenerate semiconductor

A semiconductor so heavily doped that the Fermi level lies inside the conduction band (n$^+$) or valence band (p$^+$).

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2. Quantum-mechanical tunnelling

Passage of a particle through an energy barrier higher than its energy; likely only for very thin barriers.

3. Negative differential resistance

A region of the I-V curve where the current falls as the voltage rises, $dI/dV < 0$.

4. Peak-to-valley current ratio

I_p/I_v, a figure of merit of a tunnel diode; larger is better for oscillators and switches.

4. Notes and Remarks

- **Inventor:** Leo Esaki (1958), who received the Nobel Prize for it; the device is also called the Esaki diode.
- **Backward diode:** a tunnel diode with slightly lower doping (Fermi levels just at the band edges) has almost no forward peak but a large reverse tunnelling current. It is used as a detector of small signals.
- **Zener breakdown** in a normal diode is the same tunnelling (case 5), but it needs a large reverse field because the junction is wider.
- **Depletion width numbers** were estimated with the non-degenerate V_{bi} formula, so they are approximate; the point is the order of magnitude (nm, not μm).

5.(d) Current is flowing through an extrinsic semiconductor placed in a magnetic field as shown in Fig. 1. i. What is the Hall voltage? ii. What is type of carrier? iii. What is the carrier concentration? iv. What is the mobility? Marks: 10

1. Theory

1. What is the Hall effect? (Core Idea)

A charge moving in a magnetic field feels a sideways force, the **Lorentz force**:

$$\vec{F} = q\,\vec{v} \times \vec{B}$$

where q is the charge (C), $\vec{v}$ its velocity (m/s) and $\vec{B}$ the magnetic flux density (T). The force is at right angles to both $\vec{v}$ and $\vec{B}$.

When a current flows along a semiconductor bar in a magnetic field, the moving carriers are pushed to one side of the bar. They pile up there, and the opposite side is left with the opposite charge. This charge creates a sideways electric field, the **Hall field** E_H, and a voltage across the width of the bar, the **Hall voltage** V_H. This is the **Hall effect** (Edwin Hall, 1879).

It is one of the most useful measurements in semiconductor work. From it we get three things: the **type** of the carriers (n or p), their **concentration**, and, with a resistance measurement, their **mobility**.

2. The geometry (Neamen's set-up)

The bar has length L along x, width W along y and thickness d along z. A current I_x flows in $+x$, and the magnetic field B_z points in $+z$. The voltage V_x is measured along the length, and the Hall voltage V_H across the width. Neamen defines $V_H = E_H W$, which is the potential of the $y = 0$ face minus that of the $y = W$ face.

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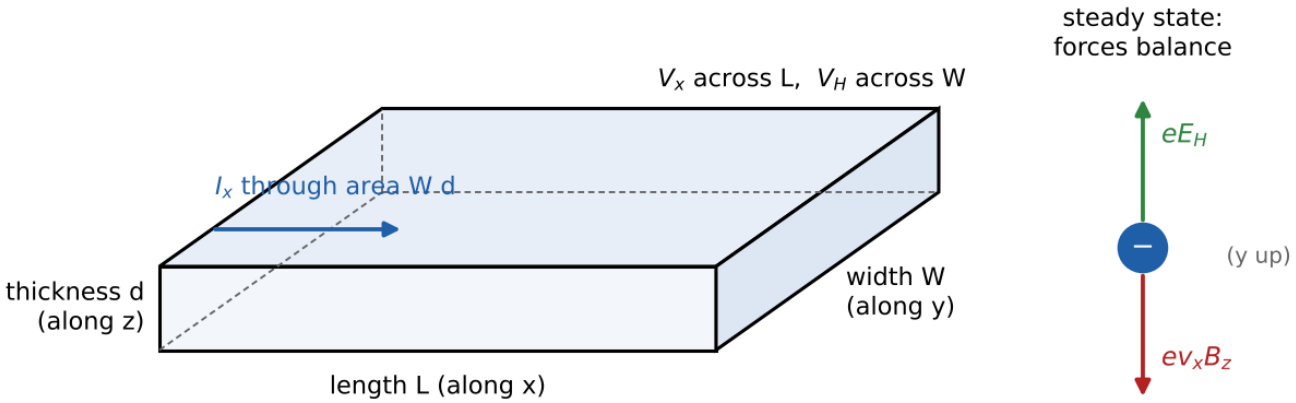


Figure 38: The bar for the Hall effect derivation: dimensions and the force balance on one electron

Reading the figure. The bar on the left has length L along x, width W along y (the depth of the drawing) and thickness d along z. The current I_x (blue arrow) crosses the area Wd. On the right is one electron in steady state, with y upwards: the magnetic force ev_xB_z (red, downwards, towards $y = 0$) is balanced by the force of the Hall field eE_H (green, upwards).

3. Which way are the carriers pushed? It is the same side for both types

Take $\vec{B} = B_z\hat{z}$ and the current in $+x$.

- **Holes** ($q = +e$) move with the current, $\vec{v} = +v\hat{x}$: $\vec{F} = e(v\hat{x}) \times (B_z\hat{z}) = evB_z(\hat{x} \times \hat{z}) = -evB_z\hat{y}$. They are pushed to the $y = 0$ face.
- **Electrons** ($q = -e$) move against the current, $\vec{v} = -v\hat{x}$: $\vec{F} = (-e)(-v\hat{x}) \times (B_z\hat{z}) = -evB_z\hat{y}$. They are *also* pushed to the $y = 0$ face.

Both kinds of carrier end up on the same face, but they carry opposite charges. So the **sign of the Hall voltage tells the carrier type**:

- **p-type:** positive charge at $y = 0$, E_H points in $+y$, $V_H > 0$.
- **n-type:** negative charge at $y = 0$, E_H points in $-y$, $V_H < 0$.

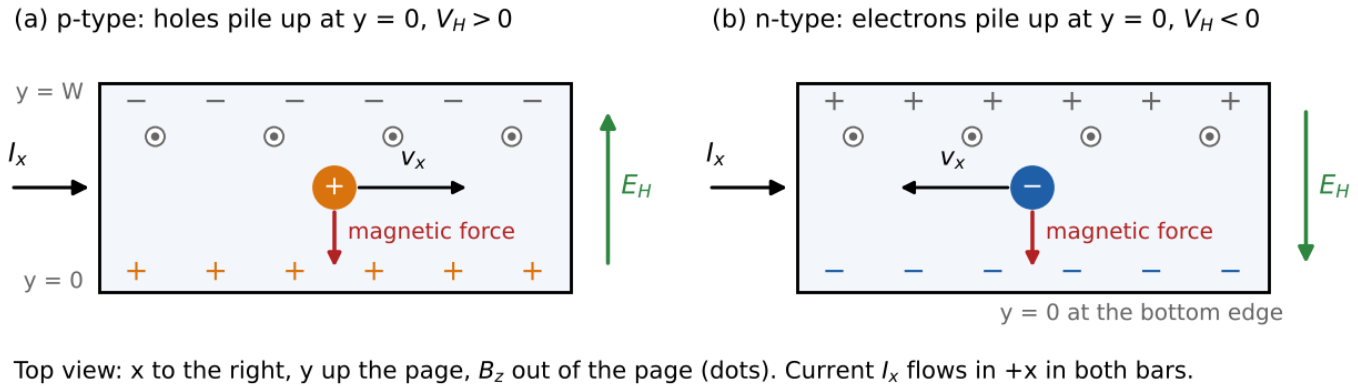


Figure 39: Top view of a p-type and an n-type bar in the Hall effect: carriers pile up on the y = 0 face

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Reading the figure. Both bars are seen from above, with B_z coming out of the page (dots) and I_x flowing to the right. In panel (a) a hole (filled circle, $+$) moves right; the red arrow, the magnetic force, pushes it down to $y = 0$, where plus signs pile up. The green arrow E_H then points up ($+y$), and $V_H > 0$. In panel (b) an electron (filled circle, $-$) moves left, is pushed down to $y = 0$ too, and minus signs pile up there. E_H points down ($-y$), and $V_H < 0$.

4. Deriving the Hall voltage

Charge piles up until the electric force on a carrier cancels the magnetic force. Then the carriers again flow straight along x (steady state):

$$q\left[\vec{E} + \vec{v} \times \vec{B}\right] = 0 \quad \Rightarrow \quad E_H = v_x B_z$$

This holds for both signs of q ($v_x > 0$ for holes, $v_x < 0$ for electrons). Since $V_H = E_H W$:

$$V_H = v_x W B_z$$

The drift velocity comes from the current. The current density is $J_x = I_x/(Wd)$, and $J_x = epv_x$ for holes or $J_x = -env_x$ for electrons. So

$$v_x = \frac{I_x}{e\,p\,W d} \text{ (p-type)}, \qquad v_x = -\frac{I_x}{e\,n\,W d} \text{ (n-type)}$$

Putting these in, the width W cancels:

$$V_H = \frac{I_x B_z}{e\,p\,d} \text{ (p-type)}, \qquad V_H = -\frac{I_x B_z}{e\,n\,d} \text{ (n-type)}$$

Turning these round gives the carrier concentration from measured quantities:

$$p = \frac{I_x B_z}{e\,d\,V_H}, \qquad n = -\frac{I_x B_z}{e\,d\,V_H}$$

A **thin** sample (small d) and a **low** concentration give a large, easily measured Hall voltage. That is why Hall samples are thin and why metals (huge n) show almost no Hall voltage.

5. Getting the mobility

The length voltage V_x drives the drift current. With $E_x = V_x/L$ and $J_x = en\mu_n E_x$ (3(c)):

$$\frac{I_x}{Wd} = en\mu_n \frac{V_x}{L} \quad \Rightarrow \quad \mu_n = \frac{I_x L}{e\,n\,V_x W d}$$

and the same with p and μ_p for p-type. Since $\sigma = en\mu_n$, this is just the conductivity of the bar divided by en.

6. Units

Use SI units throughout and convert at the end. 1 Gauss $= 10^{-4}$ T, 1 cm $= 10^{-2}$ m, 1 m$^{-3} = 10^{-6}$ cm^{-3}, 1 m^2/V s $= 10^4$ cm^2/V s. Mixing cm with tesla is the most common mistake here.

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2. Answer

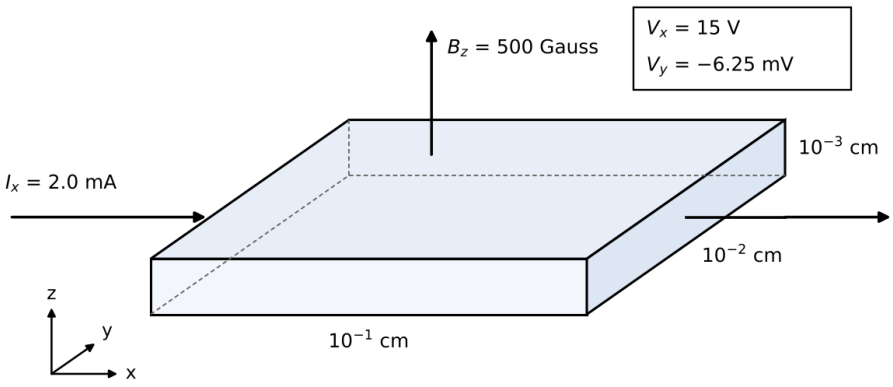


Figure 40: Repeated from the question: Fig. 1: A bar of an extrinsic semiconductor in a magnetic field (redrawn from the paper)

Given (Fig. 1, SI units):

- Current $I_x = 2.0$ mA $= 2.0 \times 10^{-3}$ A, in $+x$
- Magnetic field $B_z = 500$ G $= 500 \times 10^{-4}$ T $= 0.05$ T, in $+z$
- Voltage along the bar $V_x = 15$ V
- Voltage across the width $V_y = -6.25$ mV $= -6.25 \times 10^{-3}$ V
- Length $L = 10^{-1}$ cm $= 10^{-3}$ m (along x)
- Width $W = 10^{-2}$ cm $= 10^{-4}$ m (along y)
- Thickness $d = 10^{-3}$ cm $= 10^{-5}$ m (along z)
- $e = 1.602 \times 10^{-19}$ C

Relationships (Neamen; derived in the Theory):

$$V_H = E_H W, \qquad n = -\frac{I_x B_z}{e\, d\, V_H}, \qquad \mu_n = \frac{I_x L}{e\, n\, V_x W d}$$

where V_H is the Hall voltage across the width W, E_H the Hall field and d the thickness along $\vec{B}$.

Step 1: (i) Hall voltage

The Hall voltage is the voltage that appears across the width (y direction), at right angles to both the current (x) and the magnetic field (z). That is the measured V_y:

$$\boxed{V_H = V_y = -6.25 \text{ mV}}$$

The Hall field is $E_H = V_H/W = (-6.25 \times 10^{-3})/(10^{-4}) = -62.5$ V/m $= -0.625$ V/cm, pointing in the $-y$ direction.

Step 2: (ii) Type of carrier

With I_x in $+x$ and B_z in $+z$, the Lorentz force $q\vec{v} \times \vec{B}$ pushes both holes and electrons towards the $y = 0$ face. A **negative** V_H means that face has become negative, so the carriers that piled up there are negative:

$$\boxed{V_H < 0 \;\Rightarrow\; \text{n-type; the majority carriers are electrons}}$$

Step 3: (iii) Carrier concentration

$$n = -\frac{I_x B_z}{e\, d\, V_H} = -\frac{(2.0 \times 10^{-3})(0.05)}{(1.602 \times 10^{-19})(10^{-5})(-6.25 \times 10^{-3})} = \frac{1.0 \times 10^{-4}}{1.00125 \times 10^{-26}} = 9.99 \times 10^{21} \text{ m}^{-3}$$

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$$\boxed{n = 9.99 \times 10^{21}\ \text{m}^{-3} = 9.99 \times 10^{15} \approx 1.0 \times 10^{16}\ \text{cm}^{-3}}$$

This is a moderately doped n-type semiconductor: $N_d \approx 10^{16}$ cm^{-3} if all donors are ionized.

Step 4: (iv) Mobility

$$\mu_n = \frac{I_x L}{e\, n\, V_x W d} = \frac{(2.0 \times 10^{-3})(10^{-3})}{(1.602 \times 10^{-19})(9.9875 \times 10^{21})(15)(10^{-4})(10^{-5})} = \frac{2.0 \times 10^{-6}}{2.4 \times 10^{-5}}$$

$$\boxed{\mu_n = 0.0833\ \text{m}^2/\text{V s} = 833\ \text{cm}^2/\text{V s}}$$

Check: in steady state $|E_H| = |v_x|B_z$, so $|v_x| = 62.5/0.05 = 1250$ m/s; and $\mu_n E_x = 0.0833 \times (15/10^{-3}) = 1250$ m/s ✔. (Combining the two formulas, $\mu_n = L|V_H|/(V_x W B_z)$, which does not even need e or n.)

Final Answer:

- (i) Hall voltage $V_H = -6.25$ mV ($E_H = -0.625$ V/cm)
- (ii) V_H is negative, so the sample is **n-type**; the carriers are **electrons**
- (iii) $n = 9.99 \times 10^{15} \approx 1.0 \times 10^{16}$ cm^{-3}
- (iv) $\mu_n = 833$ cm^2/V s

3. Keyword Explanation

1. Lorentz force

The force $q\vec{v} \times \vec{B}$ on a charge moving in a magnetic field; it is at right angles to both the velocity and the field.

2. Hall voltage

The voltage across a current-carrying sample, at right angles to both the current and the magnetic field, $V_H = E_H W$.

3. Hall coefficient

$R_H = 1/(ep)$ for p-type and $R_H = -1/(en)$ for n-type; it relates the Hall field to the current density and field, $E_H = R_H J_x B_z$.

4. Hall mobility

The mobility obtained from the Hall measurement, $\mu = |R_H|\sigma$.

4. Notes and Remarks

- **Hall coefficient and resistivity of this sample:** $R_H = -1/(en) = -6.25 \times 10^{-4}$ m^3/C $= -625$ cm^3/C; resistivity $\rho = 1/(en\mu_n) = 0.75\ \Omega$ cm; bar resistance $V_x/I_x = 7500\ \Omega$.
- **Is 833 cm^2/V s reasonable?** The material is not named. For Si the lattice-limited value is 1350 cm^2/V s, and at $N_d \approx 10^{16}$ cm^{-3} ionized impurity scattering lowers it (3(c), 4(a)). 833 cm^2/V s is of the right order.
- **Sign conventions differ between books.** Neamen defines V_H so that it is negative for n-type with I_x in $+x$ and B_z in $+z$, and the paper's $V_y = -6.25$ mV matches that. Some books take V_H as the magnitude only and decide the type from which face is positive; the answer (n-type) is the same.
- **Why d and not W in the formula for n:** W cancels because a wider bar has both a larger Hall voltage per unit field ($V_H = E_H W$) and a smaller drift velocity for the same current. Only the thickness along $\vec{B}$ is left.
- **Real measurements** average over reversed current and field directions to remove offset voltages from misaligned contacts; the van der Pauw method does the same on a sample of any shape.
- **Hall factor:** a more careful theory multiplies R_H by a scattering factor r_H of about 1 to 2; Neamen takes $r_H = 1$, as done here.

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5. Final Concise Answer

(a) For n-type at low injection, $D' = D_p$, $\mu' = -\mu_p$, $\tau = \tau_{p0}$: $D_p\,\partial^2(\delta p)/\partial x^2 - \mu_p E_0\,\partial(\delta p)/\partial x - \delta p/\tau_{p0} = \partial(\delta p)/\partial t$, with $\delta n = \delta p$. A pulse of N_0 pairs at $x = 0$, $t = 0$ gives $\delta n = \delta p = [N_0/\sqrt{4\pi D_p t}]\, e^{-t/\tau_{p0}} \exp[-(x - \mu_p E_0 t)^2/4D_p t]$: a Gaussian that drifts in the direction of E_0 at the hole velocity $\mu_p E_0$, spreads (width $\sqrt{2D_p t}$) and decays ($e^{-t/\tau_{p0}}$). The excess electrons move with the holes; with $E_0 = 0$ the pulse only spreads and decays about $x = 0$.

(b) Forward active: in the base the electrons fall almost linearly from $n_{B0}e^{V_{BE}/V_t}$ at $x = 0$ to $n_{B0}e^{V_{BC}/V_t} \approx 0$ at $x = x_B$. In the emitter the holes fall from $p_{E0}e^{V_{BE}/V_t}$ at the B-E edge to p_{E0}. In the collector the holes rise from $p_{C0}e^{V_{BC}/V_t} \approx 0$ at the B-C edge to p_{C0}.

(c) Both sides are degenerately doped, so E_F lies inside the bands and the depletion layer is only about 5 to 17 nm thick. Electrons tunnel when filled states face empty states at the same energy. The current rises to a peak I_p at small forward bias, falls to a valley I_v (negative differential resistance) when the n-side conduction band faces the p-side gap, then rises as normal diffusion current. Reverse bias gives a large tunnelling current. It is used in microwave oscillators and fast switches.

(d) (i) $V_H = V_y = -6.25$ mV. (ii) $V_H < 0$, so the sample is n-type and the carriers are electrons. (iii) $n = -I_x B_z/(edV_H) = 9.99 \times 10^{15} \approx 1.0 \times 10^{16}$ cm^{-3}. (iv) $\mu_n = I_x L/(enV_x Wd) = 833$ cm^2/V s.

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6. (a) Draw the schematic of a double-gate MOS structure. What are the benefits of such structure over single-gate MOS? Marks: 5

(b) Briefly explain the terms complete ionization and freeze out. Can we achieve complete ionization in room temperature? Prove your answer. Marks: 5

(c) Explain production of inverse layer at oxide and semiconductor layer with proper sketches of energy band diagram. Marks: 5

(d) A silicon PN junction at $T = 300$ K with zero applied bias has doping concentration of $N_d = 5 \times 10^{16}$ cm^{-3} and $N_a = 5 \times 10^{15}$ cm^{-3}. Determine built-in potential, space charge width and electric field at metallurgical region ($\epsilon_t = 11.7$ [*sic*: should be $\epsilon_r = 11.7$]) Marks: 10

6.(a) Draw the schematic of a double-gate MOS structure. What are the benefits of such structure over single-gate MOS? Marks: 5

1. Theory

1. What does the gate of a MOSFET do? (Core Idea)

A MOSFET is a switch. A thin **channel** of electrons forms under the gate and joins the **source** to the **drain**. The gate is a metal (or polysilicon) plate. It sits on a thin insulating oxide, on top of the silicon body. A positive gate voltage pulls electrons to the silicon surface and makes the channel (the band picture is built in 4(b) and used again in 6(c)).

So the gate is the **control** terminal. In a good switch, only the gate decides whether current flows. The drain voltage should not be able to turn the device on.

2. What goes wrong when the channel is short?

In a long-channel MOSFET the source and drain are far apart. The gate controls almost all of the silicon under it. When the channel length L is made very short (tens of nanometres), the drain comes very close to the source. Now the electric field lines from the drain no longer end on the drain's own depletion edge. They spread through the body under the gate and reach the source end of the channel. The gate and the drain then **share** the control of the channel. This gives the **short-channel effects**:

- **Threshold voltage roll-off:** part of the depletion charge under the gate is now held by the source and drain fields, so less gate voltage is needed. V_T falls as L gets shorter.
- **DIBL (drain-induced barrier lowering):** there is a small energy barrier for electrons at the source end of the channel. The drain field lowers this barrier. So a higher drain voltage lets more current flow even with the gate off.
- **Poor subthreshold slope:** below threshold, the drain current changes exponentially with the gate voltage. The **subthreshold swing** S is the gate voltage needed for a ten times change of current. The ideal value at 300 K is

$$S = \frac{kT}{e} \ln 10 = 0.0259 \times 2.303 = 59.6 \text{ mV/decade}$$

When the gate controls the surface only weakly, S is larger (80 to 100 mV/dec or more), so the device does not switch off sharply.

- **Higher leakage (off current):** with a lower barrier and a weaker slope, current flows even at $V_{GS} = 0$. Deep in the body, far from the gate, there is a leakage path that the gate hardly controls (punch-through).

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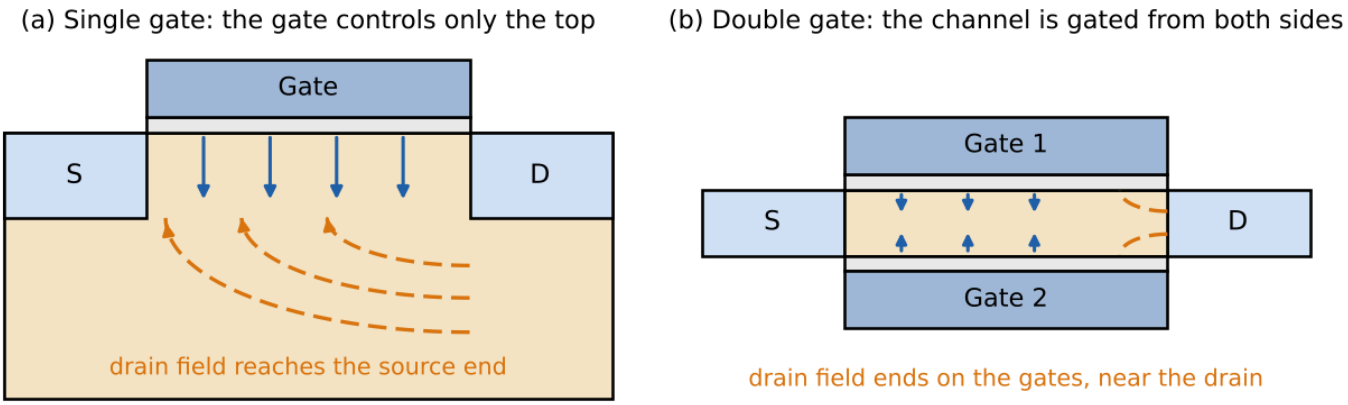


Figure 41: Single-gate and double-gate transistors: the gate field and the drain field in each

Reading the figure. The solid blue arrows are field lines from the gate; the dashed orange curves are field lines from the drain. In panel (a), a single gate sits on a thick body. The drain lines run deep under the channel and end near the source end: the drain is controlling the channel. In panel (b), the body is a thin layer with a gate on each side. The drain lines end on the two gates within a short distance of the drain, so the middle of the channel is controlled by the gates alone.

3. The double-gate idea

Old devices fought short-channel effects with heavy body doping and thinner oxides. Both have limits: heavy doping lowers mobility and gives random dopant changes from device to device; very thin oxide leaks by tunnelling.

The **double-gate MOSFET** fixes the geometry instead:

- The silicon body is made **very thin** (t_{Si} of about $L/2$ to $L/3$, a few nanometres to ~20 nm).
- A gate is placed on **both sides** of the body, each on its own thin oxide. The two gates are joined and driven together.
- The source and drain are heavily doped n^+ regions at the two ends of the thin body.

Because every point of the thin body is close to a gate, the gates hold almost all the charge in the body. The drain field cannot reach far into it. The distance over which the drain still matters is set by a **natural length** that scales with t_{Si} and the oxide thickness, so a thinner body allows a shorter gate.

4. Practical forms: planar double gate and FinFET

A planar double gate (one gate above the body, one below) is hard to make, because the bottom gate must be buried and aligned under the top one. The **FinFET** solves this. The silicon body is a thin vertical wall, the **fin**, standing on the wafer. The gate is wrapped over the fin like an upside-down U. It covers the two side walls (a double gate) and usually the top as well (a tri-gate). The fin width plays the role of t_{Si}. Current flows along the fin from source to drain. FinFETs have been used in commercial chips since the 22 nm node (2011). The three structures are drawn side by side in the Answer.

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5. The effect on switching

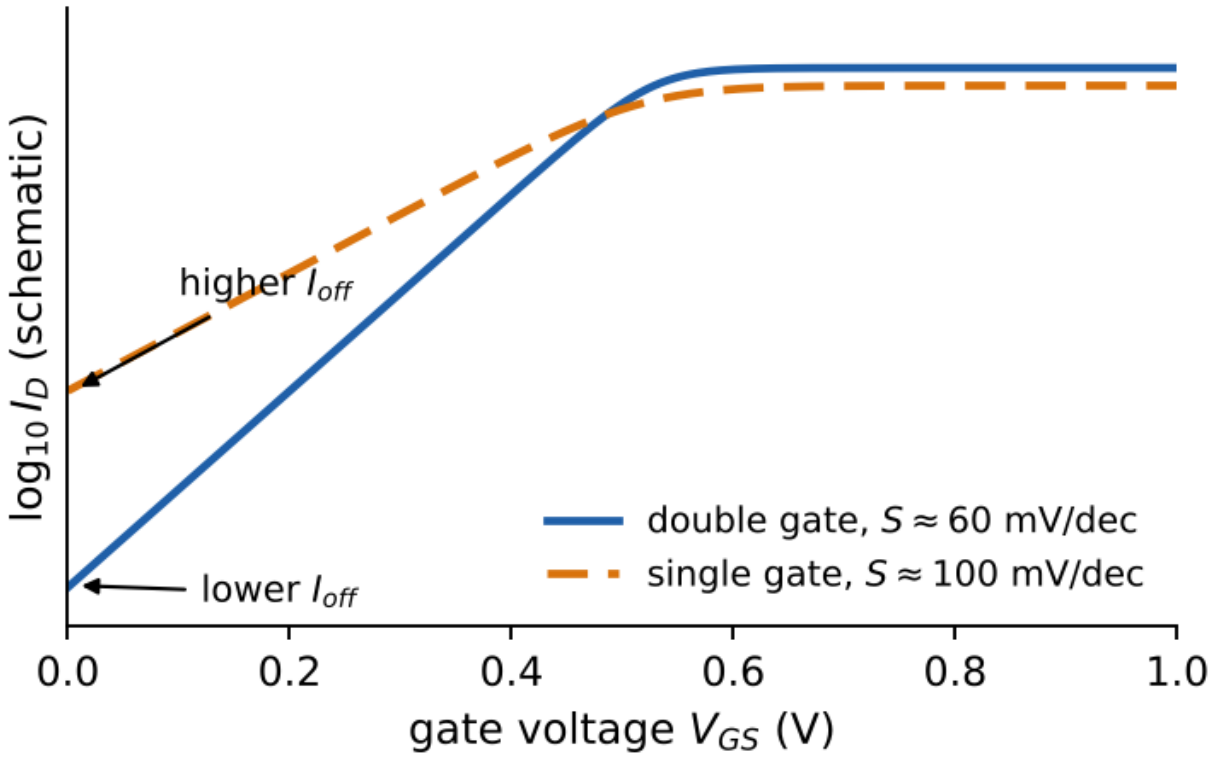


Figure 42: Subthreshold current of a single-gate and a double-gate transistor (schematic)

Reading the figure. The vertical axis is the logarithm of the drain current, so the straight part of each curve is the exponential subthreshold region and its slope is $1/S$. The solid blue curve (double gate) is steeper, close to the ideal 60 mV/dec, and so its current at $V_{GS} = 0$, the off current I_{off}, is several decades lower. The dashed orange curve (short-channel single gate) has a slope of about 100 mV/dec and a high off current. The double gate also gives a slightly higher on current (two channels).

2. Answer

Schematic. A double-gate MOSFET has a thin silicon body (undoped or lightly doped) between two thin gate oxides, with a gate on each side. Heavily doped n^+ source and drain sit at the two ends of the body. Both gates are tied together, so an inversion channel forms at both surfaces of the body. In practice it is built as a **FinFET**: the body is a thin vertical fin and the gate wraps over its two sides (and top).

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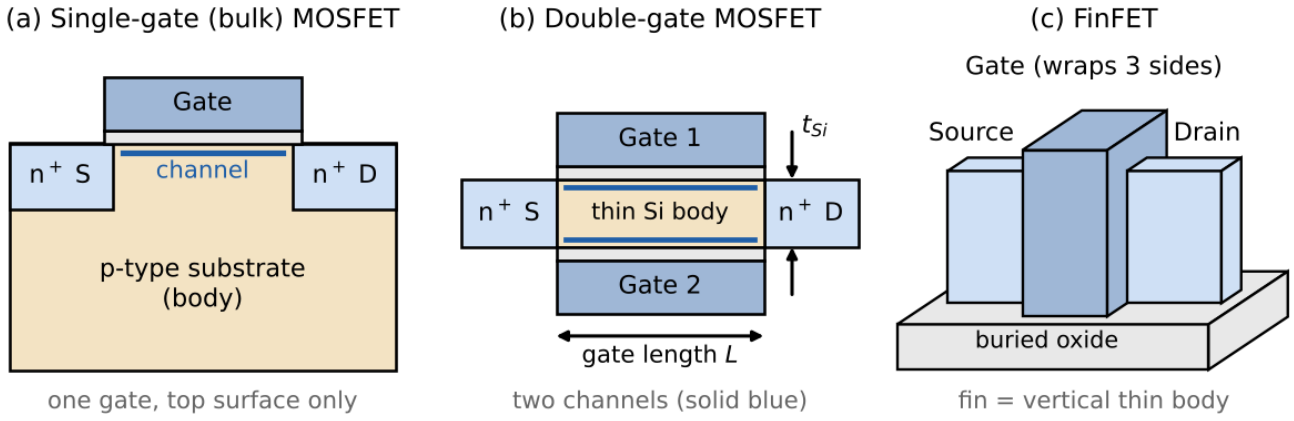


Figure 43: Schematic of a single-gate MOSFET, a planar double-gate MOSFET (two gates, thin body, two channels) and a FinFET

Reading the figure. Panel (a) is the usual bulk MOSFET: one gate on top, a thick p-type body, and one channel at the top surface (solid blue line). Panel (b) is the planar double gate: a thin silicon body between two oxides (light grey strips) and two gates; n^+ source and drain at the ends; the body thickness t_{Si} and the gate length L are marked; there are two channels (solid blue lines), one under each gate. Panel (c) is a FinFET: the fin (the lighter blocks) stands on a buried oxide; the gate (the darker block) wraps over the fin, so the part of the fin inside the gate is controlled from both sides and the top.

Benefits over a single-gate MOS:

1. **Better electrostatic control of the channel.** Every part of the thin body is close to a gate. The gates, not the drain, set the potential in the channel (field-line figure in the Theory).
2. **Reduced short-channel effects.** The drain field ends on the gates near the drain, so **DIBL** and **threshold voltage roll-off** are much smaller. The gate length can be scaled down further.
3. **Steeper subthreshold slope.** S comes close to the ideal $(kT/e)\ln 10 = 59.6$ mV/decade at 300 K, so the device turns off sharply.
4. **Lower leakage.** There is no deep body path far from the gate, so the off current and the standby power are lower.
5. **Higher drive current.** There are two channels (two or three surfaces in a FinFET) in the same footprint, so the on current and the transconductance are higher.
6. **No heavy body doping needed.** The body can be undoped. This gives higher mobility (less impurity scattering) and removes **random dopant fluctuation**, so V_T varies less from device to device.
7. **Scalability.** Together these let the transistor be scaled below about 22 nm, where planar bulk MOSFETs fail. This is why FinFETs are used in modern CMOS.

3. Keyword Explanation

1. Short-channel effects

Changes in MOSFET behaviour (lower and drain-dependent V_T, poor turn-off) that appear when the channel length is comparable to the depletion widths of the source and drain.

2. DIBL

Drain-induced barrier lowering: the drain voltage lowers the source-to-channel energy barrier, so V_T falls as V_{DS} rises.

3. Subthreshold swing

$S = dV_{GS}/d(\log_{10} I_D)$ below threshold, in mV/decade. Ideal value 59.6 mV/dec at 300 K.

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4. FinFET

A multi-gate MOSFET whose body is a thin vertical fin; the gate covers two or three sides of the fin.

4. Notes and Remarks

- **Front and back gate:** some double-gate devices drive the two gates separately (independent gates), so one gate can tune the threshold set by the other. The answer above assumes the usual tied gates.
- **Volume inversion:** in a very thin body (below ~10 nm) the two channels merge and electrons flow through the whole body, not just at the two surfaces.
- **Beyond the FinFET:** the same idea taken further gives the gate-all-around (nanosheet) transistor, where the gate surrounds the channel on all four sides.
- Neamen treats this topic only briefly (in the chapter on MOSFET non-ideal effects); the points above follow the usual device-text treatment (e.g. Taur and Ning).

6.(b) Briefly explain the terms complete ionization and freeze out. Can we achieve complete ionization in room temperature? Prove your answer. Marks: 5

1. Theory

1. Donors and their energy level (Core Idea)

A donor atom (for example phosphorus, P, in silicon) has five valence electrons. Four make bonds with the neighbouring Si atoms. The fifth is held only weakly. Its energy level E_d lies just below the conduction band edge E_c. For P in Si the **ionization energy** is

$$E_c - E_d = 0.045 \text{ eV}$$

which is less than $2kT$ at room temperature ($kT = 0.0259$ eV). A little thermal energy lifts the fifth electron into the conduction band. The donor is then a fixed positive ion, N_d^+, and the electron is free. The full statistics of a donor level are built in 1(d); here is what we need.

2. How many donors keep their electron?

The density of electrons still sitting on donor atoms is (Neamen)

$$n_d = \frac{N_d}{1 + \frac{1}{2}\exp\left(\frac{E_d - E_F}{kT}\right)}$$

where

- n_d is the density of electrons in donor states (un-ionized donors), cm^{-3}
- N_d is the donor concentration, cm^{-3}
- the factor $\frac{1}{2}$ is the degeneracy factor: a donor can take an electron of either spin, but only one

The electrons in the conduction band are $n_0 = N_c \exp[-(E_c - E_F)/kT]$ (1(a)). Dividing, and assuming $E_d - E_F \gg kT$, gives the **fraction of electrons still in donor states**:

$$\frac{n_d}{n_d + n_0} = \frac{1}{1 + \frac{N_c}{2N_d}\exp\left[-\frac{E_c - E_d}{kT}\right]}$$

where N_c is the effective density of states in the conduction band (2.8×10^{19} cm^{-3} in Si at 300 K). This is Neamen's equation for the fraction of un-ionized donors. If this fraction is small, nearly every donor has given its electron away.

3. Complete ionization

Complete ionization means that every donor has given up its electron to the conduction band (and every acceptor has accepted one from the valence band). Then

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$$n_0 \approx N_d \quad \text{(n-type)}, \qquad p_0 \approx N_a \quad \text{(p-type)}$$

This is the assumption behind almost every device calculation, for example the PN junction of 6(d).

4. Freeze-out

As the temperature falls, kT becomes small compared with $E_c - E_d$. Electrons no longer have the energy to stay in the conduction band, and they fall back into the donor states: the carriers "freeze out" onto the donor atoms. At $T = 0$ K *every* donor holds its electron, $n_0 = 0$, and the material is an insulator. This is **complete freeze-out**. Between 0 K and about 100 K only part of the donors are ionized ($n_0 < N_d$): this is the **partial ionization** or freeze-out range.

The same happens to acceptors: at $T = 0$ K no valence electron has moved up into an acceptor level, so there are no holes.

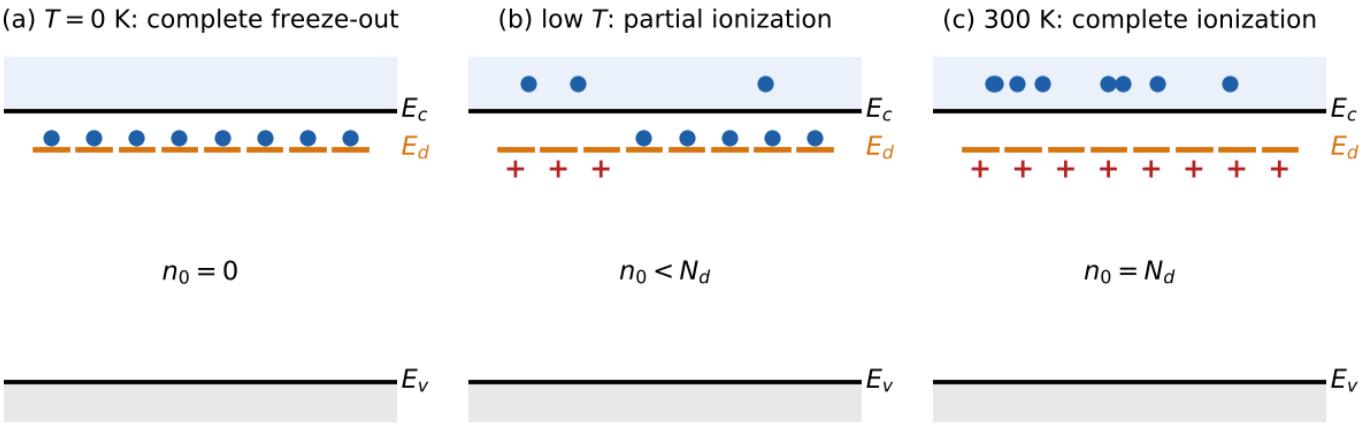


Figure 44: Donor states at 0 K, at a low temperature and at 300 K

Reading the figure. The short orange bars are the donor levels E_d, 0.045 eV below E_c (the gap is drawn larger for clarity). A filled blue dot is an electron; a red plus sign is an ionized donor. In panel (a), at 0 K, all electrons sit on the donors: complete freeze-out, $n_0 = 0$. In panel (b), at a low temperature, some donors are ionized and their electrons are in the conduction band. In panel (c), at 300 K, all donors are ionized and all their electrons are free: complete ionization, $n_0 = N_d$.

5. The full picture: n_0 against temperature

Over a wide temperature range three regions appear:

- **Freeze-out (low T):** $n_0 < N_d$; it rises steeply with T as donors ionize.
- **Extrinsic (middle):** all donors are ionized and $n_i \ll N_d$, so $n_0 = N_d$. This flat region contains room temperature.
- **Intrinsic (high T):** n_i grows as $\exp(-E_g/2kT)$ and becomes larger than N_d; the electrons made by breaking bonds swamp the donor electrons, and $n_0 \approx n_i$.

The curve below was computed from charge neutrality $n_0 = N_d^+ + p_0$ with $N_d^+ = N_d - n_d$, $p_0 = n_i^2/n_0$, $N_c \propto T^{3/2}$, and $n_i(T) = 1.5 \times 10^{10}(T/300)^{3/2} \exp\left[-\frac{E_g}{2k}(\frac{1}{T} - \frac{1}{300})\right]$ with $E_g = 1.12$ eV held constant.

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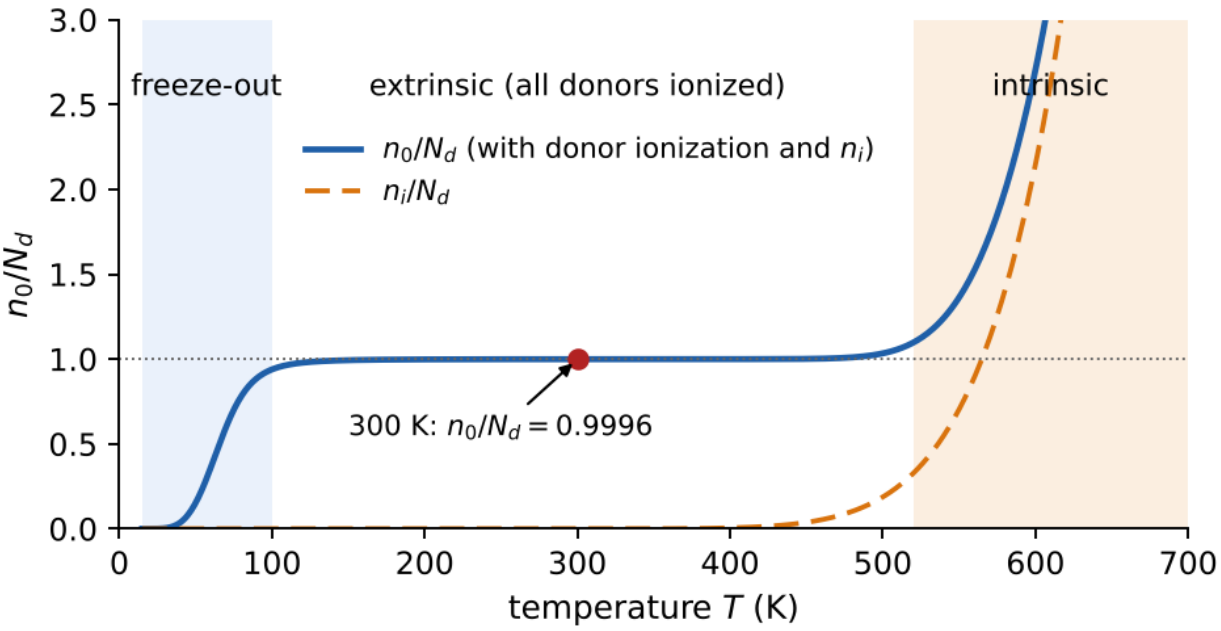


Figure 45: Electron concentration over donor concentration against temperature for phosphorus-doped silicon with $N_d = 10^{15}$ cm^{-3}

Reading the figure. The solid blue curve is n_0/N_d for $N_d = 10^{15}$ cm^{-3}. It is near zero below 30 K (freeze-out), reaches 0.15 at 50 K, 0.94 at 100 K and 0.99 at about 138 K. From there to about 450 K it is flat at 1.00: the extrinsic region, where all donors are ionized. At 300 K (red dot) $n_0/N_d = 0.9996$. The dashed orange curve is n_i/N_d; it passes 1 at about 565 K, and above about 520 K the blue curve rises steeply: the intrinsic region.

2. Answer

Complete ionization: the state in which every donor atom has given its extra electron to the conduction band (every acceptor has taken an electron from the valence band). Then $n_0 = N_d$ (or $p_0 = N_a$) and no electrons remain in the donor states.

Freeze-out: at low temperature the thermal energy kT is too small to keep the electrons in the conduction band, so they drop back into the donor states. The donors become neutral and $n_0 < N_d$. At $T = 0$ K all donors hold their electrons: $n_0 = 0$ (complete freeze-out); the semiconductor does not conduct.

Can we have complete ionization at room temperature? Yes, for normal (non-degenerate) doping.

Proof. First derive the fraction of electrons that stay in the donor states, then put in the numbers.

Step 1: Electrons in the donor states

A donor state can hold one electron of either spin, but once it holds one it cannot take a second; this halves its chance of being empty compared with an ordinary state. Its occupation is therefore a Fermi–Dirac function with a factor $\frac{1}{2}$ (Neamen, Eq. 4.50):

$$n_d = \frac{N_d}{1 + \frac{1}{2}\exp\left[\dfrac{E_d - E_F}{kT}\right]}$$

Step 2: Simplify for a non-degenerate sample

Where ionization is nearly complete, $E_d - E_F \gg kT$, so the exponential is much larger than 1, and

$$n_d \approx \frac{N_d}{\frac{1}{2}\exp[(E_d - E_F)/kT]} = 2N_d \exp\left[\frac{-(E_d - E_F)}{kT}\right]$$

(Neamen, Eq. 4.53).

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Step 3: Electrons in the conduction band

From the Boltzmann approximation (Neamen, Eq. 4.11):

$$n_0 = N_c \exp\left[\frac{-(E_c - E_F)}{kT}\right]$$

Step 4: Form the fraction and remove E_F

$$\frac{n_d}{n_d + n_0} = \frac{1}{1 + n_0/n_d}$$

and

$$\frac{n_0}{n_d} = \frac{N_c \exp[-(E_c - E_F)/kT]}{2N_d \exp[-(E_d - E_F)/kT]} = \frac{N_c}{2N_d} \exp\left[\frac{-(E_c - E_F) + (E_d - E_F)}{kT}\right] = \frac{N_c}{2N_d} \exp\left[-\frac{E_c - E_d}{kT}\right]$$

The unknown Fermi level cancels. So

$$\frac{n_d}{n_d + n_0} = \frac{1}{1 + \dfrac{N_c}{2N_d} \exp\left[-\dfrac{E_c - E_d}{kT}\right]}$$

(Neamen, Eq. 4.55).

Step 5: Put in the numbers

Take phosphorus in Si at $T = 300$ K: $E_c - E_d = 0.045$ eV, $kT = 0.0259$ eV, $N_c = 2.8 \times 10^{19}$ cm^{-3}, and $N_d = 10^{16}$ cm^{-3}.

$$\frac{N_c}{2N_d} = \frac{2.8 \times 10^{19}}{2 \times 10^{16}} = 1400, \qquad \exp\left(-\frac{0.045}{0.0259}\right) = 0.1760$$

$$\frac{n_d}{n_d + n_0} = \frac{1}{1 + 1400 \times 0.1760} = \frac{1}{1 + 246.4} = 4.04 \times 10^{-3}$$

only 0.40% of the electrons stay on donors: 99.6% of the donors are ionized

Steps to remember

Donor occupation with the factor $\frac{1}{2} \rightarrow n_d \approx 2N_d e^{-(E_d - E_F)/kT} \rightarrow n_0 = N_c e^{-(E_c - E_F)/kT} \rightarrow$ the ratio cancels $E_F \rightarrow$ fraction $1/[1 + (N_c/2N_d)e^{-(E_c - E_d)/kT}] \rightarrow 0.4$ % at 300 K.

The same formula for other doping levels:

N_d (cm^{-3})	$\frac{N_c}{2N_d} e^{-(E_c - E_d)/kT}$	in donor states	ionized
10^{15}	2464	0.041 %	99.96 %
10^{16}	246.4	0.40 %	99.60 %
10^{17}	24.6	3.9 %	96.1 %

Table 6: Summary

So at 300 K the donors in Si are essentially all ionized, and $n_0 = N_d$ is a very good approximation. The reason is that $E_c - E_d = 0.045$ eV is only $1.7\,kT$, while the conduction band offers far more states (N_c) than there are donors.

Limits. At heavy doping (10^{18} cm^{-3} and above) the same formula gives about 29 % un-ionized, so ionization is no longer complete (and the material starts to become degenerate). At low temperature it fails too: for $N_d = 10^{15}$ cm^{-3} only 15 % of donors are ionized at 50 K (freeze-out), as in the n_0/N_d curve in the Theory.

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3. Keyword Explanation

1. Ionization energy

$E_c - E_d$ for a donor (or $E_a - E_v$ for an acceptor): the energy needed to free the bound carrier. About 0.045 eV for P and 0.045 eV for B in Si.

2. Extrinsic region

The temperature range in which all impurities are ionized and n_i is still negligible, so the carrier density equals the net doping.

3. Degeneracy factor

The factor $\frac{1}{2}$ (donors) or 4 (acceptors) in the occupation of an impurity level, from the spin and band-structure counting of the states.

4. Notes and Remarks

- **Where the approximation came from:** the fraction formula assumes $E_d - E_F \gg kT$, which is true at 300 K for $N_d = 10^{16}$ cm^{-3} ($E_d - E_F = 0.16$ eV $= 6.2\,kT$). Solving charge neutrality exactly (self-consistent E_F) gives 99.60 % at 10^{16}, 96.2 % at 10^{17} and 76 % at 10^{18} cm^{-3}: the same conclusion.
- **Low-temperature example:** 1(d) works the same donor statistics at 30 K, where almost all donors are frozen out.
- **Heavy doping:** above about 10^{18} cm^{-3} the donor levels merge into an impurity band and the ionization energy falls, so the simple formula is only a rough guide there.
- **Acceptors:** boron in Si has $E_a - E_v = 0.045$ eV too, so p-type Si is also completely ionized at room temperature.

6.(c) Explain production of inverse layer at oxide and semiconductor layer with proper sketches of energy band diagram. Marks: 5

1. Theory

1. The MOS capacitor (Core Idea)

A MOS structure is a **metal** gate, a thin **oxide** (SiO_2, an insulator) and a **semiconductor** body. Here the body is p-type, with acceptor concentration N_a. No current crosses the oxide. So whatever voltage we put on the gate, the silicon stays in equilibrium and its Fermi level E_F stays **flat** (the same everywhere in the silicon). The structure, accumulation, depletion and inversion are built in 4(b); here we follow how the **inversion layer** forms step by step.

We use the **ideal** MOS: no work-function difference and no oxide charge, so at $V_G = 0$ the bands are flat.

2. The Fermi potential of the body

In p-type silicon E_F lies below the intrinsic level E_{Fi}. The distance is the **Fermi potential**:

$$\phi_{fp} = \frac{kT}{e} \ln\left(\frac{N_a}{n_i}\right)$$

where $e\phi_{fp} = E_{Fi} - E_F$ in the bulk, $kT/e = 0.0259$ V and $n_i = 1.5 \times 10^{10}$ cm^{-3}. For $N_a = 10^{16}$ cm^{-3}:

$$\phi_{fp} = 0.0259 \ln\left(\frac{10^{16}}{1.5 \times 10^{10}}\right) = 0.0259 \times 13.41 = 0.347 \text{ V}$$

3. Band bending and the surface potential

A positive gate voltage puts positive charge on the gate. The field it makes points from the gate into the silicon. It pushes the holes (positive) **away** from the surface. Near the surface the negative acceptor ions are left uncovered: a **depletion region** of width x_d with charge $-eN_a$ per unit volume.

The potential in the silicon rises towards the surface, so the electron energy bands **bend down**. The total band bending is $e\phi_s$, where ϕ_s is the **surface potential** (the potential at the surface relative to

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the bulk). With the depletion approximation and Poisson's equation (see 2(c) for the same calculation in a PN junction):

$$x_d = \sqrt{\frac{2\epsilon_s \phi_s}{eN_a}}$$

where $\epsilon_s = 11.7\epsilon_0 = 1.035 \times 10^{-12}$ F/cm.

4. Why the surface turns n-type

The carrier densities follow the distance of E_F from E_{Fi} (1(a)):

$$n = n_i \exp\left(\frac{E_F - E_{Fi}}{kT}\right), \qquad p = n_i \exp\left(\frac{E_{Fi} - E_F}{kT}\right)$$

As the bands bend down, E_{Fi} at the surface moves down towards the flat E_F. Step by step:

- **Depletion, $0 < \phi_s < \phi_{fp}$:** E_{Fi} is still above E_F at the surface. The surface has fewer holes, but it is still p-type.
- **$\phi_s = \phi_{fp}$:** $E_{Fi} = E_F$ at the surface. The surface is intrinsic: $n = p = n_i$.
- **Weak inversion, $\phi_{fp} < \phi_s < 2\phi_{fp}$:** E_{Fi} goes below E_F at the surface. Now $n > p$ there: the surface has become **n-type**, the opposite of the body. This is **inversion**. There are still few electrons.
- **Threshold (strong) inversion, $\phi_s = 2\phi_{fp}$:** E_{Fi} at the surface is as far **below** E_F as it is **above** E_F in the bulk. So the surface electron density equals the bulk hole density, $n_s = N_a$. This is the definition of the **threshold inversion point**; the gate voltage that reaches it is the threshold voltage V_T.

The band diagrams of these stages, drawn to scale for $N_a = 10^{16}$ cm^{-3}, and the charge at threshold are sketched in the Answer.

5. Where do the electrons come from?

The p-type body has very few electrons ($n_0 = n_i^2/N_a = 2.25 \times 10^4$ cm^{-3}). The inversion electrons come from **thermal generation** of electron–hole pairs in the depletion region (the field sweeps the electrons to the surface and the holes into the body), and in a MOSFET mainly from the n$^+$ **source**, which supplies them almost at once.

6. The maximum depletion width

After threshold, a small extra rise of ϕ_s gives a huge rise of the electron density (it grows as $\exp(e\phi_s/kT)$). The extra gate charge is then balanced by the inversion electrons, not by more depletion. So the depletion width stops growing at its threshold value, the **maximum space charge width**:

$$x_{dT} = \sqrt{\frac{2\epsilon_s(2\phi_{fp})}{eN_a}} = \sqrt{\frac{4\epsilon_s \phi_{fp}}{eN_a}}$$

At threshold the gate charge Q'_m is balanced by two charges in the silicon: the thin sheet of inversion electrons Q'_n and the depletion charge $Q'_{SD}(max) = -eN_a x_{dT}$.

2. Answer

Set-up. Take an ideal MOS structure on a **p-type** substrate (acceptors N_a) and apply a **positive** gate voltage V_G. In the bulk E_F is below E_{Fi} by $e\phi_{fp}$, with $\phi_{fp} = (kT/e)\ln(N_a/n_i)$. No current flows through the oxide, so E_F in the silicon stays flat.

Step 1: Depletion. The positive gate charge repels the holes from the surface, leaving negative acceptor ions: a depletion region of width $x_d = \sqrt{2\epsilon_s\phi_s/(eN_a)}$. The bands bend *down* near the surface by $e\phi_s$ (surface potential ϕ_s). While $\phi_s < \phi_{fp}$, E_{Fi} is still above E_F at the surface: the surface is depleted but still p-type (panel a).

Step 2: Surface becomes intrinsic, then n-type. At $\phi_s = \phi_{fp}$, E_{Fi} meets E_F at the surface ($n = p = n_i$, panel b). For larger V_G, E_{Fi} crosses **below** E_F at the surface. Since $n = n_i \exp[(E_F - E_{Fi})/kT]$,

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the surface now has more electrons than holes: an **n-type inversion layer** has formed on the p-type body (weak inversion).

Step 3: Threshold inversion. When $\phi_s = 2\phi_{fp}$, E_{Fi} at the surface is $e\phi_{fp}$ below E_F, the mirror of the bulk. Then the surface electron density equals the bulk hole density, $n_s = N_a$ (panel c). This is the **threshold inversion point**, reached at $V_G = V_T$. The depletion width has its maximum value

$$x_{dT} = \sqrt{\frac{4\epsilon_s \phi_{fp}}{eN_a}}$$

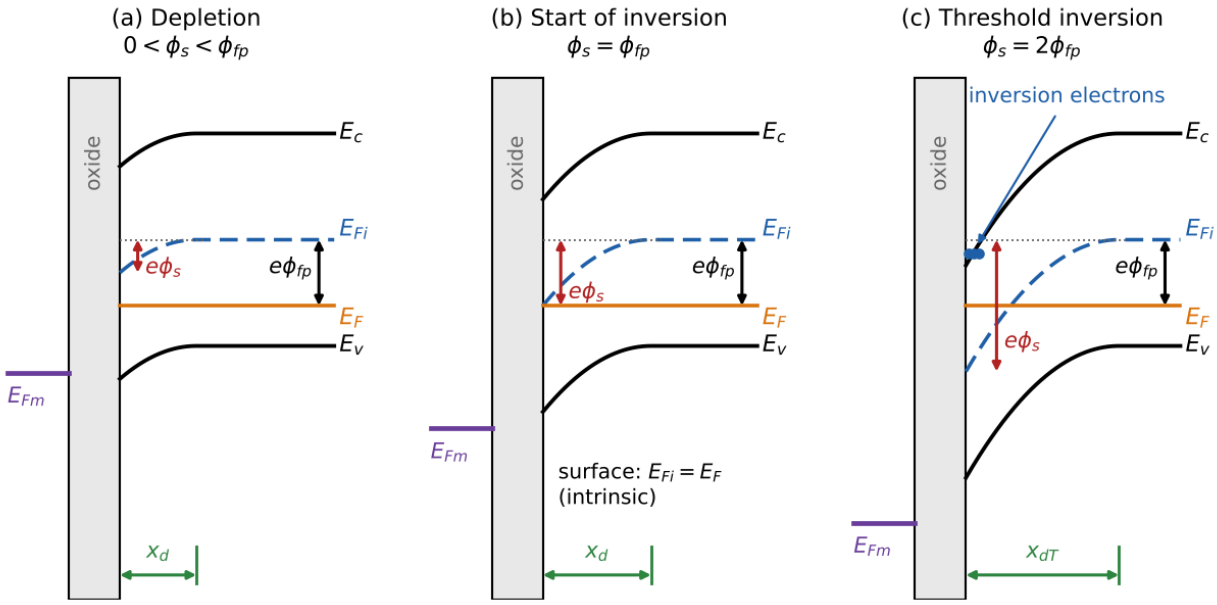


Figure 46: Band diagrams of a p-type MOS structure for $N_a = 10^{16}$ cm^{-3}: (a) depletion, (b) start of inversion, (c) threshold inversion, with E_c, E_v, E_{Fi}, E_F, $e\phi_{fp}$ and $e\phi_s$ marked (silicon drawn to scale)

Reading the figure. In each panel the gate is on the left (its Fermi level E_{Fm}, solid purple bar), then the oxide (grey box, not to scale), then the silicon. E_c and E_v are solid black, E_{Fi} is the dashed blue curve, E_F is the solid orange line, flat. The black double arrow on the right is $e\phi_{fp} = 0.347$ eV; the red double arrow at the surface is the band bending $e\phi_s$; the green arrow is the depletion width. In (a) $\phi_s = \phi_{fp}/2$ and $x_d = 0.15$ μm. In (b) $\phi_s = \phi_{fp}$: the dashed blue E_{Fi} touches E_F at the surface, $x_d = 0.21$ μm. In (c) $\phi_s = 2\phi_{fp} = 0.695$ V: E_{Fi} is 0.347 eV below E_F at the surface, E_c comes close to E_F, and electrons (filled blue dots) gather in a thin layer at the surface. The depletion width has reached $x_{dT} = 0.30$ μm. The gate Fermi level drops further below E_F as V_G grows (by eV_G).

Step 4: Beyond threshold. The electron density grows exponentially with ϕ_s, so a small extra band bending adds many electrons. Any extra gate charge is balanced by the inversion layer, and x_d stays at x_{dT}. The electrons come from thermal generation in the depletion region, and in a MOSFET from the n$^+$ source. This thin electron sheet is the **channel**.

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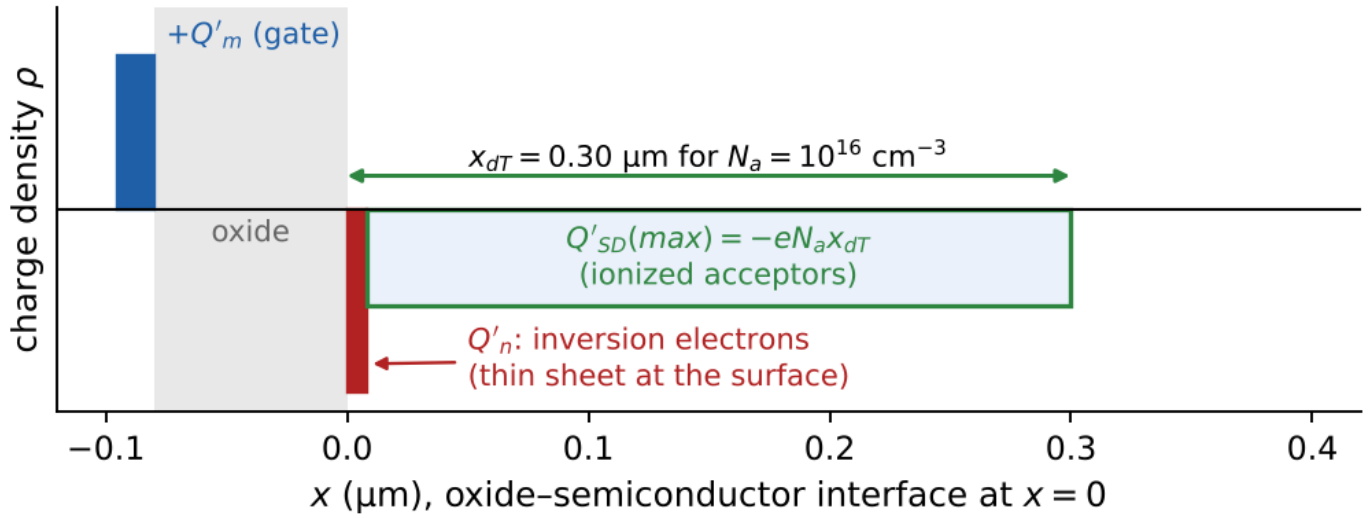


Figure 47: Charge distribution at threshold inversion: gate charge, thin inversion electron sheet, and acceptor depletion charge over x_{dT}

Reading the figure. The tall bar at the left (solid blue) is the positive charge Q'_m on the gate. The oxide (grey) holds no charge. In the silicon there is a thin, very dense sheet of inversion electrons Q'_n right at the surface (solid red bar) and the depletion charge of ionized acceptors $-eN_a$ spread over x_{dT} (green outlined box). Charge balance: $Q'_m = |Q'_n| + |Q'_{SD}(max)|$.

Example

($N_a = 10^{16}$ cm^{-3}, 300 K):

$$\phi_{fp} = 0.0259 \ln\left(\frac{10^{16}}{1.5\times10^{10}}\right) = 0.347 \text{ V}, \qquad 2\phi_{fp} = 0.695 \text{ V}$$

$$x_{dT} = \sqrt{\frac{4(1.035\times10^{-12})(0.347)}{(1.602\times10^{-19})(10^{16})}} = 3.00\times10^{-5} \text{ cm} = 0.30\ \mu\text{m}$$

So the bands must bend by about 0.69 eV at the surface to invert this substrate, and the depletion layer is then 0.30 μm deep.

3. Keyword Explanation

1. Inversion layer

A thin layer at the semiconductor surface whose carrier type is the opposite of the body (electrons on a p-type body).

2. Surface potential ϕ_s

The electric potential at the oxide–semiconductor interface, measured from the neutral bulk; $e\phi_s$ is the total band bending.

3. Threshold inversion point

The condition $\phi_s = 2\phi_{fp}$, where the surface electron density equals the bulk hole density.

4. Notes and Remarks

- **Charge at threshold:** the depletion charge per unit area is $|Q'_{SD}(max)| = eN_a x_{dT} = 4.80\times10^{-8}$ C/cm^2 for the example.
- **n-type body:** with a negative gate voltage the same steps give a p-type (hole) inversion layer on an n-type body; then the condition is $\phi_s = 2\phi_{fn}$.

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- **Real MOS:** the work-function difference ϕ_{ms} and oxide charge Q'_{ss} shift the gate voltage needed, but not the band picture in the silicon. V_T itself is treated in 4(b).
- **Weak vs strong inversion:** some texts call $\phi_{fp} < \phi_s < 2\phi_{fp}$ weak inversion and $\phi_s > 2\phi_{fp}$ strong inversion, as here; the $2\phi_{fp}$ point is a convenient definition, not a sharp physical change.

6.(d) A silicon PN junction at $T = 300$ K with zero applied bias has doping concentration of $N_d = 5 \times 10^{16}$ cm^{-3} and $N_a = 5 \times 10^{15}$ cm^{-3}. Determine built-in potential, space charge width and electric field at metallurgical region ($\epsilon_t = 11.7$ [*sic*: should be $\epsilon_r = 11.7$]) Marks: 10

1. Theory

Note on the paper

The paper writes the permittivity as $\epsilon_t = 11.7$. A permittivity of 11.7 (with no unit) can only be the **relative** permittivity (dielectric constant) of silicon, $\epsilon_r = 11.7$. So we use $\epsilon_s = \epsilon_r \epsilon_0 = 11.7 \times 8.85 \times 10^{-14}$ F/cm.

1. The PN junction in equilibrium (Core Idea)

Join a p-type region (acceptors N_a, many holes) to an n-type region (donors N_d, many electrons). Electrons diffuse from n to p and holes from p to n. They leave behind fixed ions: positive donors on the n side and negative acceptors on the p side. This region with no free carriers is the **space charge region** (or depletion region). Its charge makes an electric field pointing from n to p, which stops further diffusion. In equilibrium the Fermi level is flat, and the bands bend by eV_{bi} across the junction. The band diagram is built in 3(b) and the electrostatics in 2(c); here is a short summary.

2. The built-in potential

On the n side, E_F is above E_{Fi} by $e|\phi_{Fn}|$; on the p side it is below by $e\phi_{Fp}$. With E_F flat, E_{Fi} must drop by the sum of the two across the junction:

$$V_{bi} = |\phi_{Fn}| + |\phi_{Fp}| = \frac{kT}{e} \ln\left(\frac{N_d}{n_i}\right) + \frac{kT}{e} \ln\left(\frac{N_a}{n_i}\right) = \frac{kT}{e} \ln\left(\frac{N_a N_d}{n_i^2}\right)$$

where

- V_{bi} is the built-in potential barrier (V)
- N_a, N_d are the acceptor and donor concentrations (cm^{-3}), assumed completely ionized (6(b))
- n_i is the intrinsic carrier concentration (1.5×10^{10} cm^{-3} for Si at 300 K)

3. Charge, field and potential from Poisson's equation

With the **depletion approximation** (no free carriers inside $-x_p < x < x_n$, and sharp edges), the charge density is:

$$\rho(x) = -eN_a \quad (-x_p < x < 0), \qquad \rho(x) = +eN_d \quad (0 < x < x_n)$$

Poisson's equation $dE/dx = \rho/\epsilon_s$ gives a field that is linear in x on each side and zero at both edges:

$$E(x) = -\frac{eN_a}{\epsilon_s}(x + x_p) \quad (-x_p \le x \le 0), \qquad E(x) = -\frac{eN_d}{\epsilon_s}(x_n - x) \quad (0 \le x \le x_n)$$

The field is negative (it points from n to p, the $-x$ direction). Its magnitude is largest at the **metallurgical junction** $x = 0$. The field must be continuous at $x = 0$, which gives **charge balance**:

$$N_a x_p = N_d x_n$$

The total positive charge equals the total negative charge. So the space charge region spreads mostly into the **lightly doped** side.

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4. The width of the space charge region

The potential is the negative integral of the field. Its total change across the region is the area of the field triangle, which must equal V_{bi}:

$$V_{bi} = \frac{1}{2}|E_{max}|(x_n + x_p)$$

Combining this with charge balance gives (Neamen):

$$W = x_n + x_p = \sqrt{\frac{2\epsilon_s V_{bi}}{e}\left(\frac{N_a + N_d}{N_a N_d}\right)}$$

$$x_n = W\frac{N_a}{N_a + N_d}, \qquad x_p = W\frac{N_d}{N_a + N_d}$$

and the peak field at $x = 0$:

$$|E_{max}| = \frac{eN_d x_n}{\epsilon_s} = \frac{eN_a x_p}{\epsilon_s} = \frac{2V_{bi}}{W}$$

The three forms must agree; that is a good check on the arithmetic.

2. Answer

Given:

- Silicon, $T = 300$ K, zero bias ($V_R = 0$)
- $N_d = 5 \times 10^{16}$ cm^{-3} (n side), $N_a = 5 \times 10^{15}$ cm^{-3} (p side)
- $n_i = 1.5 \times 10^{10}$ cm^{-3}, $kT/e = 0.0259$ V, $e = 1.602 \times 10^{-19}$ C
- $\epsilon_s = \epsilon_r \epsilon_0 = 11.7 \times 8.85 \times 10^{-14} = 1.0354 \times 10^{-12}$ F/cm (the paper's ϵ_t read as ϵ_r)

Relationships:

$$V_{bi} = \frac{kT}{e}\ln\left(\frac{N_a N_d}{n_i^2}\right), \qquad W = \sqrt{\frac{2\epsilon_s V_{bi}}{e}\left(\frac{N_a + N_d}{N_a N_d}\right)}, \qquad |E_{max}| = \frac{eN_d x_n}{\epsilon_s}$$

where V_{bi} is the built-in potential, $W = x_n + x_p$ the space charge width, x_n and x_p its parts in the n and p regions, and E_{max} the field at the metallurgical junction.

Step 1: Built-in potential

$$\frac{N_a N_d}{n_i^2} = \frac{(5 \times 10^{15})(5 \times 10^{16})}{(1.5 \times 10^{10})^2} = \frac{2.5 \times 10^{32}}{2.25 \times 10^{20}} = 1.111 \times 10^{12}$$

$$V_{bi} = 0.0259 \ln\left(1.111 \times 10^{12}\right) = 0.0259 \times 27.736$$

$$\boxed{V_{bi} = 0.718 \text{ V}}$$

Check: $\phi_{Fn} = 0.0259\ln(N_d/n_i) = 0.389$ V and $\phi_{Fp} = 0.0259\ln(N_a/n_i) = 0.329$ V, and $0.389 + 0.329 = 0.718$ V ✔. Electrons on the n side see a barrier of 0.718 eV towards the p side.

Step 2: Space charge width

$$\frac{N_a + N_d}{N_a N_d} = \frac{5.5 \times 10^{16}}{2.5 \times 10^{32}} = 2.2 \times 10^{-16} \text{ cm}^3$$

$$\frac{2\epsilon_s V_{bi}}{e} = \frac{2(1.0354 \times 10^{-12})(0.7184)}{1.602 \times 10^{-19}} = 9.286 \times 10^{6} \text{ cm}^{-1}$$

$$W = \sqrt{(9.286 \times 10^{6})(2.2 \times 10^{-16})} = \sqrt{2.043 \times 10^{-9}} = 4.520 \times 10^{-5} \text{ cm}$$

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$$\boxed{W = 0.452\ \mu\text{m}}$$

Its two parts:

$$x_n = W\frac{N_a}{N_a + N_d} = 0.452 \times \frac{5 \times 10^{15}}{5.5 \times 10^{16}} = 0.0411\ \mu\text{m}$$

$$x_p = W\frac{N_d}{N_a + N_d} = 0.452 \times \frac{5 \times 10^{16}}{5.5 \times 10^{16}} = 0.411\ \mu\text{m}$$

$$\boxed{x_n = 0.0411\ \mu\text{m}, \qquad x_p = 0.411\ \mu\text{m}}$$

The p side is ten times more lightly doped, so 91 % of the width (x_p) lies in the p region. Check: $N_a x_p = (5 \times 10^{15})(4.109 \times 10^{-5}) = 2.05 \times 10^{11}$ cm^{-2} $= N_d x_n = (5 \times 10^{16})(4.109 \times 10^{-6})$ ✔.

Step 3: Electric field at the metallurgical junction

$$|E_{max}| = \frac{eN_d x_n}{\epsilon_s} = \frac{(1.602 \times 10^{-19})(5 \times 10^{16})(4.109 \times 10^{-6})}{1.0354 \times 10^{-12}}$$

$$\boxed{E_{max} = -3.18 \times 10^4\ \text{V/cm} \quad (|E_{max}| = 3.18 \times 10^4\ \text{V/cm} = 3.18\ \text{V}/\mu\text{m})}$$

The minus sign means the field points from the n side to the p side. Check: $eN_a x_p/\epsilon_s = 3.18 \times 10^4$ V/cm and $2V_{bi}/W = 2(0.7184)/(4.520 \times 10^{-5}) = 3.18 \times 10^4$ V/cm ✔.

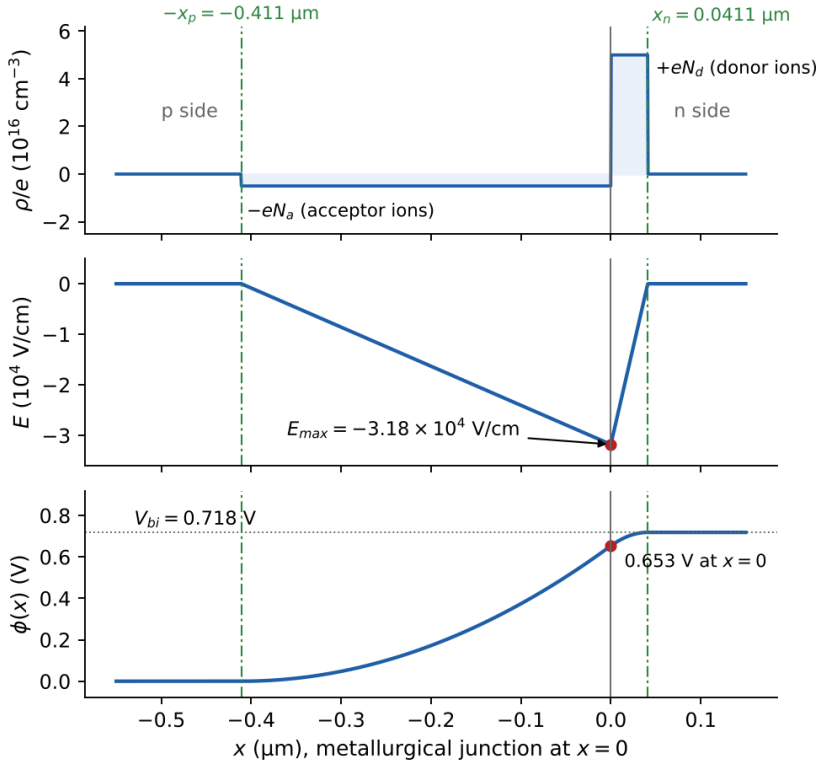


Figure 48: Space charge density, electric field and potential across the junction, drawn to scale with the computed values

Reading the figure. The dash-dot green lines mark the edges $-x_p = -0.411$ μm and $x_n = 0.0411$ μm; the thin grey line is the metallurgical junction. Top: the charge density (solid blue) is a shallow negative rectangle $-eN_a$ over the wide p part and a tall positive rectangle $+eN_d$ (ten times higher) over the narrow n part; the two shaded areas are equal. Middle: the field (solid blue) falls linearly from zero at $-x_p$ to -3.18×10^4 V/cm at $x = 0$ (red dot), then rises steeply back to zero at x_n. Bottom: the potential (solid blue) rises from 0 to $V_{bi} = 0.718$ V; 0.653 V of it (91 %) is dropped across the p side.

Final Answer:

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- Built-in potential: $V_{bi} = 0.718$ V
- Space charge width: $W = 0.452$ μm ($x_n = 0.0411$ μm, $x_p = 0.411$ μm)
- Field at the metallurgical junction: $E_{max} = -3.18 \times 10^4$ V/cm (magnitude 3.18×10^4 V/cm, pointing from n to p)

3. Keyword Explanation

1. Metallurgical junction

The plane $x = 0$ where the doping changes from p-type to n-type. The field is largest there.

2. Space charge region

The region around the junction emptied of free carriers, holding only the fixed ion charges; also called the depletion region.

3. Built-in potential

The potential difference between the neutral n and p regions in equilibrium, set by the doping; it cannot be measured with a voltmeter.

4. Depletion approximation

The assumption that the space charge region has no free carriers and sharp edges, so its charge is exactly $-eN_a$ and $+eN_d$.

4. Notes and Remarks

- **One-sided junction:** with $N_d = 10N_a$ the junction is close to one-sided. The one-sided formula $W \approx \sqrt{2\epsilon_s V_{bi}/(eN_a)} = 0.431$ μm is 5 % lower than the exact 0.452 μm, so the full formula is used here.
- **With reverse bias:** replace V_{bi} by $V_{bi} + V_R$ in W and E_{max}; the width grows as $\sqrt{V_{bi} + V_R}$.
- **Constants:** with $n_i = 1.0 \times 10^{10}$ cm^{-3} (used by some texts, e.g. Sze) V_{bi} would be 0.739 V; the answers above use Neamen's 1.5×10^{10} cm^{-3}.

5. Final Concise Answer

(a) A double-gate MOSFET has a thin (undoped) silicon body between two thin gate oxides, with a gate on each side tied together and n$^+$ source and drain at the ends; the FinFET is its practical form (a thin vertical fin with the gate wrapped over its sides). Benefits over a single gate: better electrostatic control of the channel, much smaller short-channel effects (DIBL, V_T roll-off), a subthreshold swing close to the ideal 59.6 mV/dec, lower leakage, higher drive current from two channels, no heavy body doping (higher mobility, no random dopant fluctuation), and scaling below about 22 nm.

(b) Complete ionization: every donor has given its electron to the conduction band, so $n_0 = N_d$. Freeze-out: at low temperature the electrons fall back into the donor states, $n_0 < N_d$, and at 0 K $n_0 = 0$. Yes, at room temperature: for P in Si ($E_c - E_d = 0.045$ eV) with $N_d = 10^{16}$ cm^{-3}, the fraction in donor states is $1/[1 + (N_c/2N_d)e^{-(E_c-E_d)/kT}] = 1/247.4 = 0.40\%$, so 99.6 % are ionized (99.96 % at 10^{15}, 96.1 % at 10^{17} cm^{-3}; ionization becomes incomplete only near 10^{18} cm^{-3}).

(c) On a p-type body a positive gate voltage repels holes and bends the bands down by $e\phi_s$: depletion for $\phi_s < \phi_{fp}$, an intrinsic surface at $\phi_s = \phi_{fp}$, and an n-type inversion layer once E_{Fi} falls below E_F at the surface. Threshold inversion is $\phi_s = 2\phi_{fp}$ ($n_s = N_a$), where the depletion width reaches its maximum $x_{dT} = \sqrt{4\epsilon_s\phi_{fp}/(eN_a)}$; the electrons come from thermal generation and the source. For $N_a = 10^{16}$ cm^{-3}: $\phi_{fp} = 0.347$ V and $x_{dT} = 0.30$ μm.

(d) $V_{bi} = 0.718$ V, $W = 0.452$ μm ($x_n = 0.0411$ μm, $x_p = 0.411$ μm), and $E_{max} = -3.18 \times 10^4$ V/cm at the metallurgical junction (taking the paper's $\epsilon_t = 11.7$ as ϵ_r).

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FORMULA SHEET

EEE421

Solid State Devices

Formula Sheet

Session 2020–21 paper

Every formula behind the 4th Year 1st Semester Final Examination, 2024

TOPICS

1 Carriers in equilibrium — 10 formulas

2 Dopants and charge neutrality — 6 formulas

3 Drift, diffusion and the Hall effect — 9 formulas

4 Excess carriers — 5 formulas

5 PN junction — 8 formulas

6 MOS structures — 3 formulas

7 Bipolar transistor — 4 formulas

Each topic is one table: **Title** · **Equation** · **Notation** · **Reference**.

Prepared by Nur Mohammad

October 2026

BACK OF THE COVER

Before you use the sheet

How to read a row

★ **Fermi–Dirac function**	$f_F(E) = \dfrac{1}{1+\exp\left(\dfrac{E-E_F}{kT}\right)}$	f_F = probability that a state at energy E holds an electron E_F = Fermi level	1(a)
the name you search for	the formula, exactly as derived	every new symbol with its unit	link: opens the derivation

In the exam

- Write the formula first, then put the numbers in: the formula alone often earns a mark.
- Read the symbols in the Notation column before you substitute, and check the units.
- Not sure where a formula comes from? Its Reference opens the derivation.
- The **Key formulas** page at the end gathers the rows marked ★.
- Lost a symbol? The **Symbol index** at the end links to the row where it first appears.

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1 Carriers in equilibrium **4**

2 Dopants and charge neutrality **5**

3 Drift, diffusion and the Hall effect **5**

4 Excess carriers **6**

5 PN junction **6**

6 MOS structures **7**

7 Bipolar transistor **7**

8 Key formulas **8**

9 Symbol index **10**

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Conventions (Neamen): e is the magnitude of the electron charge, $kT = 0.0259$ eV and $V_t = kT/e = 0.0259$ V at 300 K; $\epsilon_s = \epsilon_r \epsilon_0$ with $\epsilon_0 = 8.85 \times 10^{-14}$ F/cm. Si at 300 K: $N_c = 2.8 \times 10^{19}$ cm^{-3}, $N_v = 1.04 \times 10^{19}$ cm^{-3}, $n_i = 1.5 \times 10^{10}$ cm^{-3}, $E_g = 1.12$ eV, $\epsilon_r = 11.7$. References point to the solutions of the 2020–21 paper.

Columns: Title · Equation · Notation · Ref. (reference).

1. Carriers in equilibrium:

Title	Equation	Notation	Ref.
★ **Fermi–Dirac function**	$f_F(E) = \dfrac{1}{1+\exp\left(\dfrac{E-E_F}{kT}\right)}$	f_F = probability that a state at energy E holds an electron E_F = Fermi level	1(a)
Energy for a given probability	$E - E_F = kT \ln\left(\dfrac{1}{f_F} - 1\right)$	$f_F = 0.1$ gives $E - E_F = kT \ln 9 = 2.2\,kT$	2(d)
★ **Density of states**	$g_c(E) = \dfrac{4\pi(2m_n^*)^{3/2}}{h^3}\sqrt{E - E_c}$	g_c = states per unit energy per unit volume $m_n^* = 1.08\,m_0$ (Si)	1(c)
States from E_c to $E_c + kT$	$N = \dfrac{4\pi(2m_n^*)^{3/2}}{h^3} \cdot \dfrac{2}{3}(kT)^{3/2}$	Si, 300 K: $N = 2.13 \times 10^{19}$ cm^{-3}	1(c)
★ **Electron concentration**	$n_0 = N_c \exp\left[-\dfrac{E_c - E_F}{kT}\right]$	N_c = effective density of states in the conduction band	1(a)
★ **Hole concentration**	$p_0 = N_v \exp\left[-\dfrac{E_F - E_v}{kT}\right]$	N_v = effective density of states in the valence band	1(a)
Mass-action law	$n_0 p_0 = n_i^2 = N_c N_v \exp\left(-\dfrac{E_g}{kT}\right)$	n_i = intrinsic carrier concentration	1(a)
N_c at another temperature	$N_c(T) = N_c(300)\left(\dfrac{T}{300}\right)^{3/2}$	Si at 30 K: $N_c = 8.85 \times 10^{17}$ cm^{-3}	1(d)
★ **Fermi level vs doping**	$E_F - E_{Fi} = kT \ln \dfrac{N_d}{n_i}$ (n-type), $E_{Fi} - E_F = kT \ln \dfrac{N_a}{n_i}$ (p-type)	E_{Fi} = intrinsic Fermi level (near midgap)	1(a)
Fermi level from E_c	$E_c - E_F = kT \ln \dfrac{N_c}{n_0}$	$N_d = 10^{15}$ cm^{-3}, 300 K: 0.265 eV	1(d)

Table 1: Carriers in equilibrium

2. Dopants and charge neutrality:

Title	Equation	Notation	Ref.
★ **Charge neutrality**	$n_0 + N_a^- = p_0 + N_d^+$	N_a^-, N_d^+ = ionized acceptors and donors (cm^{-3})	3(a)
★ **Electrons, compensated**	$n_0 = \frac{N_d - N_a}{2} + \sqrt{\left(\frac{N_d - N_a}{2}\right)^2 + n_i^2}$	complete ionization; $n_0 \approx N_d - N_a$ when $N_d - N_a \gg n_i$	3(a)
★ **Holes, compensated**	$p_0 = \frac{N_a - N_d}{2} + \sqrt{\left(\frac{N_a - N_d}{2}\right)^2 + n_i^2}$	$p_0 = n_i^2/n_0$	3(a)
Donor occupation	$f_d = \frac{1}{1 + \frac{1}{2}\exp\left(\frac{E_d - E_F}{kT}\right)}$	E_d = donor level; the $\frac{1}{2}$ comes from the spin degeneracy	1(d)
★ **Fraction not ionized**	$\frac{n_d}{n_d + n_0} = \frac{1}{1 + \frac{N_c}{2N_d}\exp\left[-\frac{E_c - E_d}{kT}\right]}$	n_d = electrons still in donor states Si, 10^{15} cm^{-3}: 4.9×10^{-4} at 300 K	1(d), 6(b)
Freeze-out estimate	$n_0 \approx \sqrt{\frac{N_c N_d}{2}}\exp\left[-\frac{E_c - E_d}{2kT}\right]$	when $n_0 \ll N_d$; Si at 30 K: 1.33×10^{12} cm^{-3}	1(d)

Table 2: Dopants and charge neutrality

3. Drift, diffusion and the Hall effect:

Title	Equation	Notation	Ref.
★ **Drift velocity**	$v_d = \mu E$	μ = mobility ($\text{cm}^2/\text{V s}$) E = electric field (V/cm)	3(c)
★ **Mobility**	$\mu = \frac{e\tau}{m^*}$	τ = mean time between collisions (s) m^* = conductivity effective mass	3(c)
Conductivity	$\sigma = e(n\mu_n + p\mu_p)$	σ in $(\Omega\ \text{cm})^{-1}$; $\rho = 1/\sigma$	3(c)
★ **Temperature and mobility**	$\mu_L \propto T^{-3/2}$, $\mu_I \propto \frac{T^{3/2}}{N_I}$, $\frac{1}{\mu} = \frac{1}{\mu_L} + \frac{1}{\mu_I}$	μ_L, μ_I = lattice- and impurity-limited mobility N_I = ionized impurity concentration	4(a)
★ **Diffusion current**	$J_n = eD_n\frac{dn}{dx}$, $J_p = -eD_p\frac{dp}{dx}$	D_n, D_p = diffusion coefficients (cm^2/s)	1(b)
★ **Einstein relation**	$\frac{D_n}{\mu_n} = \frac{D_p}{\mu_p} = \frac{kT}{e}$	Si, 300 K: $D_n = 35$, $D_p = 12.4$ cm^2/s	2(b)
★ **Hall voltage**	$V_H = \frac{I_x B_z}{epd}$ (p-type), $V_H = -\frac{I_x B_z}{end}$ (n-type)	d = thickness along B_z; the sign gives the carrier type	5(d)
★ **Carrier density, Hall**	$n = -\frac{I_x B_z}{edV_H}$	Fig. 1: $n = 1.0 \times 10^{16}$ cm^{-3}	5(d)

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Title	Equation	Notation	Ref.
★ **Hall mobility**	$\mu_n = \dfrac{I_x L}{enV_x W d}$	L = length along x, W = width along y Fig. 1: $\mu_n = 833$ cm^2/V s	5(d)

Table 3: Drift, diffusion and the Hall effect

4. Excess carriers:

Title	Equation	Notation	Ref.
★ **Poisson's equation**	$\dfrac{d^2\phi}{dx^2} = -\dfrac{\rho(x)}{\epsilon_s} = -\dfrac{dE}{dx}$	ϕ = potential (V), ρ = charge density (C/cm^3)	1(b)
★ **Ambipolar transport**	$D'\dfrac{\partial^2(\delta n)}{\partial x^2} + \mu' E\dfrac{\partial(\delta n)}{\partial x} + g' - R' = \dfrac{\partial(\delta n)}{\partial t}$	p-type, low injection: $D' = D_n$, $\mu' = \mu_n$, $R' = \delta n/\tau_{n0}$	3(d)
n-type, low injection	$D_p\dfrac{\partial^2(\delta p)}{\partial x^2} - \mu_p E_0\dfrac{\partial(\delta p)}{\partial x} - \dfrac{\delta p}{\tau_{p0}} = \dfrac{\partial(\delta p)}{\partial t}$	$g' = 0$; $\delta n = \delta p$	5(a)
★ **Steady-state diffusion**	$\delta n(x) = \delta n(0)\exp\left(-\dfrac{x}{L_n}\right)$, $L_n = \sqrt{D_n\tau_{n0}}$	L_n = minority-electron diffusion length 3(d): $L_n = 45.8$ μm, 10^{14} cm^{-3} at $x = 179$ μm	3(d)
★ **Moving pulse**	$\delta p(x,t) = \dfrac{N_0}{\sqrt{4\pi D_p t}}\exp\left(-\dfrac{t}{\tau_{p0}}\right)\exp\left[-\dfrac{(x-\mu_p E_0 t)^2}{4D_p t}\right]$	N_0 = pairs created per unit area at $x = 0$, $t = 0$	5(a)

Table 4: Excess carriers

5. PN junction:

Title	Equation	Notation	Ref.
★ **Fermi potentials**	$\phi_{Fp} = V_t\ln\dfrac{N_a}{n_i}$, $\lvert\phi_{Fn}\rvert = V_t\ln\dfrac{N_d}{n_i}$	$e\phi_{Fp} = E_{Fi} - E_F$ (p side), $e\lvert\phi_{Fn}\rvert = E_F - E_{Fi}$ (n side)	3(b)
★ **Built-in potential**	$V_{bi} = \lvert\phi_{Fn}\rvert + \lvert\phi_{Fp}\rvert = V_t\ln\dfrac{N_a N_d}{n_i^2}$	6(d): $V_{bi} = 0.718$ V	3(b), 6(d)
Charge balance	$N_a x_p = N_d x_n$	x_n, x_p = depletion widths on the n and p sides	2(c)
★ **Depletion width**	$W = x_n + x_p = \left\{\dfrac{2\epsilon_s V_{bi}}{e}\left[\dfrac{N_a + N_d}{N_a N_d}\right]\right\}^{1/2}$	add the reverse bias: $V_{bi} \to V_{bi} + V_R$ 6(d): $W = 0.452$ μm	2(c)
Each side	$x_n = W\dfrac{N_a}{N_a + N_d}$, $x_p = W\dfrac{N_d}{N_a + N_d}$	the lightly doped side holds most of W	2(c)
★ **Peak field**	$E_{max} = \dfrac{eN_d x_n}{\epsilon_s} = \dfrac{eN_a x_p}{\epsilon_s} = \dfrac{2V_{bi}}{W}$	at the metallurgical junction; 6(d): 3.18×10^4 V/cm	6(d)

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Title	Equation	Notation	Ref.
★ **Law of the junction**	$p_n(x_n) = p_{n0} \exp\left(\frac{eV_a}{kT}\right)$, $n_p(-x_p) = n_{p0} \exp\left(\frac{eV_a}{kT}\right)$	$V_a > 0$ forward, $V_a < 0$ reverse; $p_{n0} = n_i^2/N_d$	4(d)
Majority carriers at the edge	$n_n = N_d + \delta p_n$	charge neutrality; low injection needs $\delta p_n \ll N_d$	4(d)

Table 5: PN junction

6. MOS structures:

Title	Equation	Notation	Ref.
★ **Fermi potential, p substrate**	$\phi_{fp} = V_t \ln \frac{N_a}{n_i}$	$e\phi_{fp} = E_{Fi} - E_F$ in the bulk	4(b)
★ **Threshold inversion**	$\phi_s = 2\phi_{fp}$	ϕ_s = surface potential (band bending); electrons at the surface equal holes in the bulk	6(c)
Maximum depletion width	$x_{dT} = \left[\frac{4\epsilon_s \phi_{fp}}{eN_a}\right]^{1/2}$	$N_a = 10^{16}$ cm^{-3}: $\phi_{fp} = 0.347$ V, $x_{dT} = 0.30$ μm	6(c)

Table 6: MOS structures

7. Bipolar transistor:

Title	Equation	Notation	Ref.
★ **Equilibrium minority carriers**	$p_{E0} = \frac{n_i^2}{N_E}$, $n_{B0} = \frac{n_i^2}{N_B}$, $p_{C0} = \frac{n_i^2}{N_C}$	N_E, N_B, N_C = emitter, base, collector doping	4(c)
★ **Base edges**	$n_B(0) = n_{B0} \exp\left(\frac{V_{BE}}{V_t}\right)$, $n_B(x_B) = n_{B0} \exp\left(\frac{V_{BC}}{V_t}\right)$	x_B = neutral base width; forward active: $n_B(x_B) \approx 0$	5(b)
Emitter and collector edges	$p_E(0) = p_{E0} \exp\left(\frac{V_{BE}}{V_t}\right)$, $p_C(0) = p_{C0} \exp\left(\frac{V_{BC}}{V_t}\right)$	each measured from its own junction	5(b)
★ **Four modes**	active: $V_{BE} > 0$, $V_{CB} > 0$; saturation: $V_{BE} > 0$, $V_{CB} < 0$; cutoff: $V_{BE} < 0$, $V_{CB} > 0$; inverse: $V_{BE} < 0$, $V_{CB} < 0$	NPN; $V_{CB} > 0$ reverse-biases the B–C junction	4(c)

Table 7: Bipolar transistor

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8. Key formulas:

The formulas marked ★ in the sheet, on one page. Each title links back to its row.

Title	Equation
Fermi–Dirac function	$f_F(E) = \dfrac{1}{1+\exp\left(\dfrac{E-E_F}{kT}\right)}$
Density of states	$g_c(E) = \dfrac{4\pi(2m_n^*)^{3/2}}{h^3}\sqrt{E-E_c}$
Electron concentration	$n_0 = N_c \exp\left[-\dfrac{E_c-E_F}{kT}\right]$
Hole concentration	$p_0 = N_v \exp\left[-\dfrac{E_F-E_v}{kT}\right]$
Fermi level vs doping	$E_F - E_{Fi} = kT\ln\dfrac{N_d}{n_i}$ (n-type), $E_{Fi}-E_F = kT\ln\dfrac{N_a}{n_i}$ (p-type)
Charge neutrality	$n_0 + N_a^- = p_0 + N_d^+$
Electrons, compensated	$n_0 = \dfrac{N_d-N_a}{2} + \sqrt{\left(\dfrac{N_d-N_a}{2}\right)^2 + n_i^2}$
Holes, compensated	$p_0 = \dfrac{N_a-N_d}{2} + \sqrt{\left(\dfrac{N_a-N_d}{2}\right)^2 + n_i^2}$
Fraction not ionized	$\dfrac{n_d}{n_d+n_0} = \dfrac{1}{1+\dfrac{N_c}{2N_d}\exp\left[-\dfrac{E_c-E_d}{kT}\right]}$
Drift velocity	$v_d = \mu E$
Mobility	$\mu = \dfrac{e\tau}{m^*}$
Temperature and mobility	$\mu_L \propto T^{-3/2}$, $\mu_I \propto \dfrac{T^{3/2}}{N_I}$, $\dfrac{1}{\mu} = \dfrac{1}{\mu_L} + \dfrac{1}{\mu_I}$
Diffusion current	$J_n = eD_n\dfrac{dn}{dx}$, $J_p = -eD_p\dfrac{dp}{dx}$
Einstein relation	$\dfrac{D_n}{\mu_n} = \dfrac{D_p}{\mu_p} = \dfrac{kT}{e}$
Hall voltage	$V_H = \dfrac{I_x B_z}{epd}$ (p-type), $V_H = -\dfrac{I_x B_z}{end}$ (n-type)
Carrier density, Hall	$n = -\dfrac{I_x B_z}{edV_H}$
Hall mobility	$\mu_n = \dfrac{I_x L}{enV_x W d}$
Poisson's equation	$\dfrac{d^2\phi}{dx^2} = -\dfrac{\rho(x)}{\epsilon_s} = -\dfrac{dE}{dx}$
Ambipolar transport	$D'\dfrac{\partial^2(\delta n)}{\partial x^2} + \mu' E\dfrac{\partial(\delta n)}{\partial x} + g' - R' = \dfrac{\partial(\delta n)}{\partial t}$
Steady-state diffusion	$\delta n(x) = \delta n(0)\exp\left(-\dfrac{x}{L_n}\right)$, $L_n = \sqrt{D_n\tau_{n0}}$

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Title	Equation
Moving pulse	$\delta p(x,t) = \frac{N_0}{\sqrt{4\pi D_p t}} \exp\left(-\frac{t}{\tau_{p0}}\right) \exp\left[-\frac{(x-\mu_p E_0 t)^2}{4D_p t}\right]$
Fermi potentials	$\phi_{Fp} = V_t \ln\frac{N_a}{n_i}$, $\lvert\phi_{Fn}\rvert = V_t \ln\frac{N_d}{n_i}$
Built-in potential	$V_{bi} = \lvert\phi_{Fn}\rvert + \lvert\phi_{Fp}\rvert = V_t \ln\frac{N_a N_d}{n_i^2}$
Depletion width	$W = x_n + x_p = \left\{\frac{2\epsilon_s V_{bi}}{e}\left[\frac{N_a+N_d}{N_a N_d}\right]\right\}^{1/2}$
Peak field	$E_{max} = \frac{eN_d x_n}{\epsilon_s} = \frac{eN_a x_p}{\epsilon_s} = \frac{2V_{bi}}{W}$
Law of the junction	$p_n(x_n) = p_{n0}\exp\left(\frac{eV_a}{kT}\right)$, $n_p(-x_p) = n_{p0}\exp\left(\frac{eV_a}{kT}\right)$
Fermi potential, p substrate	$\phi_{fp} = V_t \ln\frac{N_a}{n_i}$
Threshold inversion	$\phi_s = 2\phi_{fp}$
Equilibrium minority carriers	$p_{E0} = \frac{n_i^2}{N_E}$, $n_{B0} = \frac{n_i^2}{N_B}$, $p_{C0} = \frac{n_i^2}{N_C}$
Base edges	$n_B(0) = n_{B0}\exp\left(\frac{V_{BE}}{V_t}\right)$, $n_B(x_B) = n_{B0}\exp\left(\frac{V_{BC}}{V_t}\right)$
Four modes	active: $V_{BE}>0$, $V_{CB}>0$; saturation: $V_{BE}>0$, $V_{CB}<0$; cutoff: $V_{BE}<0$, $V_{CB}>0$; inverse: $V_{BE}<0$, $V_{CB}<0$

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9. Symbol index:

Every symbol the sheet defines, in the order it first appears. The last column links to that row.

Symbol	Meaning	First used in
f_F	probability that a state at energy E holds an electron	Fermi–Dirac function
E_F	Fermi level	Fermi–Dirac function
g_c	states per unit energy per unit volume	Density of states
N_c	effective density of states in the conduction band	Electron concentration
N_v	effective density of states in the valence band	Hole concentration
n_i	intrinsic carrier concentration	Mass-action law
E_{Fi}	intrinsic Fermi level (near midgap)	Fermi level vs doping
N_a^-, N_d^+	ionized acceptors and donors (cm^{-3})	Charge neutrality
E_d	donor level; the $\frac{1}{2}$ comes from the spin degeneracy	Donor occupation
n_d	electrons still in donor states	Fraction not ionized
μ	mobility ($\mathrm{cm}^2/\mathrm{V\ s}$)	Drift velocity
E	electric field (V/cm)	Drift velocity
τ	mean time between collisions (s)	Mobility
m^*	conductivity effective mass	Mobility
μ_L, μ_I	lattice- and impurity-limited mobility	Temperature and mobility
N_I	ionized impurity concentration	Temperature and mobility
D_n, D_p	diffusion coefficients (cm^2/s)	Diffusion current
d	thickness along B_z; the sign gives the carrier type	Hall voltage
L	length along x, W = width along y	Hall mobility
ϕ	potential (V), ρ = charge density (C/cm^3)	Poisson's equation
L_n	minority-electron diffusion length	Steady-state diffusion
N_0	pairs created per unit area at $x = 0$, $t = 0$	Moving pulse
x_n, x_p	depletion widths on the n and p sides	Charge balance
ϕ_s	surface potential (band bending); electrons at the surface equal holes in the bulk	Threshold inversion
N_E, N_B, N_C	emitter, base, collector doping	Equilibrium minority carriers
x_B	neutral base width; forward active: $n_B(x_B) \approx 0$	Base edges

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